

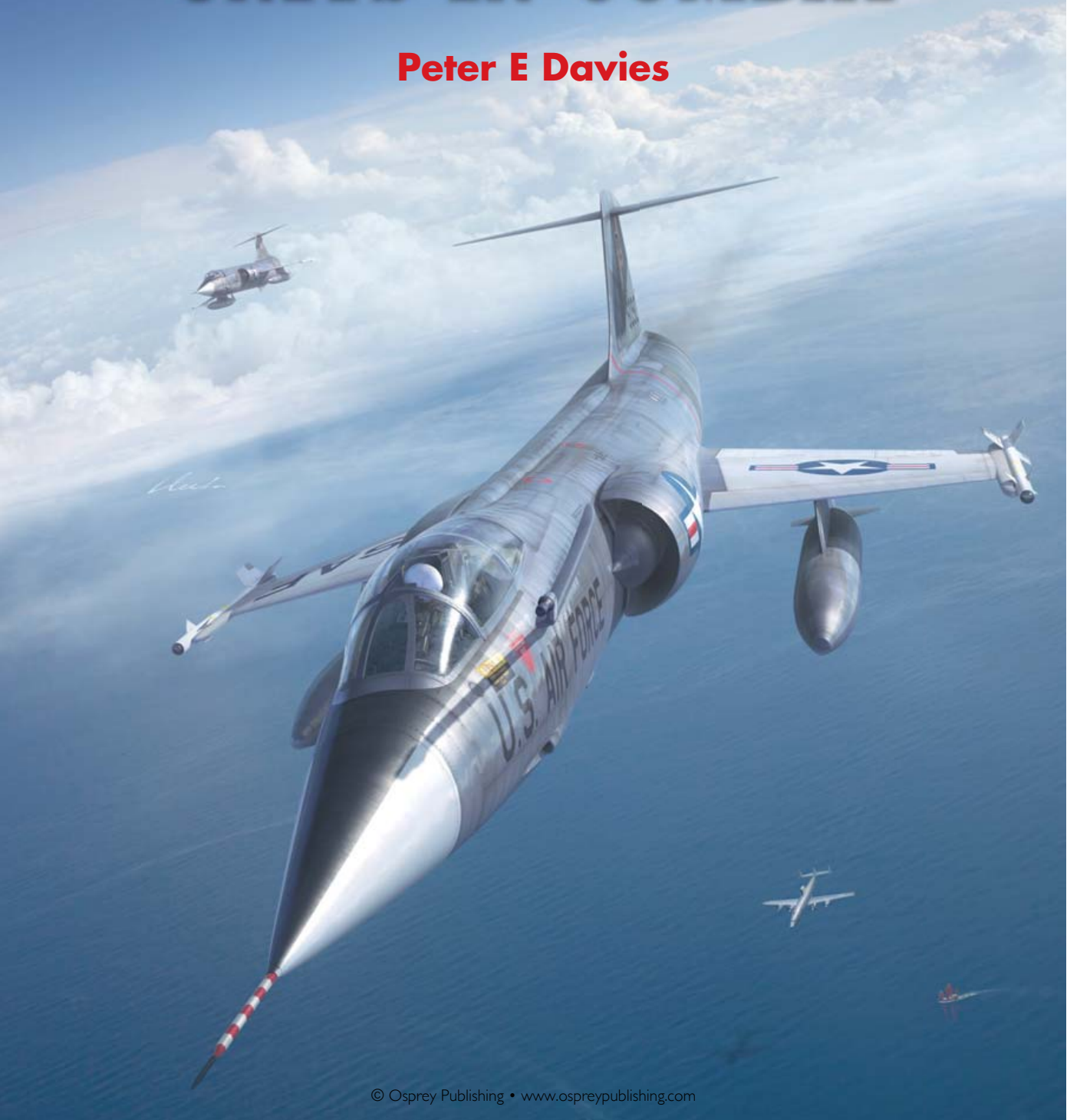


OSPREY COMBAT AIRCRAFT • 101



F-104 STARFIGHTER UNITS IN COMBAT

Peter E Davies



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FASTER AND HIGHER

In 2013 the F-104 Starfighter was still contributing to technological innovation. An early F-104A-10-LO (56-0763, once a chase aircraft for the SR-71A, XB-70 Valkyrie and North American X-15 rocket-powered research aircraft) christened the North American Eagle, fitted with a modified J79 engine, custom-built suspension and solid aluminium wheels, was being prepared for an attempt to break the 763-mph speed record on land.

Meanwhile, at NASA's Kennedy Space Center, Rick Svetkoff's Starfighters Aerospace operation uses his privately-owned fleet of nine F-104s for a variety of research and development programmes in support of the US government, academic institutions and commercial space ventures. Like NASA's now-retired F-104Ns, used for many years to train astronauts, Rick's F-104s are uniquely suited to this task. So much so that Starfighters Aerospace has the only Federal Aviation Administration-certified astronaut training programme in North America. In Svetkoff's words, 'We can go from ground to 23,000 ft as fast as some of the rockets launched here. High altitude and high speed is the key'.

These thoughts were also foremost in the mind of Clarence 'Kelly' Johnson some 61 years previously when he began to design the F-104 after USAF fighter pilots in Korea discovered that they needed much better performance for consistent air superiority over communist MiG-15s. The 1950s engendered unprecedentedly rapid development and change, both in aviation technology and defence requirements. The unexpected appearance of Soviet MiG-15 fighters in Korea in November 1950, when the Cold War turned hot, had kick-started a surge of fighter development to restore American aerial superiority. Aircraft designers had to devise complex solutions to unforeseen military situations in record time, requiring enormous expenditure and designers with vision, daring and talent approaching genius level.

Clarence 'Kelly' Johnson at the Lockheed Aircraft Company was one of a number of American designers with all these qualities. He had created the P-38 Lightning long-range interceptor, an innovative twin-boomed aircraft with a distinguished World War 2 record. Experience with compressibility early in the P-38's life showed Johnson that totally new concepts using jet power would be needed for significant increases in speed. Merely doubling the Lightning's power had increased top speed by only 17 mph. Johnson began jet fighter studies in 1939, and by 1942 he had a firm proposal, the L-133-02-01 with an advanced canard planform and two L-1000 jet engines designed by Lockheed's Nate Price. However, the USAAF considered it too challenging a project for them to sponsor.

In 1942 the Bell Aircraft Corporation produced America's first jet fighter, the P-59 Airacomet, but its low-powered General Electric J31-GE-5 engines in a high-drag airframe severely compromised its performance. The unexpected appearance of the Messerschmitt Me 262 jet fighter over Germany changed USAAF perceptions overnight, and a

First flights at Lockheed were usually timed to coincide with Kelly Johnson's birthday. The XF-104 missed it by one day after a year in the building process. Its temporary J65 engine installation limited maximum speed to Mach 1.49 in level flight. Together with the wing anhedral, the ventral fin added to production F-104s decreased the positive dihedral effect of the T-tail. It also significantly improved high-Mach directional stability and provided a convenient location for the radio antenna. The combination of ventral fin, anhedral and unswept wings eliminated 'Dutch roll' tendencies and facilitated turns above 350 knots. The first XF-104-LO is seen here with an F-100A, YF-102A and F-101A – the first four Century Series fighters (AFFTC/USAF)



request was issued for an equivalent jet fighter to be delivered within 180 days from 3 June 1943. Although the Lockheed factory was already producing 28 warplanes daily, Johnson managed to assemble a team of designers and they delivered his XP-80 prototype 37 days early. It was the first US production jet fighter, and the first to exceed 500 mph.

Although the USAF achieved a favourable kill-to-loss ratio against communist MiG-15s during the Korean War, F-80 and F-86 pilots generally requested more speed and faster rates of climb to reach the MiGs' combat ceiling when interviewed by Johnson's Lockheed team during a 21-base visit to South Korea to study operational use of USAF fighters. They also requested greater simplicity in operating their fighters' cockpits.

At a time when the USAF was demanding heavier and more complex fighters, Johnson (like his contemporary, designer Ed Heinemann with his A-4 Skyhawk) aimed for light weight, small size and simplicity. The result was a fighter with roughly three times the performance of an F-80. While designers in other countries were conceiving supersonic point-defence interceptors like the MiG-21 and English Electric Lightning, Johnson's new aircraft was intended as a clear air superiority fighter.

Johnson's team conceived the Model 83 (CL-246) in 1952 and showed it to USAF procurement decision makers Gen Don Putt and Col Bruce Holloway, who rapidly concocted a USAF request for the fighter as the XF-104, even though no operational requirement existed at the time. The

Model 83 embodied design features that were partly conventional but, in many cases, were revolutionary approaches to achieving the specialised goals that Johnson sought to achieve for America's first Mach 2 fighter, including ease of manufacturing and operation.

Most obvious among the innovations were the unprecedentedly thin, strong tapered wings, extending only 7.5 ft each side of the 54.75 ft (F-104A) fuselage and prompting the question, 'Where are the wings?' from test pilot Tony LeVier, who was one of the first people to see the highly-classified prototype. He was also the pilot for the XF-104's first flight on 4 March 1954. The overall span was a mere 21 ft 9 in, 5 ft 3 in less than the tiny 1930s Boeing P-26 'Peashooter'. One resultant advantage was a rate of roll exceeded only by the small-winged Northrop F-5/T-38.

With a thickness/chord ratio of only 3.36 per cent and a maximum depth of 4.2 in at the root and 1.96 in at the tip, there was no space within the wing for fuel or landing gear, all of which had to be contained in the sleek, tubular fuselage. The landing gear had back-up manual retraction and anti-skid brakes. Braking was assisted by an 18-ft ring-slot parachute. Later, an arresting hook was added that partially extended automatically when the drag 'chute was deployed. Additional fuel could be carried in 170-gallon wingtip tanks or in two underwing 195-gallon tanks. The heat-treated steel ailerons were powered by ten cylindrical 'piccolo' hydraulic boosters, incorporated in a metal block within a wing depth of 1.1 in. None of the control systems required external fairings. The biconvex aerofoil had a butter-knife sharp leading edge, with a 0.016 in radius, that was covered with detachable wing-guards to avoid the very real danger of injury to groundcrew.

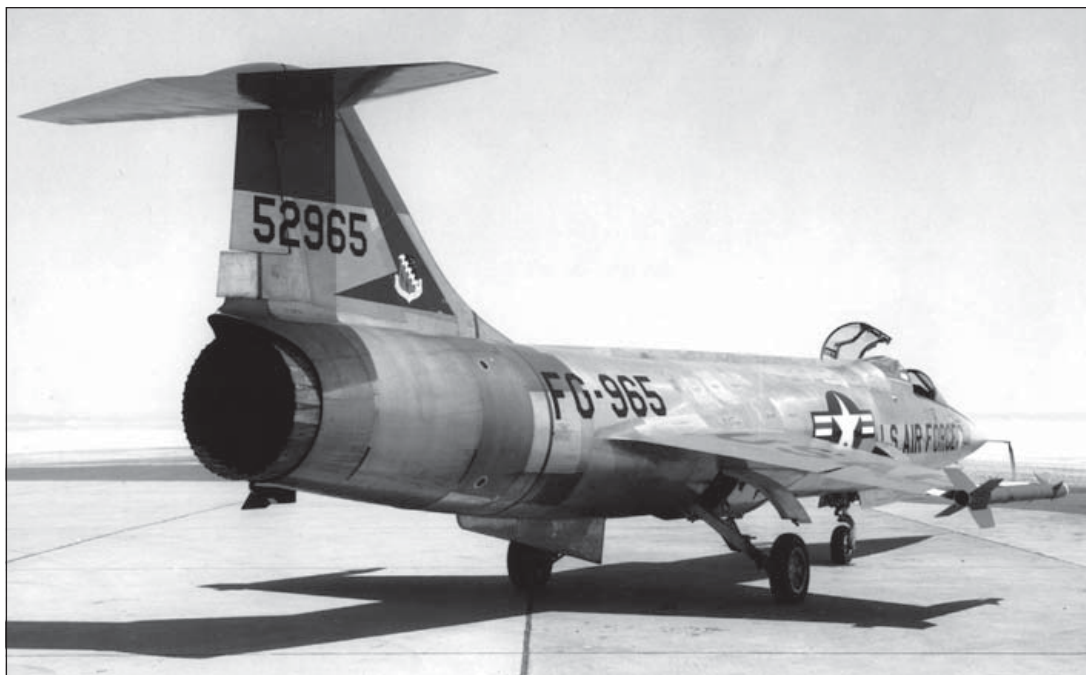
There was no 'carry through' structure joining the wings through the fuselage. The ribless wing structures were simply attached to five heavy-duty forgings that were tied into the tapered wing skins. To prevent twisting when flying controls were operated at high speeds, and to also provide adequate strength and rigidity, the wing skins were attached to a core structure manufactured from a single slab of aluminium. A similar but less rigid wing planform was used by the Douglas X-3 Stiletto experimental test aircraft that flew in 1952, although the X-3 lacked a suitably powerful engine to explore the high-speed potential of the wing.

The designers trialled various configurations before settling on a high 'T-tail' empennage arrangement. Extensive wind tunnel testing and free-flight investigation using 5-in rockets led to the conclusion that a T-tail minimised drag and provided the best stability and control across the broad speed and altitude envelope of the F-104. The placement of the horizontal stabiliser decreased interference drag and minimised trim change in the transonic and supersonic regimes. While most aircraft become nose-heavy as they transition to supersonic speed, the F-104 would experience only negligible trim changes, markedly improving manoeuvrability. The horizontal stabiliser acted as an end plate for the vertical tail, increasing the effectiveness of the relatively small tail while providing a positive dihedral effect. In fact the tail's dihedral effect proved too powerful, and after testing with Calspan's NT-33A variable stability aircraft, a ten-degree anhedral was applied to the Starfighter's wings to improve handling characteristics.

The Starfighter's handling was not without vice, however. One of the problems inherent in the overall design choices was that of pitch-up, as Lockheed test pilot G L 'Snake' Reaves explained;



Most of the early Starfighter testing was conducted by Lockheed's Chief Test Pilot, Tony LeVier (left) and Chief Engineering Test Pilot Herman 'Fish' Salmon (Lockheed)



YF-104A-1-LO 55-2965 with the USAF Flight Test Centre for AIM-9B missile tests. A straight wing provided more lift versus drag per square foot than a delta or swept wing, and appeared to give the lowest drag at Mach 2 and the lightest weight. A version swept at 26 degrees on its leading edge was decided upon after testing numerous configurations using 5-in high-velocity rockets with test wings bolted to them. For Johnson, these tests were crucial in selecting a suitable small wing that would avoid problems of flutter. The latter could not be effectively explored with wind-tunnel models without risking damage to the tunnel itself, so rocket tests were the answer. It was a pioneering wing format that was later seen, in modified form, in the Northrop F-5 and General Dynamics F-16 fighters (USAF)

'A fully developed pitch-up is best described as a non-repeatable stall manoeuvre that is inherent in aircraft with high tails. In the F-104 the choice of a thin, supersonic wing improved its resistance to stalling compared with a conventional subsonic wing. The supersonic wing continues to provide increasing lift at angles of attack [AoA] that would result in a stall for the subsonic wing. Total lift from the F-104 configuration does not decrease as the aircraft assumes extremely high AoA.'

At high AoA a large turbulent wake of air from the wing and fuselage covered the F-104's high horizontal stabiliser, causing the aircraft to lose its nose-down tendencies and become unstable in the pitch axis. Reeves and his test-pilot colleagues evolved techniques for recovery should this condition be allowed to occur, or even develop into a flat spin. Another test pilot, James 'Jake' Holliman, considered that the F-104 was 'beyond doubt the most pleasant handling piece of flying machinery that has taken the air since its grandfather, the F-80, launched the jet age [in the USA] back in 1944'. He also drew attention to the pitch-up issue, and the fact that 'the F-104 configuration continues to produce lift at AoA that would result in a fully-developed stall of a more conventional aircraft'.

To warn the pilot of a dangerously high AoA, an Automatic Pitch Control (APC) system was installed. When its indicator showed '5' on a dial in the cockpit the system would be activated, causing an automatic 'stick shaker' to vibrate the control column and emit a sinister 'machine gun' rattle. If the pilot still failed to react this was followed by an auto pitch control 'stick kicker' – a hydraulic ram that thrust the control column forwards, pushing the nose down. The system was installed at the USAF's insistence, partly due to serious pitch-up problems with the T-tailed McDonnell F-101 Voodoo and because it felt that all aircraft, including trainers, should have a 'shaker' warning system. Kelly Johnson

preferred an aural warning system (similar to the one later included in the F-4 Phantom II), as the F-104 was actually quite difficult to get into a dangerous pitch-up situation if the pilot avoided high pitch rates.

At subsonic speeds the wing itself provided enough stall warning even for novice pilots. However, at supersonic speeds there was no warning of an impending stall via the usual aerodynamic signals of heavy buffet and a tendency to roll, so a warning system was indeed necessary. The F-104 Pilot's Manual prohibited pitch-up, not because it was inherently dangerous (the F-104 refused to spin in all but the most extreme conditions), but because the aircraft's gyrations after a stall could damage the airframe. When it pitched down the aircraft could swing violently sideways into the airflow and the tail would then flip it from a right to a left roll. This sudden reversal caused several cases of twisted fuselages during F-104 flight tests to explore spin characteristics.

Early pilots emphasised that despite its unusual configuration the Starfighter essentially behaved like a conventional aircraft, but it required careful, smooth handling and would not tolerate inattentive or 'ham-fisted' piloting. In service use the findings of these early tests continued to provide an effective guide to safe flying for F-104 pilots, but failure to observe them undoubtedly became a major cause of the aircraft's unenviable safety record with several foreign air forces. As 'Jake' Holliman commented, 'Treat her rough and you may have a miserable time. Be smooth and gentle and you're in for a very pleasant experience. "Pitch-up" doesn't mean "give-up" if you have sufficient altitude to wait it out'.

MACH 2 POWER

Basic to all the F-104's qualities was the General Electric J79 variable-stator, axial-flow engine. When the first XF-104 (53-7786) made its maiden flight, it and the second prototype had to use a Buick-manufactured Wright J65-B-3 – a licence-built British Armstrong Siddeley Sapphire. This engine had an afterburner installed later in the programme as the J65-W-6, but the first XF-104 had already gone supersonic in a dive with the non-afterburning J65-B-3. Development of the revolutionary X-24A or J79 (the first engine capable of Mach 2 flight) ran contemporaneously with that of the F-104 as General Electric engineers led by Neil Burgess and Gerhard Neumann worked on the problems of achieving rapid acceleration without compressor stalling, low fuel consumption and reliable performance throughout a wide speed range.

The engine commenced test runs in June 1954 and a J79-GE-3 was installed in the first of 15 YF-104As (59-2955), which undertook its maiden flight on 17 February 1956. YF-104As had substantial structural changes, including a 5 ft 9 in fuselage extension from the XF-104's 49 ft 2 in to accommodate more fuel, a forward-retracting nose undercarriage and a revised rear fuselage with a ventral fin to aid stability.

Major revisions to the air inlets to cope with the extra power of the J79 introduced half-cone centre-bodies to generate multiple shockwaves at supersonic speeds. Incoming air had to be slowed to below Mach 1 to avoid choking the engine, and some of it by-passed the J79 and was used to cool the afterburner. The intakes were optimised for Mach 1.7 (or Mach 2 in versions with the more powerful J79-GE-19 engine) and the aircraft generated its maximum excess thrust at that speed. The J79

maintained approximately the same speed once it was wound up. Prototype J79-GE-3s had an un-modulated afterburner system that essentially gave the pilot the choice of either Mach 1.1 in full military power or Mach 2.2 with afterburner. All production aircraft had modulated afterburners, however.

On 27 April 1956 the first YF-104A reached Mach 2.2 – effectively the maximum speed for the aircraft before air intake temperature limits were exceeded. Normally, F-104As could just about reach Mach 2, and Mach 2.2 could only be attained in cold conditions. Vital to the high-speed performance of the J79-GE-3 was the ‘T2 reset’, as Maj Gen Stan Hood (a pilot with fighter experience stretching from the F-80 to the F-16) explained;

‘Air inlet air temperature determined whether T2 reset would cut in [at 100 degrees Centigrade]. It was like a second afterburner, and the engine would suddenly go to 103 per cent power. In a matter of a few seconds you were up to Mach 2 and beyond.’

High landing speed was another corollary of the small wing area, and this was ameliorated to some extent by the use of Attinello-type blown flaps from the YF-104A onwards. Bleed air from the engine’s 17th compressor stage was blown across the flaps to provide boundary layer control (BLC) at increasing pressure through 55 slots in titanium ducts when flaps were lowered between 15 and 45 degrees, reducing landing speed by more than 30 knots. Pilots also had to be ready to pull the flaps back to 15 degrees if there was a malfunction in the flap-blowing to one wing, causing an immediate roll at a dangerous moment in the landing approach. Supplying sufficient airflow to the flap-blowing system required high engine power while landing. A dead-stick landing without flap-blowing was possible, but with a touch-down speed of 240 knots very accurate judgement was crucial.

All F-104 versions had two flap settings. For take off, both leading and trailing edge flaps dropped to 15 degrees without BLC. For landing, the leading edge flaps dropped to 30 degrees and the trailing edge flaps to 45 degrees, with BLC applied progressively as the flaps extended past 15 degrees in the ‘land’ position. Pilots could also select the ‘take off’ (15 degrees) setting as ‘manoeuvre flaps’ at higher speeds to improve turn rate by about 1g. Pilots often used this facility, but with discretion to prevent undue loss of speed in combat with better-turning fighters.

The choice of a high-T tailplane out of 290 possible airframe configurations also raised the concern that no existing cartridge-driven ejection seat would be powerful enough to avoid impacting the tail at high speed up to Mach 2 without using a cartridge that would injure a pilot’s back due to its extreme acceleration. Having rejected the idea of a self-contained escape capsule, the designers, therefore, and with some reservations, included a seat that fired downwards through a hatch (which formed the cockpit floor) once the cockpit had been de-pressurised, the control column retracted and the pilot’s legs pulled back against the seat.

The first seats for the prototypes and the first 26 F-104As were Stanley B downward-ejection models with an M3 catapult initiator. These were replaced by the Stanley C and later C-1 with cables attached to ‘spurs’ on the pilot’s boots to pull his feet back safely against the seat front, while knee guards rotated downwards for further protection.

By 1960 all F-104A/Bs and subsequent models up to the two-seat F-104D were fitted with the Lockheed C-2 (licensed from Stanley) with an XM-10 two-tube rocket-catapult – one of the earliest rocket-powered seats. This was powerful enough to enable upward ejection without injurious acceleration rates, and many F-104 pilots preferred it to the Martin-Baker Mk GQ7(A) seat fitted in the F-104G and F-104S. The C-2 suited taller pilots, and its headrest allowed better rearward visibility, although the Martin-Baker seat had advantages at lower speeds and altitudes. The C-2 could be employed at ground level with 90 knots forward speed, using a zero-delay lanyard to cause immediate seat separation and parachute deployment. Canopy jettison was also activated by the seat ejection handle.

Ease of maintenance was a priority for the Starfighter's designers at a time when maintenance man-hours were constantly rising for a generation of increasingly complex aircraft. In due course this made the F-104A more attractive to users such as Pakistan and Taiwan, who were familiar with fairly basic jets like the F-86 Sabre.

Lockheed stated that the radar could be removed in ten minutes, the M61 gun could be stripped in 30 minutes via a single panel and the original downward-launching ejection seat could be removed through the lower fuselage hatch, allowing full cockpit access to a maintainer standing on the ground. Most of the engine accessories could be reached through a single access door so that servicing could be undertaken without removing the engine – a requirement on many other fighters at the time. An engine could be changed in an hour by four men after disconnecting the rear fuselage.

The electronics were ahead of their time in being mostly modular and contained in a series of similar 'Jeep cans' that fitted into a shock-mounted electronics rack in a large bay behind the cockpit. They could be replaced in minutes, even in the 'quick check' area while preparing to take off on an interception mission, from spares stocks in vans near the end of the runway. Throughout the Starfighter's career, maintenance man hours per flight hour for the F-104A/C never exceeded ten hours, and this only rose to 15-20 with the more complex F-104G. This was a major factor in its selection by so many foreign air forces.

The main undercarriage, a light-weight liquid spring design by Dowty in the UK and licence-manufactured by H M Loud/Cleveland Pneumatic, needed little maintenance. It had a manual 'gravity operated' back-up for extension. One inconvenience in early aircraft was the need to keep the nose-wheel steering button on the control column pressed in while taxiing. Hydraulics were powered by two separate systems backed up by a ram-air turbine (RAT) that extended from the forward fuselage near the starboard air intake. The cockpit's logical, practical design was widely praised at a time when many fighters had cluttered, inefficient 'office space'.



The original downward-firing Stanley B ejection seat, connected to an ejectable lower fuselage hatch, offered simplicity and light weight, but it presented problems at altitudes below 1500 ft, where the pilot was supposed to eject with the aircraft inverted – albeit with a greater risk of back injury due to negative-g. High-altitude ejections were, in any case, rare. Seen here with the second XF-104 (53-7787), the seat had an exit route that obviously left little room for error (Lockheed)

WAR WEAPONS

Although fighter design philosophy in the mid 1950s increasingly tended to prioritise the air-to-air-guided missile, the development of guns continued, and all the Century Series fighters apart from the F-102A Delta Dagger were equipped with them at some stage of their service history. For the F-104 and F-105 Thunderchief, the General Electric M61/M61A-1 Vulcan six-barrel rotary cannon, first tested in 1953, was integral from the outset.

Offering higher rates of fire and greater accuracy than the standard four 20 mm cannon or six 0.5-in machine guns of the Korean War-vintage fighters, the M61 and its ammunition container occupied a considerable space in the slender forward fuselage of the F-104 – the first operational user of the weapon. A complex two-piece gun mount structure was needed to absorb the firing loads generated by the gun and stabilise the weapon. The standard ammunition load for the storage box behind the cockpit was 725 rounds, or about 100 rounds for range practice. Ammunition cases were retained but the links were ejected downwards, even if a centreline store was carried. The system could also be configured to retain both links and cases but this restricted ammunition carriage to 400 rounds for the F-104A and 450 for later single-seat versions.

Lockheed originally specified a pair of 30 mm guns for the aircraft, but when it was adopted by the Pentagon as Weapons System WS-303A in March 1953, the T171 (later re-designated the M61) Vulcan was included instead. It proved to be an extremely accurate weapon, capable of placing all of its bullets within a three-foot circle at 1000 ft and 67 rounds per second.

The Vulcan became the F-104's primary armament for air superiority, but WS-303A also specified the GAR-8 (later, AIM-9B) Sidewinder 1A missile, which was intended mainly for use against more distant, non-maneuvring targets like bombers. Its 25-lb warhead was the controlled fragmentation type, triggered by contact or proximity fusing and capable of projecting 1500 metal fragments through an inch of aluminium from a detonation distance of some 30 ft away. Early Sidewinders, weighing only 155 lbs, had to be launched from Red Dog wingtip launchers almost directly behind the target, with the launch aircraft not exceeding 2g. They were easily distracted by other heat sources such as the sun or ground heat, by cloud cover or by being fired slightly outside the requisite tight range parameters. Launching

at altitude against a large, 'hot' target like a bomber, with a clear sky background gave a good chance of success, however.

Weapons were controlled with the lightweight RCA AN/ASG-14T1 fire control system, using a 24-in, pencil-beam radar antenna and independent optical and infrared sights. In Search Mode the radar displayed a range of 20 miles (40 miles in later versions) and a spiral scan pattern covering a 90-degree cone.

Service testing of the AIM-9B Sidewinder's compatibility with the F-104A was performed at NAS China Lake by the US Naval Ordnance Test Station using three YF-104s that had been repaired after accidents and then transferred to the US Navy for the tests, flown by pilots from the 83rd FIS and US Marine Corps. Seven other USAF F-104As and F-104Ds were used to continue the tests. It was found that Sidewinder launches at Mach 1.2 were to be avoided as the supersonic shockwave at that speed tended to rest on the seeker head nose, upsetting its coolant system. Carriage of a missile on only one wing tended to induce a rolling motion, particularly at low speeds to the extent that formation takeoffs in a one-missile configuration were banned. F-104A-10-LO 56-0748 later became a JF-104A for use in the North American X-15 test programme (USAF)



The difficulty, in the era before pulse-Doppler radars, of overcoming the degrading effects of ground return limited the use of Search Mode up to an altitude of about 3000 ft. Closing to within ten miles of the target, the radar could be switched to Track Mode, using a track button on the control column. The radar scan then narrowed to 20 degrees and a range strobe began to sweep between 300 and 3000 yards in auto-acquisition mode. Holding down the track button set the range strobe at the point where the button was released. Pressing the button again made the sweep range move back to zero. A pilot could thereby set the range to coincide with the target, and at that point a lock-on light illuminated and the radar started to send range information to the optical sight in order to establish correct lead angles. On his radar scope, the pilot got a 'paint' of the target every three seconds. A separate meter showed closure rate to the target, and the radar could be set to 'receive only' if it was subjected to electronic countermeasures (ECM) interference. It would then home onto the interference in ECM HOM mode.

To complete the attack the computing optical sight, with a single gyroscope, projected a reticle 'pipper' that was also adjustable for manual sighting.

The AN/ASG-14 radar was sufficiently powerful for the air superiority mission, performed within visual range for which it was designed. The concept of combat beyond visual range using much larger search radars was still some years away.

Pilots of the F-104 and its contemporaries relied on ground control radars for information on more distant targets. Lt Col David Andrews recalled that in the F-104A as an interceptor 'we flew routinely with wingtip fuel tanks. Other external tanks were used for ferry or cross-country purposes, with a consequent reduction in maximum speed. We had a very small "window" in which to locate, attack and return to base. I suspect the "powers that be" considered an F-104A intercept as a one-time shot. We operated with and depended on ground-controlled interception [GCI] for initial target acquisition, and we practised both high altitude and low-level interceptions and we were not considered a visual meteorological conditions-only fighter'.

Quick-reaction interceptions were set up by a GCI controller, with the F-104 pilot guided into a supersonic stern attack on an intruder at about 120-150 miles from base. He would then start his return home with around 1500 lbs of fuel still remaining.

An infrared sight fitted below the windscreen was intended for use at night or in other situations where visual sighting was impossible, although at shorter range than the radar offered. It had a 14-degrees field of view



The Starfighter's well appointed cockpit, configured here for the F-104C variant (USAF Museum via D Menard)

The first F-104B, 56-3719, flew on 7 February 1957, and the 83rd FIS received its initial examples early in 1958. The nose-wheel assembly retracted rearwards (as on the XF-104) rather than forwards like the units for single-seat versions. The F-104's manually operated canopy (seen here with the original minimal centre-section) was much lighter than the powered clamshell-type versions used on most contemporary fighters, although it only allowed access from the aircraft's right side (Lockheed)



and a photocell that received infrared energy through two slits in a scanner disc. This was converted electrically into a cross-shaped target indication on the combining glass of the visual sight. A large bomber would appear as a single cross at a range of about 14 miles, and at closer range each of its engines would be indicated by a separate cross. In its early tactical fighter configuration the F-104 was also required to carry and launch an MB-1 (AIR-2A Genie) nuclear rocket missile at Mach 2 using an extending trapeze launcher. One aircraft (56-0749) tested this installation, but it was not adopted for service use, although it was used later by NASA for launching test rockets.

INTO UNIFORM

As YF-104A testing continued into 1956, the first USAF contract was placed on 2 March for 146 F-104As and six F-104Bs for Air Defence Command (ADC), with 56 externally similar F-104Cs for Tactical Air Command (TAC). A further 21 F-104Cs were added on 26 December. Lockheed's hot new fighter entered ADC service with the 83rd Fighter-Interceptor Squadron (FIS) at Hamilton AFB, California, on 26 January 1958, two years after the anticipated introduction date. For ADC, the fighter's spectacular climb and speed performance figures were compelling at a time when the USA was thought to be well behind the USSR in the size of its bomber forces. This was an illusion dispelled in 1961 by another Lockheed product in the form of the U-2, which conducted flights that revealed much smaller Soviet air forces and no evidence of the 'nuclear-powered bomber' that was believed to exist in 1958.

By then, however, ADC had ordered the long-delayed Convair F-102A Delta Dagger followed by a developed version of the aircraft, the F-106A Delta Dart. With a more powerful search radar and Genie missile armament, this was ADC's intended interceptor, but it too was delayed, with production aircraft unavailable until May 1959. ADC therefore decided to use the F-104A (and a two-seat conversion of the F-101 Voodoo) as an interim interceptor rather than as an air superiority fighter, since its performance was clearly far superior to the existing F-102A and older types.

Some concerns were expressed by senior figures like Brig Gen J Stanley Holtner, who criticised the aircraft's short interception range using only internal fuel. He estimated that this was only 150 miles against a target at 45,000 ft, but much less for those at higher

altitude. In fact, although the F-101, F-102 and F-106 had better subsonic interception radii, the F-104 was the only fighter that could make a Mach 1.5 (or better) interception up to a distance of 150 miles with wingtip Sidewinders. Its interception times from being scrambled were superior to those of the F-106A, and it could perform interceptions at a greater range than any comparable fighter. The myth of the F-104's lack of range may have originated in the tendency of senior officers and politicians to seek fuel-exhausting Mach 2 back-seat rides in the two-seat F-104B, which only carried 73 per cent of the fuel load of the single-seat aircraft.

The F-104B had two full cockpits and the AN/ASG-14T1 system with Sidewinder provision, but the Vulcan cannon was deleted in favour of a 120-gallon fuel tank in the gun bay. Reduction of basic fuel tankage to accommodate a second cockpit cut around 350 nautical miles from its maximum range, and after the first six aircraft (with F-104A vertical tails) a substantially enlarged, hydraulically boosted rudder was installed to maintain directional stability. This was retained for subsequent Starfighter variants after the F-104C.

COMBAT CAPABILITY

The combination of the short, straight wing and T-tail made the Starfighter a very responsive aircraft, albeit one that would only 'wake up' at speeds above 300-350 knots. A good F-104 pilot would not 'get slow', but would instead exploit the aircraft's high thrust/weight ratio to accelerate quickly into a fight (often unobserved due to the F-104's small visual signature) and use the excess thrust in the vertical dimension. Because the jet could out-accelerate every contemporary enemy aircraft and could readily use its top speed advantage, especially at medium and low altitudes, a pilot could choose to engage and disengage combat at will.

Also, the F-104's low-drag, efficient engine/airframe match and generous internal fuel capacity gave it a good combat endurance so it could normally outlast an opponent. An F-104 could fly over enemy territory using the fuel in its drop tanks, jettison those tanks for aerial combat, engage in low-altitude combat for five minutes at maximum thrust, fly home and have sufficient reserves for a safe landing. Notably, the Starfighter's low drag gave it a high combat cruise speed at intermediate throttle settings.

Indeed, an F-104A with two wingtip AIM-9Bs could easily reach Mach 2, whereas with two tip-tanks speed was limited to around Mach 1.7. In either case, the fighter could provide very effective air superiority cover.

After the F-104's service entry air-to-air tactics were developed, ranging from simple single-ship, 'slash-and-run' attacks or high 'yo-yos' against less-capable adversaries, to more sophisticated coordinated double attack tactics, which at the time were used exclusively by the Starfighter community. Pilots could tailor their tactics to their opponents. Against a lower-performance aircraft like the MiG-17, the F-104 pilot could afford to let speed bleed down to 350 knots in manoeuvres, while a MiG-19 would demand at least 450 knots and a MiG-21 would dictate a 600-knot minimum. Because of the proximity of George AFB, in California, to Nellis AFB, Nevada, George's F-104Cs (assigned to TAC's 479th Tactical Fighter Wing) often supported the USAF's Fighter Weapon School (FWS) at Nellis as a wildcard threat.

F-104C-5-LO 56-0883, the first F-104C, later became the only Starfighter to be shot down in air-to-air combat during the Vietnam War. The four-hour morning CAP in which Capt Philip E Smith was hit by Chinese MiGs on 20 September 1965 when he strayed too close to Hainan after a series of navigation system failures was to have been followed by a bombing sortie over North Vietnam that afternoon. China has always defended Hainan Island fiercely, and before the arrival of 479th TFW F-104Cs communist Chinese Navy Shenyang J-6 (MiG-19) fighters often flew harassment sorties against US aircraft patrolling offshore from the island. As late as 1 April 2001 a US Navy Lockheed EP-3E Aries II on an ELINT mission 70 miles off its coast was intercepted by Chinese J-811 Finbacks and forced to land on the island's Lingshui air base after a mid-air collision with the fighters (USAF)

The F-104Cs also participated in the groundbreaking Project *Featherduster*, which pitted F-86Hs in the role of MiG-17s and F-5As and F-106As in the role of MiG-21s against all USAF fighters for the purpose of evaluating relative performance and tactics development in dissimilar air combat. The F-104's performance came as a shock to many. On 1 July 1965 the USAF's 6002nd Standardization and Evaluation Group published the results of Project *Featherduster*. The aim was to produce updated Series 3 Tactical Manuals for pilots facing MiGs in Vietnam. F-105D, F-100D and F-4C fighters were also included in the 200 aerial engagements.

Pilots flying the F-86H found that the F-104 was the most challenging of all these adversary fighters even in some high-g turning situations. In 16 F-104C sorties it was found that a diving escape manoeuvre in afterburner was the pilot's best method of defeating an initial gun attack. As an aggressor, the best results came from high-speed surprise attacks, taking advantage of the Starfighter's low head-on visibility but avoiding attempts to enter a turning fight with the target aircraft. For Sidewinder attacks the surprise factor was considered crucial, as the missile's restrictive launch parameters 'in a manoeuvring situation result in a severe operational limitation for effective utilisation by tactical fighters'. However, in most tests the F-104C showed that it could cope with MiG-17-type aircraft better than the other fighters in the programme, and it regularly beat all the other aircraft involved except in turning engagements at high altitudes.

Against the 'dogfighting' F-86H the F-104's excess thrust gave it better sustained turn rates, and it could turn inside a Sabre in a climb above 300 knots. At low altitude it could turn with the jet at speeds above 270 knots, and although the F-86H pilot could execute snap turns at lower speeds to try and foil an F-104's tracking, his aircraft rapidly bled off energy, allowing the Starfighter pilot to use the vertical plane to reposition for another attack. This was known as a 'high yo-yo', a basic manoeuvre taught to US fighter pilots. An F-86 pilot could use a 'low yo-yo' as a counter-move, requiring sufficient altitude to execute it. As the less powerful F-86 bled off energy the pilot had to trade altitude for speed. The dogfight would then gradually move to lower altitudes, where the F-104's thrust-to-weight ratio advantage increased and the F-86's larger wings, so effective at higher altitudes, produced excessive drag. These tests produced invaluable data for F-104 pilots in several air forces who would later face MiGs, Hunters, Mystères and other fighters.



Capt (later Lt Gen) Robert Bond, a Super Sabre FWS instructor pilot, commented on the F-104;

‘It is a nice little bird. Goes like a bat and turns well as long as speed is maintained. They don’t, by the way, seem to have any problem doing this. What do our pilots at the FWS think of the airplane after hassling with it? We are considering starting a new course here at the FWS. The title? “How to Hide Envy”.’

F-104s WITH ‘NUKES’

The F-104C version was originally intended to replace some of TAC’s F-100 Super Sabres, offering much higher speed in Low Altitude Bombing System (LABS) ‘toss bomb’ attacks with nuclear weapons (increasing its chances of safe escape) and superior air-to-air qualities. Equipping fighters to enhance Strategic Air Command’s dominant nuclear influence was also a way of sustaining TAC’s budget at that time.

Development of TAC’s Starfighter, substantially funded by Lockheed in the hope of larger orders, took 18 months. It had the up-rated J79-GE-7A engine, with a two-inch increase in turbine diameter, generating 15,800 lbs of afterburning thrust. Blown flaps were installed from the outset and range was increased by plumbing in an external, detachable refuelling probe which had been tested on F-104As ahead of the Operation *Jonah Able* deployment to Taiwan in September 1958. Its nozzle was placed conveniently within the pilot’s left-side vision, making refuelling from the temperamental trailing basket of a KC-97 tanker a relatively simple five-minute operation. As former F-104C pilot Norm Filer explained;

‘The boom was just a pipe with the nozzle at one end and a fitting to replace the fuel cap at the other. The refuelling light was in a small fairing at the base of the boom and the fairing that covered the pipe from the light aft was a simple, somewhat flimsy, fibreglass part with a million screws into the fuselage.’

For its tactical roles the F-104C was wired for a 2000-lb ordnance load such as a Mk 28 or Mk 43 nuclear store on a short centreline pylon. Two 195-gallon tanks or ordnance, typically two 750-lb M117s or (for the F-104G) 1000-lb bombs, could be hung on the two underwing pylons, as could other TAC stores including BLU-1 fire-bombs or LAU-3 rocket pods. The M61 cannon was re-installed, and from October 1961 the ordnance upgrade programme Project *Grindstone* added a twin-Sidewinder ‘catamaran’ launcher beneath the fuselage, thus allowing the carriage of wingtip fuel tanks. This facility was not used operationally by USAF F-104Cs as the launcher increased drag and could not be jettisoned.

An upgraded AN/ASG-14T-2 fire control system (FCS) gave improved night target acquisition, but like other TAC fighters of the time, the F-104C was limited by its visual-only weapons. However, the FCS ‘switchology’ remained very simple to operate, particularly in comparison with the systems in the F-105D Thunderchief and F-4C Phantom II, leaving the pilot free to concentrate on both flying the jet and on keeping an adversary in sight. An overall structural beef-up to cope with the extra weight (23,590 lbs maximum) required 36 new metal forgings for the airframe.

There were extended delays in the F-104C delivery process caused mainly by development problems with the J79 engine and M61 gun installation. Engine compressor stalls, afterburner failure due to the jet nozzle 'petals' sticking in the open position and reducing thrust, together with engine overheating, were all encountered during the YF-104 test programme. A grounding order for all F-104As was enforced in April 1958 while improved J79-GE-3B engines were installed, further postponing the type's service introduction. TAC's interest in the F-104 had by then begun to decline at a time of rapidly changing defence policies. From 1956 it sought larger, longer-ranged fighters with bigger ordnance loads. Further emphasis on TAC's conventional warfare capability at the end of the 1950s reinforced its need for fighters that could carry nuclear weapons or a more substantial conventional weapons load, rather than the F-104C with its limited ground attack role.

The Starfighter order was, therefore, limited to 77 aircraft – enough for only one TAC wing. If the aircraft had become embroiled in a European nuclear war scenario, on which much of TAC's planning centred, it is likely that the F-104Cs would have been used in the air superiority role while the nuclear missions would have been allocated to F-100Ds or F-105Ds.

The first F-104C flew on 24 July 1958 and deliveries to the 476th TFS/479th TFW at George AFB began in September of that year, replacing the unit's F-100 Super Sabres. Although its use was limited to the four George-based squadrons, the Starfighter had a far more useful, active and varied combat career than TAC planners might have guessed in 1958. Twenty-one F-104D trainers were also delivered, with a two-cockpit nose added to the F-104C airframe without the M61 gun or centreline stores provision. A revised canopy had a larger fixed centre-section and windscreen between the two opening transparencies, later retro-fitted to some early F-104Bs to coincide with the installation of upward-firing ejection seats.

F-104Ds also had in-flight refuelling probes and 15,800-lb thrust J79-GE-7 engines. The latter was the subject of a programme by General Electric, TAC and the Oklahoma City Air Materiel Center in 1960-61 to improve the reliability of the F-104/J79 combination using re-designed compressor blades.

Each F-104C squadron usually had two F-104Ds on strength for transition and proficiency training, and they had the speed and climb performance to match the single-seaters.

Lockheed had invested heavily in the F-104, and the USAF orders for 296 aircraft – fewer than those for the other Century Series fighters – were a disappointment. The company, therefore, urgently sought new ways of modifying the aircraft to suit changing requirements, principally by turning it into a more sophisticated fighter-bomber. This contravened Kelly Johnson's original 'light, fast and simple' doctrine, but he could also see that the airframe could be adapted for the interceptor, fighter-bomber and reconnaissance roles that NATO countries, principally West Germany, were looking to fill. In these air forces, and those of Japan, Canada and other nations, there was an increasing need to replace first-generation fighters like the F-84 and F-86.



In ways that are beyond the scope of this volume the basic F-104C/D was quickly turned into a stronger, heavier (by about 1000 lbs) and more sophisticated warplane in the form of the F-104G, which became the company's greatest sales success. By the mid-1960s Starfighters were in use with 15 air forces – more than any other US fighter – with licence production in seven countries whose aviation industries benefited enormously from the experience. 'It brought many air forces into the modern age and founded production consortiums that went on to produce the European combat aircraft that are in service today', explained Lt Gen Tom Baker (who taught many Luftwaffe pilots to fly the Starfighter, and who went on to command the 48th TFW at RAF Lakenheath). 'The conferences set up between European F-104 users were a great means of bringing together air forces throughout Europe'.

In all, 2578 were produced, and the last, an F-104S for the Italian Air Force, was rolled out in March 1979. The US Air National Guard continued to use some of the early USAF versions until 31 July 1975, and the Republic of China Air Force (RoCAF) operated Starfighters until May 1998. Another eight years of operational service lay ahead for Italy's F-104s until their final sortie was flown on 27 July 2005, completing 42 years of active service for a fighter that the USAF had initially seen as a stopgap interceptor. This considerably surpassed the service longevity of all other Century Series aircraft including the F-106A Delta Dart, which was phased out in 1988.

The F-104 had fulfilled a unique role as TAC's best air superiority fighter, and the only one that was entirely suitable for the fighter-versus-fighter roles to which it was assigned during the Cuba Crisis and in Vietnam. Its comparatively short USAF service was in part a consequence of pressure to fund the F-15 Eagle, the USAF's ultimate air-superiority fighter for more than three decades.

The USAF accepted 663 out of the total F-104 production of 2579 aircraft. Only 296 of these were for the USAF's own use, with 367 Starfighters being acquired for MAP. Low production of the F-104A/C for USAF service severely under-used a production line that was intended to turn out up to 20 aircraft daily. F-104G variants accounted for most of the remaining production worldwide. Germany provided 110 F/TF-104Gs for a 20-year training programme for Luftwaffe pilots at Luke AFB, Arizona, beginning in 1963. The heavier, more complex F-104G, with four external tanks and a reconnaissance pod, could still cruise at 600 knots at low altitude for an hour and land with 1000 lbs of fuel remaining. These F-104Gs flew with USAF markings and serial numbers, although they were also allocated 'shadow' German serials. This F-104G 67-14892 (shadow serial 2448) was later transferred to the RoCAF, modified into RF-104G configuration and lost in a crash on 4 March 1993 (USAF)

COLD WAR CLASHES

Hamilton AFB resounded with the distinctive, ululating moan of Starfighters for the first time in January 1958, and the base's 83rd FIS was operational by 20 February. Its sister squadron, the 84th FIS, traded in its 'gravel gobbler' Northrop F-89J Scorpions in June 1959 to become the first ADC unit with the two-seat F-101B Voodoo. Seventeen squadrons eventually re-equipped with this big McDonnell fighter, which was ADC's principal all-weather interceptor for four years.

While the F-104A's gun and two Sidewinders were effective at short range, the F-101B could carry two AIR-2A Genie nuclear-tipped missiles with a range of more than six miles, or up to six AIM-4 Falcon infrared missiles. F-104As were inevitably short-term occupants of the interceptor alert 'pads' because they could not accommodate the electronic equipment required to integrate fully into America's complex SAGE (semi-automatic ground environment) defence network.

However, within ADC, the 83rd FIS's primary task was to meet and destroy high-altitude intruders. Training involved sorties against SAC bombers and even Lockheed U-2 reconnaissance aircraft. The Starfighter excelled in this role, and although F-104As were confined to four squadrons, including the 83rd FIS, their first, short period of ADC service was eventful. Tom Delashaw famously intercepted a U-2 at 72,000 ft and performed a roll around the spyplane to show its pilot that he was not the only inhabitant of those lofty heights. He also zoom-climbed an F-104 to 92,000 ft over West Germany while on a deployment during the Berlin Crisis – his way of demonstrating that any Soviet high-flyers were also vulnerable to the Starfighter.

Service entry of 83rd FIS F-104As was an ideal opportunity to demonstrate the aircraft's performance and enhance its reputation at a time when accidents during the fighter's test programme were still making unwelcome headlines. Maj Harold C Johnson from the Hamilton unit reached 91,249 ft over Edwards AFB, California, on 7 May 1958, while Capt Walter W Irwin attained 1404.9 mph over a 15-mile (25 km) course at the same location on 16 May. The F-104A thus became the first aircraft to hold both the World Airspeed and World Altitude records simultaneously. Further time-to-climb records were set on 10 and 13

Several YF-104A-1-LOs including 55-2969 were used in the training programme for the first operational unit, the 83rd FIS. This aircraft was also flown by Capt Walter W Irwin on 16 May 1958 when he set a world air speed record of 1404.19 mph, making the F-104 the holder for both World Airspeed and Altitude records. Like most USAF aircraft the F-104 acquired nicknames, although by a fairly devious route. Initially, pilots called it the 'O-4', but then the F-100 Super Sabre became known as the 'Zip Zilch' (for 'O-Zero'). Thus the 'O-4' evolved into 'Zip-4' and then to 'Zipper' (Lockheed)



December, and on the 14th Capt Joe Jordan flew an F-104C to 103,389 ft, beating the existing high-altitude balloon record and making the F-104 the first aircraft to exceed 100,000 ft entirely under its own power.

CHINA CRISIS

Within months of establishing operational readiness, the 83rd FIS was tasked with peacekeeping in a potential international conflict thousands of miles from its California home. On 23 August 1958 the army of the People's Republic of China (PRC) began a devastating artillery assault on the disputed offshore islands of Quemoy and Matsu – up to 450,000 shells were fired. The islands had been occupied by the Nationalist Chinese after the civil war that had brought about their Chinese People's Republic, centred on the island of Formosa (later called Taiwan).

Artillery duels between mainland communist Chinese forces and those on Quemoy had continued since 1954, but the 1958 tensions included aerial battles between the Chinese Nationalist Air Force (CNAF, later Republic of China Air Force) and communist Chinese MiGs, 200 of which had been moved to two airfields on the mainland opposite Taiwan. Overflights of Quemoy by MiG-17s began on 18 August. On 24 September a group of experienced CNAF F-86F Sabre pilots claimed the destruction of 11 out of a wave of 100 MiG-15s and MiG-17s, including four shot down with AIM-9B Sidewinders – the first combat use of this missile. Within six weeks they had claimed 31 MiGs for the loss of only two Sabres.

To demonstrate support for the new republic, substantial US forces were deployed to Formosa, including F-100D, F-101C and B-57B attack aircraft together with US Marine Corps F4D Skyrays and US Navy Seventh Fleet carriers offshore. The early arrivals were expected to provide air defence for Taiwan, and the 83rd FIS was a late addition to the original *X-Ray Tango* plan, but it became the primary fighter unit for this purpose, supporting the CNAF's three wings of F-86Fs and defending US assets in the area. The decision to send Starfighters was taken so that the communist Chinese would, as Col Howard 'Scrappy' Johnson put it, 'track them on their radar screens' and 'sit back and scratch their heads in awe'. In fact, more far-reaching plans were drawn up for strikes on mainland Chinese airfields and military control centres with conventional weapons, neutralising an airfield every four days. If these plans had been activated the F-104As would surely have fought the defending MiGs.

In Operation *Jonah Able*, 12 F-104As of the 83rd FIS that were under ADC control as part of the 13th Air Task Force (Provisional) were partially disassembled, crated and loaded into Military Air Transport Service (MATS) C-124 Globemaster II transport aircraft, which made 20 flights in all.

Not so much 'the jet that came apart in their hands' as the careful removal of an F-104A's nose-cone as it is loaded into a C-124 at Hamilton AFB, California, in September 1958 for the airlift to Formosa (USAF)





TSgt Fred Maynard from Santa Rosa, California, checks an AIM-9B on 83rd FIS F-104A 56-0828 at Tai Yuan AB, Taiwan, in October 1958. Conditions were very primitive at first, with tented accommodation, a field kitchen and an ancient World War 2 hangar for maintenance. It was a good test of the F-104A's simple maintenance requirements. Hamilton pilots were also rather perturbed by the local RoCAF procedure that allowed its alert F-86s to takeoff from the opposite end of the runway to the F-104s, but at the same time! This was made more alarming by the 'hump' in the runway that prevented pilots from seeing all the way along it (USAF)

Lt Col John Bennett, 83rd FIS commander, is strapped into F-104A-25-LO 56-0842, the first *Jonah Able* Starfighter to be re-assembled at Tao Yuan AB in September 1958 (Kenneth L Hackman/USAF)



The first Starfighter was inside a C-124 within 48 hours of the official notification, and it arrived at Tao Yuan AB, Formosa, on 10 September. *Jonah Able* saw the first long-distance movement of high-performance aircraft in this way. Aircrew had no warning of the deployment or indication of the destination. The squadron commander, Lt Col John W Bennett, was on leave and enjoying a fishing trip when the 'return' order was received. Following the usual tradition of having aircraft operational within 24 hours of arrival was more challenging in this case as the F-104As had to be reassembled, but the first (flown by 1Lt Crosley J Fitton) was in the air 30 hours after reaching Formosa. All Starfighters were in place by 19 September, ready on five- and 15-minute day or night alert status.

One F-104A, 56-0791, was recovered to the base in the first operational 'dead stick' landing in a Starfighter. 1Lt Larry Dube was engaged in practice interceptions with his wingman when his oil pressure dropped almost to zero and the engine seized up. Extending his RAT, 1Lt Dube was able to slow the aircraft to around 170 knots on touchdown at Tao Yuan. Vibration had caused a small oil line to fracture, and the resulting drop in oil pressure prevented the engine exhaust nozzles from opening correctly, resulting in a loss of thrust. The oil line was given a reinforcing brace on subsequent J79s and a manual back-up nozzle control system was later installed.

Five months prior to the *Jonah Able* deployment, the 337th 'Falcons' FIS at Westover AFB, in Chicopee, Massachusetts, had received Starfighters to replace its F-86L Sabres. Squadron commander Col James Jabara, a strafing ace in World War 2 and the world's first jet ace in Korea (where he finished with 15 MiG-15 kills to his credit) flew to the Lockheed factory to collect his unit's first F-104B on 13 April. The date proved to be unlucky as the aircraft was grounded due to the type's recurrent engine intake stall problem. F-104B 57-1296 finally arrived on 26 April, and it appeared at the base's May 1958 Open House nicknamed *Double Trouble*. Single-seat models soon followed, and the area around Westover became

used to sonic booms (permitted below 30,000 ft under contemporary USAF regulations) as pilots practised high-altitude interceptions from five-minute alerts under the control of the Boston Air Defence Sector. These seem to have been accepted as a necessary price for effective air defence, except when they occurred in the early hours of the morning or left residents with broken windows.

A weapons training deployment (the first by an F-104 unit) to Tyndall AFB, Florida, included AIM-9B firing against drone targets and gun-firing on Delmar targets, towed by B-57s, both day and night at altitudes between 1000 ft and 50,000 ft. The squadron became operational without losses despite a comparatively inexperienced complement of pilots,

and it completed an accident-free year on 26 April 1959 with 3000 flying hours. The crash of F-104B 56-3724 on 19 October ended that record, however. Both pilots (Capt James Akeley and 1Lt Peter McBurnie) attempted to eject at too low an altitude and they were killed when the aircraft crashed on approach for unknown reasons. A

similar accident to an 83rd FIS F-104B took the life of 1Lt Ken Mason, on his first check ride in the aircraft. Having to downward-eject into San Pablo Bay on approach, he drowned close to the shore when his parachute dragged him under water. A similar fate befell the squadron's commander, Col Ray Evans, when his engine failed at 500 ft on an over-water approach.

The Quemoy dispute rumbled on, and in October 1958 ADC sent orders for the 337th FIS to despatch more than half of its personnel, including 18 pilots, to Tao Yuan for a 179-day temporary deployment (TDY) commencing in the first week of December to relieve the 83rd FIS and fly its F-104As. The unit's own aircraft, with distinctive yellow squadron flashes, were transferred to Hamilton AFB to replace those of the 83rd FIS, which had retained only ten jets when the squadron had been sent to Taiwan. Thirteen pilots remained at Westover as Det 1 of the 337th FIS for upgrade training to operational status.

On arrival in Taiwan, the 'Falcons' were treated to slightly better tented accommodation than their predecessors, albeit in some challenging weather conditions, while the aircraft were parked in World War 2 earth-walled revetments. Better aircrew quarters were constructed during the deployment.

A ceasefire was agreed on 6 October 1958, and although the Starfighters had not been required to engage communist Chinese aircraft directly, it was judged that they had acted as a very potent deterrent to possible attacks on Nationalist forces. Part of their task was to fly supersonic runs between Taiwan and the mainland at speeds up to Mach 2 as a demonstration of air superiority. 1Lt Crosley J Fitton made the first Mach 2 run, which was probably a source of great interest to communist Chinese radar operators. Capt George Martin was one of the 83rd FIS pilots who saw the vapour trails of MiGs while he was patrolling over the Formosa Strait, but the communist pilots did not approach their potential USAF adversaries.



F-104A-20-LO 56-0791 rolls out to the runway at Tao Yuan AB while 56-0828 is prepared for a sortie. The former aircraft was eventually passed on to the RJAF while 56-0828 remained in Taiwan with the RoCAF until it was lost in an accident in April 1962 (USAF)

F-104A-25-LO 56-0842, ready for reassembly at Tao Yuan AB after its long transit flight (USAF)





F-104As from the 538th FIS based at Larson AFB, Washington, took part in the October 1959 William Tell II Meet at Tyndall AFB, together with entries from the 337th FIS. Their impressive results left little doubt as to the F-104's effectiveness as an air superiority fighter (USAF)

After the USAF F-100s, B-57Bs and F-101s were withdrawn from the area it was decided to retain the F-104As on Taiwan for a few weeks longer. Gen Laurence Kuter (commander-in chief of Pacific Air Forces) and Adm Harry Felt (Pacific forces commander) wanted them to remain 'for psychological reasons', and because communist China was likely to receive MiG-19s and MiG-21s. It was hoped that the RoCAF would eventually receive its own Starfighters, both Kuter and Felt recommending to the US government that the USAF F-104As still in Taiwan should be given to the RoCAF, but that decision was postponed.

At the end of the deployment the F-104As had their wings, aft fuselages and radomes removed once again for loading into C-124 Globemaster IIs and the return journey to the USA. Back at Westover, Col Jabara reassured local journalists that the deployment had 'convinced the communists that we mean business', and it had shown the Republic of China that 'this country would support [Taiwan] against any threat of invasion'. Gen Kuter reported that the F-104As 'made a tremendous impression on both sides of the Taiwan Strait'. The presence of the Sidewinder missile on F-104s and F-100s, unmatched by anything in the communist Chinese inventory, also had a powerful deterrent effect. This reputation as a fearsome potential adversary, established during the Taiwan deployment, would serve the F-104 well when it was introduced to other combat scenarios several years later.

The 337th FIS continued to fly Starfighters until its deactivation on 8 July 1960, the unit providing an entry (F-104As 56-0847 and 56-0876) in the 1959 William Tell Fighter Weapons competition. Sadly, the last F-104A to leave the squadron, 56-0806, crashed on its departure flight on 13 June 1960 after a loss of oil pressure and Maj Emmett Barrentine ejected safely.

The 56th FIS operated F-104As from Wright-Patterson AFB, Ohio, within the Detroit Air Defense Sector from May 1958 until its deactivation in March 1960. F-104A-25-LO 56-0856 was later passed to the 331st FIS at Webb AFB, Texas, where it was lost in an accident in July 1965 (J Aaron via T Panopolis)



The 538th FIS became the third ADC F-104A unit in May 1958, trading up from F-86L Sabres at Larson AFB, Washington. The final unit was the 56th FIS at Wright-Patterson AFB, which acquired F-104As in July 1958. Deliveries of the remaining examples of the 153 F-104As assigned to ADC continued until December of that year.

STATE MILITIA STARFIGHTERS

This brief 'stop gap' period of ADC service ended with the phase out of its F-104As and their transfer to Air National Guard (ANG) squadrons. However, ADC still had access to them as ADC-gained units, and this made them frontline squadrons in an emergency, ready to deploy abroad if necessary. Westover AFB's aircraft began to transfer to Knoxville for service with the 151st FIS/134th FG, Tennessee ANG, under the command of Lt Col Robert Akin on 26 May 1960. By 1 July that year the unit had completed its conversion from the F-86L and thus become an ADC-gained unit.

At Congaree AB, South Carolina, the 157th FIS/169th FG (South Carolina ANG) also relinquished F-86Ls in favour of F-104A/Bs from February 1960, as did the 197th FIS/161st FIG of the Arizona ANG at Skyharbour Airport, Phoenix, in July 1960. The latter unit was manned by no fewer than 750 personnel and equipped with 22 F-104As, some of its aircraft coming from the disbanded 538th FIS.

From the original ADC allocation 24 F-104As were converted into QF-104 target drones for the 3205th Drone Squadron and three were passed to NASA and converted into NF-104As with Rocketdyne AR-2 rocket engines for training astronauts. Maj Ken D Luedeke was a member of the initial QF-104 test team at Eglin AFB, Florida, from 9 January 1961, developing an effective aerial target to train fighter pilots, many of whom would eventually meet real MiGs over North Vietnam;

'I had the honour of controlling the first QF-104 pilotless takeoff and flight on 17 October 1961. Chuck Dildine and I were the rudder and elevator [control] team that landed many of the drone missions. As safety pilot I would fly a functional test flight the day before a pilotless flight, set the QF-104A up for takeoff on the end of the runway, get out of the aircraft and then man the elevator position as back-up transmitter during takeoff. Then came the long wait to see what shape the QF-104A would be in for landing. I really loved that airplane. Every flight was a thrill, and I was hooked and couldn't get enough flying time in the bird. The only "sass" I got from it was a nose-gear collapse during a manned remote-control flight, a faulty fuel gauge that almost made me a glider pilot (in an aeroplane that does not like to glide) and a few engine stalls, which were exciting.'

Ken almost got the chance to test the F-104 in combat in Vietnam when he received orders for the 435th TFS/479th TFW flying F-104Cs from Udorn Royal Thai Air Force Base (RTAFB), but the squadron's return to the USA in July 1967 curtailed these instructions. He went instead to Tuy Hoa AB, South Vietnam, to fly 114 combat missions in the F-100D Super Sabre.

A neat quartet of F-104As from the South Carolina ANG. Only one aircraft has the usual USAF marking above its right wing, painted on to the white finish that was applied to the carefully prepared upperwing surface to ensure the smoothest possible airflow. All but the furthest example (the only one with an arresting hook) later flew with the RJAF. The South Carolina ANG returned its F-104As to ADC in mid-1963, 56-0856 being the last to leave the 157th FIS at McEntire ANGB (Lt Col David Andrews)





F-104A-25-LO 56-0844 in 'SCANG' service in May 1961. Previously, it had served with the 538th and 83rd FISs (participating in Operation *Jonah Able* to Taiwan with the latter unit) prior to deploying to Morón with the 157th FIS. The jet was later re-engined with a J79-GE-19 and transferred to the RoCAF, where it was coded 4243. Once again, this aircraft clearly lacks a Vulcan cannon. Removal of the gun allowed space for the fitment of a 100-gallon tank, although this was not used by SCANG F-104As. Fitting two wingtip tanks with 340 gallons of fuel added to the 900 gallons held in fuselage tanks. Another 390 gallons in underwing pylons took the maximum total capacity to 1630 gallons (*D Lucabaugh via T Panopolis*)

Lt Col David W Andrews flew F-104As with the South Carolina ANG after 1000 hours on USAF F-100D/F Super Sabres. He recalled, 'There were very few restrictions on breaking the sound barrier in those days, and it was done rather routinely by our jocks. We were supposed to do it off the coast, which was about 100 miles away, but many pilots chose not to travel that distance to punch through the Mach'. 'Ted' Andrews' brother was also a 'SCANG' pilot, and he gave Ted his introductory flight in a squadron T-33. 'I think the flight lasted 30 minutes, and at no time did we have less than 3g on the aeroplane, with some low-level railroad following mixed in with the acrobatics. My brother was a "hot rod" pilot who was always hesitant to pull the throttle off maximum power. By the time we landed I had a helmet full of that day's lunch' (*Lt Col D W Andrews*)

Lt Col David W 'Ted' Andrews transitioned to the F-104A with the South Carolina ANG after 1000 hrs on F-100s with USAF's 50th TFW. New pilots were sent solo in a Starfighter following two rides in the unit's two F-104Bs, as Andrews recalled;

'It didn't take long to become qualified to sit alert duty since they had two full-time 24/7 alert slots and they needed pilots to pull alert. I recall during the checkout being sternly warned that on the takeoff roll, just before lift-off, I should grab the landing gear handle so as to yank the gear up immediately after leaving the ground so that I would not exceed the gear-down limit. It was a very fast aircraft in every flight condition. We did not have the "Gatling" gun mounted in our F-104As, so we didn't use the "dart" target for gunnery. We didn't have the 100-gallon gun-bay fuel tank either.'

Stan Hood added, 'They took the gun out of the F-104A as they didn't think we were going to need it. People in the USAF and the industry thought it was going to be strictly air-to-air missiles only'.

Without external tanks the F-104A's Mach performance obviously limited its endurance, as 'Ted' Andrews experienced at first hand;

'I was assigned to take an aircraft to Knoxville, Tennessee, for a static display. After the show was over I asked ground control to issue me an unlimited climb-out on the flight back to McEntire ANGB [formerly Congaree AB, in South Carolina]. The clearance was approved, as was a high-speed pass along the runway on departure. After clearing the airfield limits I pulled the stick back, held the mike in the "keyed" position and announced the 5000-ft stages in the climb as requested by departure control. There were only a few seconds between transmissions. After

levelling out at around 38,000 ft, I pushed the throttle up and used the afterburner to push through Mach 2.

'En route control was by Ground Controlled Intercept in those days, and I was talking with them. I asked the controller my ground speed and he reported something like 1150 knots. Also on that radio frequency was a "bug smasher" [multi-engined military aircraft]. I heard him say to the controller, "I don't believe what you have just told that other aircraft. What is he flying?"'



‘Ted’s’ flight plan for that journey showed 15 minutes for the 194 nautical miles between Knoxville and McEntire ANGB. ‘If an instrument approach or go-around had been required at McEntire I probably would have flamed out prior to landing.

‘As a very new pilot to the F-104, I was following my flight lead on a low-level intercept mission on another occasion. After several practice intercepts he finally enquired as to my fuel status, and I remember his alarm when I told him. We immediately climbed to medium altitude to return to base. As we approached the airfield I could detect that he was going to enter a normal initial approach for a routine landing. I took exception to this and declared myself on a base turn to final approach. Later, the crew chief told me I had less than 50 gallons of JP-4 in the aircraft when I shut down. In the F-104 that fuel would have lasted about three minutes during low-altitude operations.’

GUARDING EUROPE – THE BERLIN CRISIS

Infuriated by the outflow of more than two million East Germans to the West in search of better lives, and by the refusal of America, Britain and France to accept an end to the four-power treaty that allowed them access to West Berlin, Soviet Premier Nikita Khrushchev threatened to use force to deny any possibility of a re-united Germany. On 13 August 1961 he sealed off most of the border-crossing points between East and West Berlin and constructed a wall to divide the city. America’s military strategy was focused on a nuclear response, but in this case it was clear that a show of conventional force was required.

The lack of frontline conventional tactical air power meant that ANG squadrons were required to deploy to Germany in a situation in which President John F Kennedy wanted to ‘have a wider choice than humiliation or all-out nuclear action’. Eleven ANG fighter squadrons were sent to Germany, France and Spain to support nuclear-capable USAFE tactical units, preceded by elements of the 479th TFW with nuclear-capable F-104Cs. Another 11 ANG squadrons were mobilised on short-notice alert to deploy as part of Operation *Stair Step* (the codename for the rapid aerial movement of the fighters to Europe). Among them were the three F-104A units, mobilised on 1 October.

Once again the MATS C-124 transport force was called in to airlift 60 F-104As to West Germany in November in Operation *Brass Ring*, as it seemed likely that Khrushchev would blockade West Berlin. The 197th FIS took 18 F-104As and two F-104Bs to Ramstein AB, West Germany, where it was joined by a similar number from the 151st FIS of the Tennessee ANG. They were both assigned to the 86th Air (Defence) Division of the Seventeenth Air Force and the Tennessee crews assumed alert duty from 19 December. The Arizona pilots flew patrols towards and along the border with the Soviet bloc until August 1962. Most pilots had to visit Wheelus AB, in Libya, to qualify on live-firing the AIM-9B missile before they were able to perform alert duties in Germany. Capt Gene West, who flew the F-104 for 17 years and amassed more than 3000 hours in the aircraft, was in the Arizona-based contingent;

‘I flew the F-86 with the Arizona ANG from 1954 to 1960, when news of replacing the Sabres with F-104As reached us. You can imagine the excitement throughout the squadron. The ANG always got the “hand me



The Globemaster II transportation method was also used to deploy F-104As to Europe during the Cold War crises. Douglas C-124C-DL 53-0027 is caught in the act of 'swallowing' F-104A 56-0862 of the 157th FIS for its Morón deployment on 10 November 1961. This particular Starfighter was lost near Córdoba, Spain, on 2 June 1962 after suffering engine failure (USAF)

down" aircraft, but Mach 2 "Zippers" – Wow! A few months after their arrival our unit was activated and sent to Ramstein. Many of the pilots had minimal time on the aircraft and the weather in Germany did not resemble what we were accustomed to. After returning to Phoenix, Arizona, the USAF or ANG decided we had had too much fun flying the "Zippers" and replaced them with Boeing C-97 "trash haulers". I immediately returned to the regular USAF and was stationed at Luke AFB as the F-104G test pilot.'

South Carolina ANG's 157th FIS deployed to Morón AB, in Spain, under the control of the 65th Air Defence Division on 24 November 1961. Maj Gen Hood (who was then a captain) recalled;

'I went with the advance party on a C-124, which was carrying an F-104A and a couple of extra engines, but we hadn't been told whether we were going to Germany or Spain. When we got to Spain the USAF sent over civilian employees to put the aeroplanes back together, but I insisted that our master sergeants had the last word on whether or not they were properly ready to fly.'

Budget-driven shortages of spares and ammunition combined with little forward planning meant delays in achieving full readiness for many of the ANG units, and there were some general personnel shortages because one-eighth of the Guardsmen who were mobilised had chosen to remain in their Stateside jobs. However, the presence of Starfighters was a powerful air superiority deterrent, as it had been in Taiwan. They demonstrated very fast reaction times and stellar acceleration, which made them superior to all other available fighters in practice interceptions. This was a major factor in ADC's decision to re-admit some of the F-104As to national control the following year.

Pilots adapted to the crowded European airspace environment and the weather (very different from Arizona conditions), but there were mishaps. F-104A 56-0809 was lost after an aborted takeoff from Ramstein AB on 18 June 1962 and two aircraft (56-0813 and 56-0848) collided on



approach to the base on 17 March. One of the effects of the small F-104 wing was that pilots in formation had difficulty at night or in poor visibility in immediately seeing whether the lead aircraft had initiated a roll. At close quarters this occasionally left insufficient time for avoiding action, and a solution eventually came in the form of extra formation lights. The Tennessee Guardsmen lost another jet (56-0881) near Wiesbaden when it exceeded pitch-up limits and began an unrecoverable flat spin, obliging Capt Pierce to eject.

At Morón, the F-104A's Bendix fuel control system caused ongoing problems and was associated with some losses both there and in the USA, as Stan Hood explained;

'General Electric developed the J79 with only one fuel control system. Even the F-86 Sabre had two systems. We lost two aircraft in the USA and four in Spain. The first [in the USA] was flown by Michael Miller, later a Thunderbirds pilot. His engine failed at 26,000 ft and he stuck with it and ejected at 1000 ft, not knowing whether the aircraft had an upward or downward-ejecting seat! Our planes were being modified at the time, and it so happened that his was the last to receive an upward-ejecting seat.'

The second crash took the life of the South Carolina ANG's first commander, Brig Gen Bernie B McEntire, whose aircraft (56-0853) had engine failure on takeoff. He steered it away from the Harrisburg area towards the Susquehanna River. As his successor, Maj Gen Hood, explained, 'he was at 2500 ft and could have ejected, but he was concerned about trying to save the J79 engine so that the fault could be identified.

ANG units looked after their jets, as evidenced by gleaming F-104A-20-LO 56-0793. Assigned to the 86th Air Defence Division of the Seventeenth Air Force at Ramstein AB, West Germany, during the Berlin Crisis, this Starfighter took part in the largest aerial deployment in the ANG's history. During its time in western Europe the ANG stood ready to intercept any Soviet aircraft that attempted to deny air access to West Berlin (Lt Col D Nichols via T Panopalis)

Arizona's 'Copperhead' F-104As were joined at Ramstein AB by the 151st FIS of the Tennessee ANG. Both units relied on their AIM-9B missile armament as their jets had no M61 guns installed. Other ANG fighter units deploying to Europe had to make do with obsolescent F-86H Sabres and F/RF-84Fs (Author's Collection)



They had just come out of a conference about it that day'. Quickly choosing to make a water landing, McEntire did not have time to get his flaps down. 'When you lose your engine in an F-104 it unwinds in about 15-20 seconds, and if you don't get your takeoff flap switch down in that time you have no flaps. He had a full fuel load, no flaps and a high AOA. When he hit the water the aeroplane just slapped down and broke up. He was decapitated'. Congaree AB was re-named in McEntire's honour.

In Spain, one aircraft (56-0864) had to divert from a fog-bound runway and ran out of fuel en route to its diversionary airfield, but three were lost because of engine failure. Two were F-104As (56-0861 and 56-0862), and F-104B 57-1311, en route to a flight safety conference at Madrid, was written off when a severe nose-wheel shimmy on takeoff threw rubber debris into the engine. 'The shimmy vibration made a lot of circuit breakers pop before the pilot could get the nose-wheel up', Maj Gen Hood recalled. 'He did a closed pattern [approach] and on the down-wind leg he didn't jettison his 300-gallon wing tanks. He was almost lined up on the runway when the engine quit'. The pilot ejected first at 300 ft, but his 'zero' ejection lanyard was not connected correctly and his parachute did not open in time. The back-seat instructor pilot sustained spinal damage that terminated his flying career.

TAC's STARFIGHTERS

The first three F-104Cs (56-0883, -0884 and -0887) were delivered to the 476th TFS/479th TFW at George AFB in September 1958, being handed over to the unit by Lockheed's Vice-President, Burt C Monesmith. Their transformation from short-range interception to nuclear strike made them capable of flying two-hour missions with four external fuel tanks (plus in-flight refuelling) and a single 'nuke' at average speeds of around 500 knots. M-2 bombing computers enabled low-altitude delivery or over-the-shoulder LABS toss bombing attacks, followed by a very rapid escape.

TAC wished to demonstrate that the F-104C was not 'short-legged' and could be deployed globally without having to hitch rides on transport aircraft. Operation *Star Blazer* tested this capability when the 479th flew eight F-104Cs to Morón AB via Myrtle Beach, South Carolina, on 25 August 1959. The 3672-nautical mile flight took 7 hrs 22 min, and was a first for the F-104. It also presented obvious risks for a single-engined fighter, as Brig Gen Avelin P Tacon, commanding George AFB's 831st Air Division (AD), observed;

Under its flamboyant commander Col George Laven, the 479th TFW decorated its F-104Cs with colourful 'shooting star' markings when its four squadrons were established from 1958. The 436th TFS, the first to receive F-104Cs in September, used dark blue when decorating its jets. The squadron made several Cold War deployments to Morón AB, where this F-104C-5-LO was lost following engine failure on 15 March 1960 (via T Panopalis)



‘Nothing can set a programme back faster than dumping an aircraft in the Atlantic somewhere off the Azores on its maiden deployment flight. This line of reasoning was not entirely objective either – there were eight pilots who took a rather subjective viewpoint of this aspect of the exercise! Our greatest problem with the aircraft had been afterburner failures while in flight. A fix had been made, which an accelerated flight programme on selected aircraft had verified. Takeoff was on time. The weather was clear, with a last-quarter moon. This was just as well as the F-104 accelerated rapidly on takeoff and was extremely hard to spot in the dark, which made night join-ups difficult.’

Two air-spares returned to Myrtle Beach, leaving six to make three successful refuellings from Boeing KB-50 tankers that demonstrated how ‘air-to-air hook-ups in the F-104C were comparatively easy’. In Vietnam, these aircraft would conduct similar refuelling sessions on very many occasions, and verifying the technique at this early stage was reassuring. For European deployments the jets’ KB-50 tankers had to be pre-positioned in set orbits which the F-104 pilots could meet and join. Norm Filer recalled that with the F-104C, ‘plugging [into the drogue “basket”] was easier than in the F-100C Super Sabre, but the speed difference between the tanker and you was much more noticeable. While the F-100C would just wallow around before dropping off [the tanker], the F-104C just gave up and fell off suddenly. It did not like to go slow. The only way to get full tanks if you were carrying all four was to call for a toboggan descent when things started getting heavy. If you had more than a flight of four to refuel, and everyone needed a descent to get full tanks, the last guy on the tanker was almost down in the weeds’.

When jet-powered KC-135A tankers became available they could keep pace with the fighters and make the crossing with them. Refuelling from the faster KC-135As was also considerably easier than working with KB-50s, although a failure of the ‘disconnect’ mechanism once refuelling was complete could be lethal. The 435th TFS lost one of its F-104Ds (used for squadron check flights) during a night refuelling session over Spain. Instead of releasing the nozzle at the end of the F-104D’s stalky refuelling probe, the tanker’s Flying Boom wrenched it off. The heavy nozzle promptly flew back into the fighter’s port intake and the engine caught fire, obliging Maj John Thomas and his back-seat passenger (the squadron flight surgeon) to eject.

On their own, Starfighters could cruise at around Mach 0.9 with four external tanks, using an omni-directional VHF radio navigation aid and their search radars to locate tankers within a 40-mile search range. The return journey on 2 September was also uneventful, and the deployment was the first of several to Morón AB by F-104s. For pilots, the journey (which could last more than 10.5 hours on the non-stop crossings in the mid-1960s) was made less pleasant by the need to wear bulky, waterproof rubber ‘poopy suits’ and other survival gear – including a bottle of stimulant ‘go-pills’ to keep them awake. They also knew that if their single engine failed the chances of survival in the freezing Atlantic were minimal.

George AFB’s 436th TFS returned to Morón on 10 August 1960 as part of a regular rotation of 479th TFW squadrons to the base to provide air defence for SAC B-47 Stratojets operating from both Morón and Zaragoza. By 1963 there were also incidents involving harassment of SAC



Desert Strike was a massive USAF and US Army exercise in the Mojave Desert throughout most of May 1964. It involved more than 90,000 men and 53 USAF squadrons with 400 aircraft, including some specially-marked Starfighters of the 479th TFW whose F-104C-10-LO 56-0922 is seen here over the battle area (USAF)

B-52s flying airborne nuclear alert sorties over the Mediterranean by fighters that originated from North African air bases. These aircraft were possibly MiG-17s and MiG-21s from Algeria. At a time of considerable international tension, two F-104Cs were kept on five-minute alert to intercept any potential threats to the orbiting B-52s, but the Starfighters' credibility as a deterrent prompted hostile intruders to back off before contact could be made.

In order to maintain mission readiness for any likely action there was considerable air-to-air combat practice flown between the F-104C flights. TAC also required the deployed squadrons to maintain currency in gunnery and ordnance delivery, so they spent time on the Zaragoza range. And when the 65th AD F-86D unit at Torrejon AB converted to the F-102A Delta Dagger, the 436th FIS also had to provide an alert flight at that Spanish base. The 479th TFW's Cold War deployments to Morón continued until 1963.

The 479th also shared alert status with ANG F-104As by transferring squadrons to Hahn and Ramstein air bases during the Berlin Wall Crisis. The prospect of actual conflict was seen as a real one, exemplified by the fact that 435th TFS F-104Cs that staged through Morón in September 1961 were armed at the end of the runway prior to flying on to Ramstein. Pilots had been warned that their arrival might trigger a response from the East German MiGs that were based a few miles across the border from them. George Wells, who had recently transitioned to the F-104C, joined the 435th TFS deployment;

'When we arrived at Ramstein an air force colonel from the area headquarters gave us a briefing on the current state of the conflict. We were told that the Russians were threatening to shoot down some of the cargo aeroplanes that were supplying Berlin. He added that if they were to go so far as to down one of our air force aeroplanes we were going to go in and "shoot the bastards down".'

As a further demonstration on the wing's combat capability, the 479th TFW won the 1962 William Tell Fighter Weapons competition. Capt Charles Tofferi, alternately flying F-104Cs 57-0914 and 57-0915, beat 14 F-100 and F-105D pilots in conventional and nuclear delivery and in gunnery. He attained perfect scores (3000 points) in air-to-air gunnery, air-to-ground rocketry, strafing and napalm delivery, and drew considerable publicity to the F-104's weapons capability. Tofferi would subsequently be killed in action over Laos in October 1966.



By 1963 the USAF had revised its plans for the F-104, cutting its initial orders for 722 jets down to 296. It also decided to return the ANG F-104As to ADC national control. Maj Gen Hood attributed this decision substantially to the outcome of a high-altitude sortie on which he was launched from five-minute alert status to intercept an 'intruder'. Not informed of its identity, Hood's 'target' was being trailed by an F-101 that was flying at 56,000 ft – the Voodoo's maximum altitude;

'I went by the F-101 at Mach 2. If I'd had communication with that aeroplane I could have contacted him, but I had no reason to. I could have gone up another 10,000 ft with no trouble at all, but I slowed down with a split-S manoeuvre at 63,000 ft and returned to base, as I was getting too far south. When I got back I called the F-101 pilot and he said, "Was that you that went by me?" I said, "Where was the real target then?" He replied that it was 15,000 ft above him and 20 miles in front. We were sworn to secrecy, but were later told that the target was a U-2, probably a CIA one, and we had been sent, without partial pressure suits, to establish what it was.

'Later, Gen Curtis LeMay from the USAF operations staff said, "What is the Air National Guard doing with our only high altitude interceptor? I want those aeroplanes back on active duty with or without the pilots". We had recently returned from a nine-month tour in Europe, and as reserve pilots we didn't want to go back on active duty just so he could have our aeroplanes. He was told that he could have the aeroplanes without the pilots, so they gave us the F-102A Delta Dagger instead.'

The 319th FIS converted from F-102As to the F-104A and transferred to Homestead AFB, Florida, with the 32nd AD in March 1963. It was the last USAF unit to operate F-104As, which were retained until the squadron disbanded in December 1969. The aircraft had their J79-GE-3A engines replaced by the J79-GE-19 models in the mid-1960s, the latter featuring an improved jet nozzle and high compression ratios. The 'Dash 19's' spectacular power increase from 12,850 lbs to 18,900 lbs thrust in afterburner considerably enhanced supersonic acceleration. With the J79-GE-3A engine the F-104A took up to four minutes to accelerate from high subsonic speed to Mach 2, but the improved J79, which was similar to the version used in the final production Starfighter (the F-104S) and also the F-4E Phantom II, halved that time in a quarter of the distance

This 337th FIS F-104A-30-LO (56-0880) has its hydraulic service panel lowered, allowing easy access to the two hydraulic systems. Behind this an arresting hook was later installed, for the first time in a Century Series fighter. The hook was a last resort in an emergency landing since the 18-ft diameter braking parachute was the usual method employed for slowing the aircraft down, reducing the landing roll by about 15 per cent. The brake-chute induced a slight nose-down attitude so the pilot had to wait until all three wheels were on the runway before pulling the drag-chute control. Variable-position speed brakes, extending from the rear fuselage, could be used throughout the aircraft's speed range, although extension above Mach 1.6 could cause an abrupt nose-down attitude and compressor stall. Leading and trailing edge flaps were used to increase lift during takeoff and landing, and the takeoff setting could be employed as a manoeuvre flap to improve low-speed turn radius
(*T Cuddy via T Panopalis*)



F-104A-20-LO 56-0824 CITY OF KEMP from the 331st FIS at Webb AFB, Texas, with a 'Roadrunner' emblem on the nose and an Outstanding Unit Citation Award ribbon proudly displayed above the U.S. AIR FORCE titling on the forward fuselage. One of the three 32nd AD units to 'regain' F-104As within ADC, the 331st FIS transferred its aircraft and personnel to the 4760th Combat Crew Training Squadron in 1967 where they trained RJAF pilots. This aircraft was passed on to Jordan's No 9 Sqn in 1972 as 921/R and was lost in an accident two years later (USAF)

with much better fuel economy. It could reach supersonic speed at around 25,000 ft without afterburner, and 'on the deck' the jet was capable of a sustained Mach 0.9 cruise with a 750-mph limit in military power.

Wearing partial pressure suits, pilots flew practice interceptions of U-2s above 60,000 ft, meeting their targets at around Mach 1.8 and sustaining Mach 2 above 70,000 ft. In later tests, F-104s fitted with 'Dash 19' engines were able to defeat USAF-flown MiG-17s and MiG-21s even in turning fights if pilots kept their speed up. The aircraft also had their M61 guns re-installed and received an upgraded cockpit pressurisation system.

In March 1963 the 331st FIS at Webb AFB, Texas, also relinquished its F-102As for F-104As, retaining them until 1967 and deploying to Puerto Rico during the Dominican Republic crisis of 1965, which required seven TAC wings to stand alert.

CUBA

ADC's 're-possession' of its F-104As was a recognition of the aircraft's usefulness in meeting the threats associated with the Cuban Missile Crisis. The sighting of two SA-2 SAM sites in that country on 29 August 1962 began a stand off between President John F Kennedy and Soviet Premier Nikita Khrushchev that required numerous TAC units to deploy to the southeastern USA. The 479th TFW also recalled two of its Europe-deployed units early in 1962, leaving one at Morón, and relocated them to Naval Air Station (NAS) Key West (previously known as Boca Chica), Florida, to counter possible air attacks by the 125 Soviet-supplied MiGs believed to be based on Cuba.

The prospect of meeting Cuban MiGs in the 'hot-rodged', lightweight F-104 gave rise to some intensive air-to-air practice with Tyndall AFB-based F-106A Delta Darts. Realising that subsonic turning fights were to be avoided, the F-104 pilots relied on sustained high speed and use of the vertical dimension wherever possible. Tactics evolved whereby a flight of F-104s could vary the flight lead position, depending on which pilot had the best chance of a shot, rather than adhering to the standard USAF 'fighting wing' with one 'shooter' leading the four-aircraft flight and taking the firing opportunities. This approach required a training programme that qualified all new pilots as flight leaders. In Vietnam, this flexibility was slowly adopted by some F-4 and F-105 units, although it



was used by US Navy fighter squadrons in their MiG duels from the late 1960s. For Starfighter pilots over Florida it proved to be a successful way of defeating missile-armed F-106As.

Pilots also became very proficient in aerial gunnery through the use of towed dart targets, with good scores at ranges of up to 4000 ft being attained. A successful 'attack' on a pair of F-106s or F-4s involved a zoom climb directly behind the targeted jet and a simulated AIM-9B launch or a gun pass, followed by a vertical zoom and a high-speed re-attack from above, before escaping from the second fighter in another vertical manoeuvre.

A total of 28 F-104Cs from the 479th TFW arrived at NAS Key West to share 'hot pad' alert duties with a succession of US Marine Corps F-8 Crusader and F-4B Phantom II units, at least one of which (VMFA-531) intercepted several of the MiG-17s based on the island and turned them back towards the Cuban coast. The 479th TFW pilots arrived in style, flying a series of formation passes over the base, but the squadron commander, the colourful Lt Col George Laven, making the first landing, blew both main landing gear tyres, as did the other members of the flight! The F-104's relatively small main wheels and tyres were susceptible to this kind of damage. George Wells, arriving with the 435th TFS noticed his oil pressure warning light glowing red as he approached Key West.

'This light was telling me that I possibly had about five or six minutes before the engine would very likely seize. If that happened, both the aeroplane and I would stop flying very shortly thereafter. I informed the tower that I had an emergency and flew a wide traffic pattern at low power so that if my engine quit I would still have enough altitude to make it to the runway without assistance from the engine. This type of approach is flown at a considerably increased speed and a reduced degree of extended flaps. The runway at Boca Chica was a bit of a challenge since it was only 7000-ft long. My engine continued to operate and I landed my aeroplane right where it should have been on the runway.'

George received some comments from the wing commander, who was clearly unaware of the emergency, concerning some 'sloppy landing patterns' on the squadron's arrival, but he headed off with the other pilots to their crew quarters at the nearby *Holiday Inn*.

Crusader pilots escorting RF-8A reconnaissance flights over northern Cuba were often practice-intercepted by F-104s as they returned to US airspace. It was quickly established that if the Starfighter pilots maintained

An F-104C of the 479th TFW stands alert at NAS Key West during the Cuban Missile Crisis in 1962. The jet's two AIM-9s and single M61 gun made a hard-hitting air-to-air combination. The Vulcan weapon was bore-sighted using a Weaver scope, mounted on the firing mechanism. Cannon shells in early F-104As were fed into the aircraft's ammunition cans via a manually operated loader, without removing the cans from the fuselage. Both the loader and ammunition were delivered to the aircraft aboard an F2A trailer, which was also used to transport pre-loaded ammunition boxes to later-serial F-104s. This F-104C-10-LO (57-0929) later flew a combat tour from Udorn RTAFB with the 435th TFS in 1966 (USAF)



F-104A-15-LOs from the 83rd FIS over San Francisco Bay in 1959. During the Cuban Missile Crisis pairs of F-104s made many head-on supersonic interceptions of 'bogies' that were all cleared as 'friendly'. One F-104 would pass about 500 ft from the suspect aircraft to identify it and be ready to return for an attack if necessary. The second F-104, the 'shooter', lagged about four miles behind, ready to open fire if the initial identification revealed a hostile intruder (via *T Panopolis*)

high speed they could control the fight and disengage at will from the F-8s, which had the advantage in a turning fight, particularly at speeds below 400 knots indicated air speed (KIAS).

F-104C pilot Capt Tom 'Sharkbait' Delashaw explained that he, like the ADC pilots at Homestead, 'used a slashing attack with the cannon at 600-750 KIAS, a quarter-roll with a 0.25 to 015 "unload" for separation outside missile [and typically visual] range, then a 2g spiral climb to re-position for an unobserved re-attack. Unlike the F-4, the F-104 didn't need Military power to sustain Mach 0.9+, so it didn't drag a smoke trail everywhere it went. It fought in afterburner, which was smokeless, making the jet very hard to spot. With the slashing attack it projected a nose or tail aspect while within visual range, and only showed its planform [view] during the spiral climb while at long range'. Capt Delashaw and other George AFB pilots would find this experience valuable when they deployed to Vietnam a few years later.

During the Cuba affair the F-104s were selected partly because, in addition to Il-28 bombers and numerous nuclear-tipped missiles, Soviet MiG-21s had been sighted on the island. From more than 500 tactical fighter and fighter-bomber aircraft poised to defend the southern areas of the USA, including F-105 Thunderchiefs and F-4 Phantom IIs, the F-104 was thought most capable of meeting and defeating these opponents. Pilots in the 435th TFS were told that they could easily be involved in a 'hot' situation, as George Wells recalled;

'For the first couple of days we were assigned to fly down to designated locations on the northern coast of Cuba and be prepared to fly over Cuba itself it proved necessary to protect either USAF RF-101 Voodoo or US Navy RF-8 Crusader reconnaissance aeroplanes that were taking photographs of the missile sites. The F-104 was very fast – we could accelerate above the speed of sound in an incredibly short time, so perhaps we could be of assistance if we were needed. We were not allowed to fly

over Cuba unless we were called upon to do so. I did make a couple of turns at about 15,000 ft over Havana on one flight, but stopped doing so when we were informed that anti-aircraft guns were being installed on the roofs of hotels and other tall buildings.'

After President John F Kennedy's 22 October television address to the nation, in which he showed photographs of the Soviet missile sites in Cuba, a US invasion of that country seemed imminent. The 435th TFS pilots had been led to believe that the crucial time would be 0600 hrs the following morning, and the alert signal would be telephones ringing continuously in all the pilots' rooms at the hotel. As George Wells explained;

'We all did a little hyperventilating when the phones in all our rooms rang steadily at 0500 hrs. We literally jumped into our flying suits and ran out of our rooms, still zipping up the zippers. We were greeted by about a dozen reporters taking our pictures as rapidly as they could. I guess the operation was not as big a secret as we had imagined. I rolled out of my vehicle and almost fell on the ramp as I scrambled for the ladder up to the cockpit. My crew chief was already there with the auxiliary power unit fired up. I transmitted on the designated frequency that I was ready to go.

'I recall this whole event as being sort of surreal. I could not comprehend the enormity and the implications of what might happen, but I was mentally prepared to charge ahead into whatever the day had in store for me. After about 15 minutes we were informed over the radio that the operation had been cancelled. No matter how willing and even somewhat eager I was to launch my aeroplane into an unknown possible nuclear confrontation, I was equally relieved when I was informed that it was not going to happen, at least at that time.'

President Kennedy's strategy had forced the Soviet leader, Nikita Khrushchev, to back down, and put on hold USAF Chief of Staff Gen Curtis LeMay's determination to press for an immediate invasion following the shooting down of a U-2 by a Russian SAM over Cuba. However, the 435th pilots remained at Key West until 8 December, completing a 50-day deployment.



A 'SCANG' pilot sprints to his F-104A for an alert scramble. The aircraft's two AIM-9Bs were its sole armament for this purpose, as the M61 cannon had been removed and the gun port sealed off (via *T Panopolis*)

VIETNAM

George Wells took his first 'dollar ride' as a cadet in a Lockheed T-33 at Bergstrom AFB, Texas, in 1955, and it was 'the single most impressive event that I had ever experienced up to that point in my life'. Six years later, after a tour on F-100s with the 48th TFW, George reported to Victorville, California, to join the 435th TFS/479th TFW flying F-104Cs. His first reactions to the aircraft were extremely positive;

'There were very few fighter pilots in the air force who would not have given an arm and a leg to be assigned to an F-104 squadron. I would hear many stories of how political influence and other extraordinary efforts were used to get the assignment. As I taxied out on my first solo flight, I did not experience the usual feeling of apprehension. Instead, it seemed that this airplane was built specifically for me. I felt totally comfortable and I had not even arrived at the takeoff end of the runway yet. I found that its flying characteristics and performance were far superior to the F-100. The F-104 was really fun to fly, and it was just about the fastest thing in the air back in the early 1960s.'

F-104C production reinstated the M61 cannon, and its use became part of the syllabus at George AFB. Wells was impressed by the weapon's accuracy while he qualified in air-to-air gunnery;

'This was accomplished by shooting at a silver honeycombed dart that was towed 1500 ft behind another F-104. My instructor, Ray Krasovitch, who was one of the best F-104 pilots, in my estimation, in the world, explained the techniques to shoot holes in the dart and how the radar worked. The radar in my aeroplane did work, which was pretty rare I was told.

'Ray instructed me that when a little blip on the sight reticle appeared it was telling me that I was 3000 ft from the dart. Since the gun was certified to be accurate at 3000 ft I could start shooting any time I had my sight lined up properly on the target. The thought crossed my mind that I should wait until I was a little closer to the target, but since every parameter had been met I elected to hose off a few rounds just to get a little practice in pulling the trigger on the control stick. To my great surprise I not only scored hits on the dart but completely destroyed it. The honeycomb structure was covered with an aluminium coating, and the dart formed a sparkling, glittering cloud in the bright sunshine as it descended to the desert below.'

His early experience of firing live GAR-8 (AIM-9B) missiles was less fortunate. He and Ray Krasovitch were told to down a T-33 target drone for a TAC film called *Full Scope*, which was intended to show the command's capability and release much-needed funds at a time when the majority of USAF dollars were going to SAC. The pilots were required to fire at the target, which was flying steadily at around 1000 ft to make it more visible to the ground-based movie cameras. Distracted by ground-heat, the old-stock missiles missed the target every time during two days of filming until one of Ray's grazed the T-33's rudder without exploding and the target drone crashed – out of the cameras' view.

As the conflict in Vietnam gathered momentum in 1964 the deployment of American air assets also increased. F-105 Thunderchiefs were sent to Thailand from mid-1964, and when Operation *Rolling Thunder* began in March 1965 two F-105 wings were involved in attacks on North Vietnam. Their protection against the MiG-17s that opposed them from April 1965 initially came



from F-100 Super Sabre escorts. On 19 April 1965 the 476th TFS's F-104Cs, commanded by Col Darrell Cramer, arrived at the crowded Da Nang AB to share these duties and provide a more credible MiG deterrent, particularly after the confirmation of MiG-21 deliveries to the Vietnamese Peoples' Air Force (VPAF).

The choice of the 479th TFW and its F-104Cs was logical as the unit's air-to-air combat skills had been demonstrated at William Tell competitions and Fighter Weapons School tests. Starfighters had been deployed to Florida, Taiwan and Europe for similar reasons, and Vietnam also seemed to require the same undisputed air superiority capability to keep MiGs at bay.

435th TFS pilots had flown most of the Project *Featherduster* F-104C sorties in which they used John Boyd's and Ed Riccioni's 'double attack' tactics, where pairs of Starfighters flew as a mutually supporting team, swapping roles as 'shooter' in order to deny the target aircraft the chance of escape or the opportunity to turn and engage his attackers. They also concentrated on vertical manoeuvring, using the F-104's superior rate of climb to attack from below and then reversing for a second attack from above. This ran counter to the USAF's standard 'fluid four' fighter formation in which the flight lead was the 'shooter', passing the opportunity to one of the other three pilots at his discretion. Nevertheless, 'double attack' was the tactic that the 435th TFS intended to use in Vietnam, and the 479th TFW was the first USAF wing to adopt this approach.

In April 1965 George Wells joined the 476th TFS on its first F-104 deployment to Southeast Asia as part of Operation *Two Buck*. He had to organise tankers for 18 fighters on one of the longest over-water flights ever made by TAC aircraft, via Hickam AFB, Hawaii, with seven in-flight refuelling hook-ups. They staged through Anderson AFB, Guam, before finally reaching their main operating and maintenance base at Kung Kuan, in Taiwan. Four aircraft had to abort en route with mechanical problems. Once in Taiwan following the four-day transit, the squadron awaited further orders. These arrived on 19 April when 14 F-104Cs proceeded to Da Nang AB and flew escort sorties the following day.

One of the unit's first missions foreshadowed many that the 479th TFW would fly during two follow-on deployments to Southeast Asia. Four F-104Cs were each loaded with two M117 bombs to participate in a Vietnamese Air Force (VNAF) attack by A-1 Skyraiders in the II Corps area. Led by the flamboyant VNAF commander Nguyen-cao-Ky, 12 A-1s

***Swift Strike III* was another major pre-Vietnam War US Army-USAF joint exercise, involving 75,000 troops and large numbers of TAC squadrons in August 1963. These F-104C-10-LOs from the 479th TFW were based at Myrtle Beach AFB, South Carolina, during the exercise. TAC units had to be capable of deploying rapidly to trouble spots across the globe. The 479th TFW was arranged in four squadrons with 18 F-104Cs each, rather than the usual three-squadron TAC structure, so that each unit could make a four-month deployment to trouble spots with a year's gap between these foreign assignments (USAF)**



F-104C-5-LO 56-0892 with two M117 bombs uploaded for a ground attack sortie – the most common mission type for many Starfighter pilots in Southeast Asia. Harold Alston's logbook recorded 67 interdiction and route-reconnaissance missions during two tours at Da Nang and Udorn out of a total of 136 combat sorties. He also logged 20 missions escorting EB-66 Destroyer electronic intelligence aircraft over the North and witnessed his first SAM launch when one was shot down by an SA-2. This 476th TFS aircraft returned to George AFB on 18 December 1965 and later flew with the Puerto Rico ANG. It was eventually put on display at Luke AFB to commemorate the F-104Gs used by the 58th TFTW's training programme (USAF)

targeted a suspected barracks area near Dong Hoi. The VNAF bombs fell in the town, missing their objective, leaving the F-104Cs, led by Lt Col Howard Dale, with the task of destroying the target with a series of direct hits in dives from 14,000 ft.

Da Nang was a frequent target for Viet Cong sapper and rocket attacks, as the 476th TFS crews soon learned. Although three

F-102A Delta Daggers were destroyed by sappers throwing hand grenades into their open cockpits and firing guns up their tailpipes on 1 July 1965, the F-104Cs all escaped harm. However, a few days after its arrival the squadron sustained its first hits by ground fire when an F-104C returned from a strike with substantial wing damage.

BIG EYE PROTECTORS

A primary role for the F-104 in Southeast Asia was the protection of the Lockheed EC-121D Warning Star aircraft that patrolled off the coast of North Vietnam, providing MiG warnings to US pilots via their radar and monitoring equipment. The USAF had received 74 of these high-value aircraft, supplemented by ex-US Navy examples from 1966. The 552nd Airborne Early Warning and Control Wing (AEW&CW), based at McClellan AFB, California, was responsible for operating them in the Far East. The piston-engined 'Pregnant Guppy' or 'Connie' (after its Constellation airliner origins) radar surveillance aircraft had been active during the Cuban Missile Crisis, and in April 1965 the USAF established its Project *Big Eye* Task Force in Southeast Asia three days before the arrival of the first F-104Cs. Four EC-121Ds were sent from Tainan Air Station, Taiwan, to Tan Son Nhut AB, South Vietnam, and two began radar orbits on 16 April 1965.

Originally a US Navy aircraft, the EC-121 was designed to protect the fleet by detecting airborne intruders over the sea. It had difficulty, therefore, in achieving clear radar imagery over land due to terrain interference, particularly over mountainous North Vietnam. Its AN/APS-45 altitude-measuring radar in its large 'conning tower' radome was also only effective up to a distance of 70 miles from the EC-121. This meant that *Big Eye* aircraft had to fly their radar orbits only 30 to 50 miles offshore at altitudes as low as 50 ft so that their AN/APS-20 (or AN/APS-95) search radar signals could be bounced off the water, thus extending their detection range to 150 miles – a technique developed during the Cuba crisis missions.

Flying at low altitude also helped to protect the vulnerable EC-121s from detection by enemy radars. And even when they were detected *Big Eye* aircraft were usually left alone as VPAF controllers kept their fighters away from EC-121Ds. Any VPAF pilot that ventured out over the Gulf of Tonkin in search of patrol or tanker aircraft knew that he would also face US Navy carrier-borne fighters as well as F-104C patrols.

Escorting EC-121Ds proved a challenge for F-104 pilots nevertheless. Flying at low altitudes in poor weather or at night was difficult enough for the *Big Eye* crews, operating at speeds below 290 mph. Close escort by considerably faster F-104Cs was clearly impossible, so separate barrier combat air patrol (BARCAP) orbits by up to three flights of Starfighters and their KC-135A tankers at altitudes of between 15,000 ft and 20,000 ft were established to keep MiGs away from the two *Big Eye* patrol tracks.

Ethan Alpha Track was flown by one EC-121D at low altitude, close to the coast, while the Ethan Bravo Track – some 50 to 60 miles long, and at an altitude of 10,000 ft – was serviced by the second aircraft (also acting as an airborne spare and providing radar cover for the Alpha aeroplane) further out to sea. The Bravo track was replaced in October 1966 by an Ethan Charlie orbit over Laos, which, from August 1966, had the secondary purpose of identifying US pilots who pursued MiGs across the Chinese border. The chances of interception by terrain-hugging VPAF fighters, which would have a short journey to reach their *Big Eye* targets, could not be discounted, and the F-104Cs' spectacular acceleration enabled them to intervene rapidly.

Usually four F-104Cs covered each EC-121D, with a third flight cycling to a tanker before moving in to replace the flight escorting the Bravo Track fighters, which would in turn move to replace the Alpha Track escort flight, freeing them up to refuel. Each mission therefore required 12 F-104Cs and two tankers, increasing to four tankers when the Starfighters were later supplemented by thirstier F-4C Phantom IIs. Starfighter cover occasionally had to be reduced to two-aircraft flights if insufficient jets were available. The patrol areas were divided into a grid of ten nautical mile squares, and each was allocated a two-letter code, while MiGs had a colour code – blue for MiG-21s, red for MiG-17s and white for MiG-19s. 'Blue, Alpha Golf' therefore indicated a MiG-21 flying over Hanoi, and two CAP F-104s would be directed to head them off while a pair of jets on the tanker moved in to replace them if necessary.

The F-104's performance was already well-known to the North Vietnamese, and its reputation helped to deter the VPAF from attempting to attack the scarce, costly and essential EC-121Ds. However, because MiG pilots did not anticipate being vectored into engagements with F-104s, they had no specific training in air-to-air strategies for use against them. As Dr István Toperczer noted in his interviews with VPAF pilots, 'They planned their aerial manoeuvres against USAF F-4s, F-105s, B-52s and US Navy aircraft. They did not mention tactics against the F-104'.

479th TFW pilots felt that the MiGs deliberately avoided tangling with F-104s in much the same way

Escort for EC-121D Super Constellations (like 55-0127, built as an RC-121D-LO) occupied much of the 479th TFW's combat time, although these two F-104As are involved in pre-war training for that role. With a wing design derived from Kelly Johnson's P-38 Lightning, the EC-121 was a crucial part of the Vietnam air war. Although no EC-121Ds were lost to North Vietnam's fighters during the conflict one aircraft with its crew of 31 was shot down by a North Korean MiG-17 in April 1969 (USAF)





F-104Cs each carried fuel for about 2.5 hours, and they could replace this from a KC-135A in around five minutes via the hose and 'basket' attachment bolted onto the KC-135A's Flying Boom refuelling system. The use of underwing tanks allowed the carriage of AIM-9Bs on the wingtip launch rails (USAF)

that the US Navy F-8 Crusader community believed that VPAF pilots were told to steer clear of combat with them. Protection of the vulnerable *Big Eye* was seen from the outset as vital, and most strike missions were cancelled if the F-104C escort was unavailable, or curtailed if the fighters had to abort their BARCAP. *Big Eye's* value to the USAF strike packages over North Vietnam became evident in July 1965 when a 522nd AEW&CW aircraft vectored an F-4C Phantom II crew to the first EC-121-controlled MiG kill. Twenty-four others followed in the 13,931 combat missions flown by *Big Eye* crews, and the F-104C pilots could legitimately claim a share of those successes.

Airborne opposition to the *Big Eye* 'Connies' was not confined to North Vietnam. Communist China sometimes despatched fighters from Hainan Island (Hai Nan Tao) to intercept US aircraft that ventured too close. On one occasion George Wells was leading an escort flight when the EC-121 crew announced that they had picked up a potential enemy interceptor heading their way from Hainan. Wells and his wingman initiated a sharp turn into the bogie, which promptly headed back in the direction of the island. The F-104s returned to their patrol station, only to be told a short while later that the bogie had resumed its course towards them. George Wells explained what happened next;

'I accelerated to around Mach 1.6 as I was informed that my target was at "12 o'clock" and 12 miles. "Your target is now at six miles. Maintain heading" [was the instruction given by the *Big Eye* controller]. I knew that I should be able to pick the target up visually very soon at this range. I was probably running pretty high on adrenaline as all the training and fighter experience that I had accumulated over the years was hopefully going to be put to use by shooting down a MiG.

'As my wingman and I were eagerly searching ahead of us I noticed that we were very rapidly approaching the southern coast of China. At about a mile or two from the Chinese shoreline I executed a 6g turn in the opposite direction. My almost instantaneous reaction was a result of the realisation that this was very likely a Chinese MiG, and it would be prudent not to fly over Chinese mainland territory and shoot down one of their aeroplanes. There probably wouldn't have been much said if I had shot him down over the Gulf of Tonkin – at least the evidence would have been covered up. I reported this incident to Intelligence after I landed at Da Nang and never heard a word about it from my superiors. I guess we accomplished our mission, for the EC-121 did not get shot down.'



STRIKE ESCORT

The main purpose of the F-104Cs' presence during strike escort sorties was to keep MiGs away from the F-105 formations, proving once again the Starfighter's effectiveness in establishing air superiority. Escort duties were shared with F-4C Phantom IIs of the 45th TFS/8th TFW, which worked with F-105 strike packages during a 120-day TDY from 9 April 1965. As an escort fighter the F-104C had a considerable speed advantage over the F-100D that it replaced when Super Sabres were withdrawn from CAP and flak suppression duties after 4 April following the loss of three F-105Ds and an F-100D in two days of attacks on the Ham Rong (Dragon's Jaw) bridge.

After the destruction of two F-4C Phantom IIs by a trio of SA-2 'Guideline' SAMs on 24 July President Lyndon Johnson authorised strikes on two missile sites in the Hanoi and Haiphong areas three days later. Eight F-104Cs were included among 58 support aircraft for a formation of 48 'Spring High' F-105Ds. However, the loss of no fewer than six Thunderchiefs that day could not have been prevented by any F-104C CAP. All were hit by AAA, which was by far the most common cause of F-105 losses in the conflict.

One year later, F-104Cs from the 435th TFS were included in the first wave of devastating attacks on major petrol, oil and lubricant (POL) sites on 29 June 1966. Sharing CAP duties with 24 F-4Cs for a strike force of 25 F-105Ds, they were challenged by four MiG-17s of the 923rd Fighter Regiment. One MiG was duly shot down, but it fell to an F-105D pilot rather than the MiGCAP aircraft. Strike force pilots often reported that the VPAF's highly effective ground controllers were able to manoeuvre MiGs around the MiGCAPs to attack the F-105s. Sometimes this could be done by vectoring the MiGs in at low altitude within 'ground clutter' that hid them from the EC-121Ds or US naval radars.

As MiGCAPs were usually positioned near MiG bases to block their access to the strikers, the F-104Cs or F-4Cs were consequently at some distance from the incoming F-105s when enemy jets attacked from other directions, delaying their intervention. There were also frequent, distracting communications problems between flights or the ship-borne *Red Crown* or EC-121D radar controllers, often as a result of over-loaded radio channels.

Sightings of MiG-17s at that time were very rare, and F-104C MiGCAPs were sometimes located unfavourably. One of Capt George Wells' CAP missions placed him as No 3 in a flight led by Maj Walt Irwin, orbiting 30 miles south of Hanoi to block MiGs from reaching an F-105 strike in the Dien Bien Phu area, more than 100 miles northwest of his position. 'The

Two F-104Cs head out on a *Big Eye* mission. Enemy aircraft flying above 8000 ft could be plotted by *Big Eye* EC-121Ds for the benefit of American aircrew in the area, but their altitudes could not be established. US pilots could also be warned by EC-121D controllers that they were about to enter the buffer area separating China and North Vietnam. A crew of up to 30 men in the EC-121D's cramped, poorly air-conditioned interiors worked in temperatures above 120 degrees Fahrenheit that were raised by the vacuum tube-driven electronics crammed into the aircraft. Flight surgeons were often carried to deal with cases of heat exhaustion. The 522nd AEW&CW's EC-121s flew 98,000 combat hours during 13,921 combat missions in nine years in Vietnam without loss or accident. The fact not a single aircraft fell victim to VPAF fighters whilst orbiting just off the enemy coastline proves just how effective Starfighter (and later F-4 Phantom II) units were in protecting them on sorties that could last for up to 17 hours. F-104C crews could expect sorties of up to five hours, with 40 to 90 minutes on station before having to refuel, and a transit flight of around 250 miles from Da Nang (USAF)



F-104C-10-LO 57-0916 receives attention from its Da Nang maintainers in 1965. One of the regular tasks involved a technician crawling right into the afterburner section with the Jet Cal testing gear to check the ignition system. The narrow air intake, with a shock cone that moved forward as Mach increased, originally required a midget to crawl into the inlet and inspect the front of the engine for cracks. Paul Klatt, a four-foot tall nightclub entertainer, performed this task for Westover AFB's F-104As, and a similar character worked at Hamilton. The 479th TFW line chief in charge of maintenance at Da Nang was MSgt 'Doc' Blanchard (USAF)

MiG base was located north of Hanoi. They had a shorter distance to fly to the area than we did. Fortunately we did not get a call for help from the Thunderchiefs'. The two elements of the F-104C flight were about 1.5 miles apart for good defensive positioning and heading west in their orbit, poised for any approach by MiGs. As George Wells recalled;

'I noticed a mirror-like flash in the blue sky at quite some distance – a small black dot that appeared to be coming in our direction. I called this out to Maj Irwin and he responded that he had the bogie in sight. We all armed our guns and watched intently as the black dot developed into the form of an aeroplane. As this aeroplane, heading eastbound, passed just to the south side of the lead F-104C, all four of us punched our microphone buttons at precisely the same time and, as I remember, not too calmly and with a somewhat elevated decibel level, stated, "IT'S A MiG!" It was very apparent that the MiG pilot had not see us until he was practically abreast of our position. As we lit our afterburners and started a hard left turn towards the MiG he jettisoned all his fuel tanks. They were spinning and oscillating with raw fuel misting up the area around them.'

The encounter was over in only 15 seconds. George's role as element leader was to check for any other threats that might prevent his leader from engaging the MiG. He was aware of the VPAF tactic of flying a single



En route to the 436th TFS flightline for another mission in 1965 are (from left to right) Capt Walt Harris, 1Lt Harold Alston and Capts John Olson and Jack Gale (Harold Alston)

aircraft with a second 'trailer' about eight miles behind it to make a rear attack on any fighters that attempted to close in behind the lead aircraft.

'Another factor that played a part in our failure to shoot the guy down was that both our flight and the MiG were flying at about 0.9 Mach in opposite directions. There was so much distance between us when we completed the turn that none of us could locate him. At that time both of our wingmen called "bingo" fuel so we had to depart the area.'

The next day the members of George's flight were astonished to receive a note from headquarters in Saigon stating that, 'Any more breaches of discipline of this nature will be met with severe repercussions'. It became clear that the flight had been seen to exceed the western limits of its designated patrol orbit while dealing with the MiG. An incredulous George observed, 'If we had turned to the east the MiG would have been at our "six o'clock" position, and he would have shot at least one of us down!'

At several post-war reunions of River Rats Vietnam veteran pilots, where the experiences of 479th TFW personnel were sometimes discussed, George frequently heard an opinion expressed that helped him to understand the official attitude to his MiG encounter;

'The powers that be in Saigon purposely kept the F-104s flying missions that resulted in us having few opportunities to engage MiGs. According to the storytellers, this was because the F-4 Phantom II was replacing the F-105, and an effort was being made to prove what a superior aeroplane it was thanks to its advanced radar and missile system. It would not have looked good if an old aeroplane such as the F-104 was shown to be just as effective in combat as the latest technology.'

Harold Alston also experienced a brief encounter with communist Chinese MiGs during his 57th combat mission on 26 September 1966. In a two-ship escort flight, with Capt Donnie Tribble as wingman, he rode 'shotgun' on a 6091st Reconnaissance Squadron (RS) EC-130E-II *Silver Dawn* aircraft that was flying an electronic and communications 'comint' monitoring sortie just offshore from North Vietnam and China. This sortie was 'somewhat boring because the EC-130 was much slower than us. Our job was to clear the area, keep him in sight and deter any airborne threats. The F-104 did not like to fly slowly so we developed a "weave" manoeuvre near to Mach 0.9.



Packaging the F-104's electronics in easily accessed modular 'Jeep Cans' within one pressurised compartment under a single hatch considerably eased maintenance, particularly in the restricted environments of Vietnam, where these 436th TFS F-104Cs were based in 1965. The hatch was usually left open during refuelling to prevent the build up of fumes. Lockheed's innovative modular approach also meant that the plug-in 'cans' could be quickly replaced, or varied in later versions like the F-104G to suit fighter, attack or reconnaissance needs. F-104C-5-LO 56-0892 returned for a second tour at Udorn as *Miss Bevie J. VI/My Darlin' Dorothy*. (1352nd Photo Group/USAF)

F-104Cs under a stormy sky at Da Nang in 1965. After a few weeks some aircraft had their intake cones painted black to kill off reflections that might have revealed the jet's presence at a distance. Although the 479th TFW took 'catamaran' twin-Sidewinder launchers to Vietnam they were not used, pilots preferring the lower drag of tip-tanks and with the gun as armament for some missions. The F-104C in the centre is nicknamed *26 Booms* (USAF)





F-104C-5-LO 56-0903 at Tan Son Nhut AB in 1965 with a smoke-stained M61 gun port and two red bomb symbols on its nose. Operating the gun required at least 96 per cent of engine power to provide enough electrical power to fire it. With 725 rounds, or seven seconds firing time, it occupied much of the F-104's nose area. Bullet cases were retained to avoid airframe or engine damage but links were ejected downwards. The gun took 0.5 seconds to wind up to speed and then emitted 100 rounds in 1.5 seconds. To prevent jamming, dud rounds were ejected automatically. The weapon was situated a few inches from the pilot's left arm, leaving him in no doubt about whether or not it was firing. This F-104C was destroyed in a stall accident near George AFB on 27 July 1966 (1352nd Photo Group/USAF)

F-104C-10-LO 57-0927, which has acquired an ace of spades marking forward of its port engine intake, is armed with two BLU-1 napalm tanks for a CAS mission in 1965. The Starfighters' FG- 'buzz' numbers were removed from the aft fuselages for the Vietnam deployment, these markings having been officially phased out by January 1965 (USAF)

'Not long into the mission we headed north fairly near Hainan Island at 25,000 ft and I spotted four MiGs coming from there at our "one o'clock" position. GCI confirmed them as bandits and I turned towards them. At about five miles off the Hainan coast they reversed course to remain over land. I read this as trying to drag us over their territory so that their AAA could shoot at us. No thanks! We turned back to *Silver Dawn*. As predicted, the MiGs also turned back towards us, so back we went – with the same results. Obviously if they had wanted to engage us they would never have attacked while leaving contrails. With our last turn back towards *Silver Dawn* our cat and mouse game ended. Besides, they were probably at bingo fuel after climbing that high.

'It is every fighter pilot's dream to shoot down enemy aircraft and become an ace. Unfortunately, this was the only MiG sighting I made in 136 combat missions, 100 of which were over North Vietnam. No runs, no hits, no errors and *Silver Dawn* remained safe, so I guess our mission was a success.'

STAR BOMBERS

476th TFS pilots frequently had the chance to hit targets of opportunity during their 1965 deployment, but these were usually forbidden by the highly restrictive Rules of Engagement (RoE) imposed by the White House. George Wells was typical in experiencing 'an especially frustrating aspect of this incredibly stupid method of fighting a war. We would be sent up North to act as clay pigeons on missions that rarely had any effect on the outcome of the war, and then when we would debrief we would again be threatened with court-martial if we took any initiative of our own.

'Many times we would fly south over the Gulf of Tonkin after a mission and we would observe junk-type boats plying their way towards South Vietnam. It was acknowledged that these vessels were most likely crammed





with arms and ammunition for the insurgents in the south. We could have sawn them in half with our 20 mm guns, but sinking them was prohibited.'

Sadly, on one of the few occasions when a 476th TFS flight attacked a tempting target the result was a disaster for the four pilots concerned. A navigational error convinced the flight leader that he was 40 miles north of the demilitarised zone when he saw and strafed 'enemy' bulldozers clearing a forested area, causing numerous casualties. In fact, the troops were South Vietnamese operating American equipment within their own borders. The survivors reported that they had been attacked by MiGs, but examination of 20 mm rounds at the site incriminated the F-104C crews, who all left the USAF shortly thereafter.

The squadron was scheduled for ground-attack sorties on about 18 per cent of its missions, often against dubious 'suspected supply depots' rather than the more obvious targets that pilots often detected.

The 476th TFS's Da Nang deployment ended on 11 July 1965 after 96 days and 1182 combat sorties. More than half of these flights involved *Big Eye* escort and a quarter were MiGCAP or BARCAP missions. F-104 utilisation had been intense, and the ten aircraft retained at Kung Kuan were frequently rotated to Da Nang to replace jets needing deep maintenance. One aircraft was lost when Capt Richard Cole had to eject from F-104C 56-0937 during a 29 June CAS sortie after his hydraulic systems ran down and control of the aircraft became impossible. Recovered by helicopter near Tri Dao, Cole later took part in a SAM site attack, hitting it with M117 bombs and 20 mm fire after F-105s had launched 2.75-in rockets at the missile support vehicles.

The squadron was replaced on 12 July by the 436th TFS, which was able to take on a reduced proportion of EC-121D escort and MiGCAP missions partly due to the deterrent effect on MiGs established by the 476th TFS. The number of CAS assignments increased beyond 50 per cent and a 'hot pad' alert flight was kept ready, jets armed with M117 bombs or napalm, to fly to the immediate assistance of troops in contact with the enemy. F-104Cs could reach targets 250 nautical miles from Da Nang in around 30 minutes from takeoff. Their two M117 bombs were often delivered separately for greater accuracy, and BLU-1 napalm canisters were sometimes loaded for CAS sorties.

Two F-104Cs from the 479th TFW equipped with underwing fuel tanks depart Da Nang in a formation takeoff. An alternative pylon load included BLU-1 napalm canisters. Finned napalm could be dropped from a 30-degree dive for greater accuracy, whereas un-finned canisters usually required a low-altitude, level drop. Although ground attack had been very much a secondary role in the F-104 designers' intentions, the aircraft proved to be a very stable bombing platform. Furthermore, the excellent cockpit visibility allowed the pilot a good view of his target (USAF)

Capt Roy J Blakeley became the 436th TFS's first casualty in Southeast Asia on 22 July 1965 when his attempted dead-stick landing at Chu Lai ended in tragedy. He did, however, manage to steer his damaged F-104C clear of a US Marine Corps ammunition dump on approach prior to the aeroplane being destroyed (USAF)





Capt Willard 'Mac' MacFarlane talks tactics with Capt Larry Shassetz (left) and 1Lt Harold Alston (right, complete with parachute, helmet and ejection seat 'spurs' on his boots) under the 435th TFS squadron banner at Da Nang. The clinking of the pilot's leg-restraint spurs on the ground was a unique feature of the pre-flight ramp inspections of F-104s equipped with Lockheed ejection seats. Long-legged pilots were at some risk of knee injury with the C-2 seat should they have to 'punch out' in flight (*Harold Alston*)

Like their USAF colleagues in F-100s and F-4s, or US Marine Corps Phantom II and Crusader crews, F-104 pilots were managed over the target by forward air controllers who found that they delivered ordnance with considerable accuracy despite their lack of sophisticated bombing systems and multiple bomb-racks to increase their ordnance load. They also ran the gauntlet of ever-increasing AAA in virtually all the areas of Vietnam where the missions were scheduled.

The first 436th TFS casualty was Capt Roy Blakeley from Wink, Texas, who was lost on 22 July 1965. 1Lt Harold Alston was part of that mission, which was targeting a Viet Cong build-up west of the US Marine Corps base at Chu Lai with M117s. The flight originally included John Olson as Lead, Larry 'Shass' Shassetz as No 2, Seb Arriaga as No 3 and Jack Gale as No 4. Harold Alston was in the spare jet, and he recalled;

'At the last minute Maj Arriaga took himself off the schedule and I was inserted as No 3, with Roy as spare. After takeoff "Shass" had a hung landing gear and had to return to Da Nang, so Roy scrambled in the aeroplane that I was originally scheduled to fly. He joined up with us over Chu Lai as No 2. After two successful passes that cleaned off the external ordnance, we made our first strafing pass.

'Coming off the target Roy reported that he had lost his left leading-edge flap. I immediately went into afterburner and quickly joined on his left wing. I reported that when the flap separated it had hit the upper fuselage and torn off some aluminium skin, leaving a gash above the engine. Roy reported that he had lost oil pressure. Now we had a serious problem so we turned towards Chu Lai, about 20 miles away. I stayed on his wing in a close chase position so I could watch the approach better, since it would be a left turn to land on the newly constructed pierced steel planking runway at the austere Marine base that had not long been operational.

'I went to Guard channel and notified Chu Lai tower of our situation. Roy's landing pattern was excellent regarding altitude, descent rate and airspeed. I maintained my position, which was about one wingspan separation off his right side. It looked like a normal approach, but I became concerned that he had not lowered his landing gear. I told him so, thinking he was concentrating on making the landing under emergency conditions. There was no response and we were getting close to the runway. I repeatedly said, "Get the gear down Roy" but there was no response.

'He made a perfect landing, except that the gear was still up. Unfortunately the bomb pylons were what touched the steel planking, causing him to swerve to the left. He went off the runway and ploughed into a small hill of sand. The aeroplane exploded into a fireball. I executed my go-around from about ten feet above the right side of the runway with a sick feeling permeating through my body. I knew immediately that I had lost my flight commander and a good friend with whom I had flown many times. Olson



and Gale were spread out on the opposite side of the runway, so I quickly joined up on them and we returned to Da Nang in silence.'

The AAA damage to Blakeley's F-104C (56-0908) after he had expended 750 rounds of 20 mm cannon fire in his strafing pass had knocked out the jet's hydraulics, preventing landing gear extension. He was awarded the DFC posthumously for attempting to recover the aircraft rather than ejecting and risking casualties on the ground.



479th TFW F-104Cs lined up in Da Nang's sandbag revetments in 1965. Aircraft 56-0908 was Capt Roy Blakeley's aircraft, lost in an emergency landing at Chu Lai on 22 July 1965. Some of the 436th TFS CAS missions hit targets within ten miles of Da Nang, and Capt Blakeley flew one such mission close to the base against Viet Cong (VC) who had 'just finished ambushing a convoy, and we were diverted from our original target. We caught about 200 Viet Cong in the open and really "ate" them up. The Marines really appreciated the quick response to their call for "AIR". The "Zip 4s" were there immediately'. On another CAS sortie he reported, 'We had just got airborne and an Army unit was being ambushed and pretty well cut up [66 men were lost] and they called us and asked for help. We spotted them and their VC attackers and rolled in on them. We killed 166 VC in ten minutes' (USAF)

A Da Nang armament crew replenish a 436th TFS F-104C with 20 mm ammunition (Harold Alston)

MiG-19 COMBAT

Towards the end of its 90-day TDY the squadron responded to increased MiG activity following the combat introduction of the MiG-21, the formation of a second MiG-17 squadron and the adoption of more aggressive tactics by VPAF pilots. More MiGCAP and escort missions were therefore scheduled, and in one pilot's opinion the squadron was 'flying the wings off its airplanes'. Escort duties for the 479th TFW included protection for 6091st RS EC-130E-II *Silver Dawn* 'comint' aircraft flying electronic and communications monitoring sorties just offshore from North Vietnam and China.

One of these missions on 20 September 1965 had terrible consequences for the F-104C pilots when three aircraft from the 436th TFS, which as previously noted had replaced the 476th TFS at Da Nang on 12 July, were lost. Capt Phillip Eldon Smith, the 479th TFW Wing Weapons Officer, took off in F-104C 56-0883 (the first production F-104C) as 'Venus 5' flight's 'ground spare'. He replaced Capt Harvey E Quackenbush, whose Starfighter had aborted ten minutes into the flight to the escort orbit area with a damaged in-flight refuelling probe that prevented him from taking fuel from a KC-135A.

On this, his 80th combat mission, Smith was driven to his aircraft by his squadron commander, who told him to be sure to stay away from Hainan Island. He picked up his tanker at a range of 17 miles on an unusually efficient radar set and refuelled over the Gulf of Tonkin alongside the deputy flight leader, R E Smith. Flight leader Ken Kerwin, with Don Madonna, were meanwhile holding the fort on the patrol orbit. Kerwin then ordered Capt P E Smith to meet them, leaving R E Smith to protect the tanker so that the flight could resume its original refuelling sequence.

As he set out from the tanker, Smith received a series of incorrect and confusing directional instructions from Panama ground control, indicating that either the Panama radar or the 'parrot' IFF unit in Smith's F-104C were not working correctly. Calculating that he was about to approach Hainan at around 540 knots, he turned south and detected another aircraft, high and behind him. Smith reported this to Panama and turned to face the unidentified aircraft. After a short pursuit at 580 knots he had to abandon the chase as he was nearing China again, although solid cloud cover prevented him from seeing sea or terrain.

Panama made repeated attempts to find him on radar, but Smith then discovered that both his heading indicator and standby compass were inoperative. A request to the tanker for a fix on his position produced inconclusive results. Shedding his underwing tanks and becoming increasingly concerned about his fuel state, which had fallen to 3000 lbs, Smith declared an emergency and hoped that Da Nang's 'alert pad' F-102A Delta Daggers with their long-ranging radars might be launched to find

F-104Cs on a December 1965 *Big Eye* escort mission. When F-4C units took over some of the *Big Eye/Silver Dawn* duties from the 479th TFW they employed the same routine of four-aircraft flights, with two F-4Cs on the tanker and two flying above the picket aircraft. The wing also flew pairs of F-104Cs over North Vietnam as weather reconnaissance aircraft. Their speed enabled weather information to be available to a strike formation well ahead of the day's attacks (USAF)



him in time. Desperate to establish some idea of his position Smith was reduced to using a pencil as a rudimentary sundial, but with the mid-day sun directly overhead even that was useless. He spied a coastline through a gap in the cloud base and descended to 9000 ft to investigate but nothing looked familiar. Suddenly, his aircraft's right wing was hit by gunfire, blowing away the AIM-9B and launcher. His right arm was also struck by 37 mm bullet fragments.

Capt 'Smitty' Smith had been intercepted by a pair of Chinese J-6s (MiG-19s), with the lead jet flown by Capt Gao Xiang (commander of the 4th Division's 10th Group) and deputy commander Huang Feng Sheng at the controls of the second aircraft. They had dived at Smith from the clouds above him and fatally damaged his F-104C with gunfire at close quarters. Many warning lights appeared in the cockpit, but Smith also saw a fleeting hope of retaliation as the leading MiG passed ahead of him with its afterburners blazing. He focused on gaining enough speed to get behind the MiG, but his engine stalled. As he re-started it he also heard the growl of the remaining AIM-9B, indicating that it had acquired the target. Smith began a turn behind the MiG to launch the missile but at that moment his control column suddenly failed to respond. Both of his hydraulic systems had been knocked out.

With his aircraft in a steep dive towards the sea several miles off the coast, Smith ejected close to a Chinese fishing fleet. He was immediately captured and spent seven-and-a-half years in a Peking prison, mostly in solitary confinement, as the only USAF pilot known to have been incarcerated in China after the only air-to-air shoot-down of an F-104 during the Vietnam War. Smith's release came in March 1973, and he resumed his USAF career, finally commanding Bergstrom AFB. When he met his attacker in 1989, Smith learned that Gao's J-6 was also damaged when debris from the F-104C knocked out one of its engines.

The loss of Capt Smith's jet highlighted the lack of advanced navigational equipment such as INS, Doppler or UHF/ADF in the F-104C, although the 479th TFW's training compensated for this. Smith's unfortunate communications and navigation problems were the only example of this kind of difficulty experienced by the wing during its wartime deployment.

This shoot-down resulted indirectly in the loss of two more 436th TFS aircraft. Capt 'Q-Bush' Quackenbush and Capt Dayle W Carlson took off to look for 'Smitty' Smith, escorting a pair of Grumman SA-16 amphibious rescue aircraft and supported by a US Marine Corps KC-130F tanker. As darkness approached they made a low-altitude search of much of Hainan's coast but eventually had to turn for home.

Two F-104Cs depart the runway at Da Nang in December 1965. The lead aircraft has a Phantom II 'spook' zap on its forward fuselage. In the foreground is F-104C-5-LO 56-0928, which was shot down by a SAM on 1 August 1966 whilst serving with the 435th TFS. It was escorting an F-105F *Iron Hand* flight near Thai Nguyen when struck by an SA-2. Pilot Capt John Kwortnik was killed (1352nd Photo Group/USAF)





436th TFS F-104C-5-LOs at Da Nang in 1965. 56-0886 *fannie* in the background was among the few jets to boast nicknames during the Starfighter's initial combat tour. Extra markings were usually limited to squadron colours on external fuel tanks. Harold Alston recalled that being the 'ground spare' pilot for an F-104 flight was 'not our most favourite position. We had a low abort rate, so all the spare pilot had to do was to brief with the rest of the flight and suit up with a g-suit, survival vest, firearm, helmet and goody bag with check-list, maps, line-up card, etc. We sat in the cockpit with auxiliary power on, listening to the radio to check that all the jets started up, taxied out, took off and climb away before we were released, or scrambled, by the flight leader. It was always hot and muggy at Da Nang, so if you did not fly we returned to Operations soaking wet from sweat that left a white salt ring on the back of your flying suit' (1352nd Photo Group/USAF)

In the twilight they realised that two of their aircrafts' three sets of external navigation lights and Carlson's interior cockpit illumination were not functioning. The squadron had flown exclusively in daylight until then, and these lighting systems had not been required or prioritised in maintenance routines. Unable to see his instruments for landing, Carlson had to follow Quackenbush for a formation landing. As they began a left turn on approach Carlson lost sight of his leader in thick cloud and requested a burst of afterburner flame to locate the other F-104C. Seconds later the two F-104Cs (56-0911 and 57-0921) collided and both pilots ejected as their aircraft disintegrated, landing close to each other in the sea just off Da Nang.

On 14 October 1965 the 435th TFS briefly took over the Da Nang assignment from the 436th TFS, performing escort flights for EC-121s and EC-130s as its primary mission. The 436th had returned to Kung Kuan AB by 22 November, and from there to George AFB by the end of December. Escort duties then passed to the 390th TFS/366th TFW, whose F-4C Phantom IIs arrived at Da Nang on 29 October.

UDORN – THE SECOND PHASE

An increase in MiG activity over North Vietnam in early 1966 initiated a new 435th TFS wartime deployment to reinforce the USAF's ability to escort its valuable EB-66C ECM assets, its scarce F-105F *Wild Weasel* anti-SAM strikers and the hard-pressed F-105 bomber force. At George AFB the 479th TFW had already begun to transition to F-4D Phantom IIs, with its 476th and 434th TFSs (from July and November 1965, respectively) having relinquished their Starfighters a few months after returning from Da Nang.

The 435th TFS's first eight F-104Cs, arriving at Udorn RTAFB on 6 June 1966, were duly attached to the 8th TFW 'Wolfpack', which had also been based at George AFB since July 1964 until its move to Ubon RTAFB in the summer of 1965. Udorn was the most northerly Thai base, placing it closer to northern Laos and North Vietnam and reducing the tanking requirement for F-104Cs. After initial successes in the escort role, the eight Starfighters were joined by 12 more from the 435th TFS on 22 July. All arrived safely despite the exhaust gas temperature indicator in F-104C 56-0898 periodically dropping to zero on the long over-water flight.

The 'Wolfpack' devised tactics that would make the best use of the F-104C's speed and the F-4C's longer-ranging search radar under RoE that required pilots to acquire visual identification of potentially hostile aircraft, rather than being allowed to use their long-range radar-guided

AIM-7 missiles to knock them out beyond visual range. If 'bogies' were picked up by F-4Cs' radars, often at more than 40 miles, the Starfighter CAP would accelerate forward to make a visual identification and then turn away. The F-4C pilots would then attempt a long-range Sparrow launch and the F-104Cs could then return for a Sidewinder or gun attack from the rear quarter. In practice, no MiGs put themselves in a position to be attacked in this way.

IRON HAND STARS

Also arriving in Thailand in May 1966 were the first five F-105F *Wild Weasel III-1* SAM destroyer versions of the Thunderchief fighter-bomber. Paired with F-105D bombers, they embarked upon some of the most hazardous missions of the war, deliberately flying into the enemy's heaviest defences to draw SA-2 'Guideline' SAMs and AAA into firing at them, thereby revealing the locations of the defences and opening them up to attack. This made *Weasels* primary targets for the VPAF and its scarce but most potent fighters, the MiG-21F-13s of the 921st Fighter Regiment, which had first attacked US fighters on 23 April.

The F-104C's high speed enabled it to keep pace with the F-105s, which were the fastest US strike fighters at low altitude until the arrival of the F-111A in 1968. F-4C Phantom IIs were less able to keep up with the *Wild Weasel* crews, who relied on speed for survival. With F-104Cs ready to intercept incoming MiG-21s, the 'SAM slayers' could concentrate harder on the demanding task of identifying and attacking active SAM sites. Harold Alston flew some of these missions;

'Our job was to maintain visual contact with the *Weasels*, keep MiGs from shooting them down and protect ourselves from enemy fighters. I had visited Korat and Takhli F-105 bases to discuss tactics with the *Weasels*, and I expressed my concern at their having the most dangerous mission in the war. They acknowledged that it was dangerous, but not as dangerous as ours as escorts, being "trolled" like a fish lure behind them, and probably the first to be engaged by the enemy. We had great respect for each other.'

Providing MiGCAPs at relatively low altitudes close to some of the most heavily defended territory in history was clearly made harder by the F-104C's lack of internal electronic countermeasures at that point, and reliance on the over-worked *Wild Weasels* for SAM warnings.

One of the most intense concentrations of SAM and AAA batteries in the North was encountered in the vicinity of the Thai Nguyen industrial and transport complex, which was a frequent and costly target for USAF strike forces. On 1 August 1966 two F-104Cs were destroyed near this target, one of them during Harold Alston's 86th combat mission – a four-ship escort to protect two 355th TFW *Wild Weasels* led by Maj Ed Rock and his electronic warfare officer Maj Curt Hartzell. Alston recalled;



The 435th TFS's F-104Cs received TO-1-1-4 camouflage for their second wartime deployment, which commenced in June 1966 at Udorn RTAFB. Duties involved *Big Eye* and *Silver Dawn* CAP sorties, for which plenty of food, drink and 'piddle packs' were required in the cockpit as missions could last for up to 12 hours if there was a problem with supplying replacement aircraft. These missions were considered so important that other activity over North Vietnam was often cancelled if the CAP was not available (USAF)

'Leading our flight was Maj Charlie Wood, 435th TFS Assistant Operations Officer. No 2 [in 56-0928] was Capt Jack Kwortnik on about his fifth mission. He was a former C-140 JetStar pilot at Clark AFB and, ironically, I had crewed with him on his first flight in an F-104D. Understandably, he was a little apprehensive, and came to my room the night before the mission to talk tactics, specifically about SA-2s. He knew this was my second tour, and that I had come to Udorn with the advanced party, flying up North for two months before Jack and the rest of the squadron arrived in-theatre. I told him everything I had experienced, missiles that I had seen and what worked to evade them. Additionally, I stressed that he should write to his family often, not that fighter pilots expect to be shot down, but the possibility of being hit by a "golden BB" [AAA round] was present on every mission.

'Capt "Howie" Sargent was No 3 in the flight, with me on his wing – we felt this would put critical experience where we could keep an eye on the least experienced pilot. We flew in combat formation at between 4000-8000 ft for a little terrain masking from enemy radar, but mostly to have the *Weasels* in position and ready to fire their anti-radar missiles or strafe SAM sites. The only problem was that this exposed us to almost every gun and SA-2 in the area. Also, there was not a lot of manoeuvring space below us.

'We entered our patrol area a little north of "Thud Ridge", northwest of Hanoi. Almost immediately the *Weasels* received enemy radar activity. We were in an easy right turn eastwards when they got a SAM launch signal. They called it out to us, saying "Take it down!" The first SAM came from our right rear "four o'clock" position. We were in a left-hand fingertip formation – each aircraft about 1500 ft from an element leader, with me as No 4 on the left – about 4000 ft above a horseshoe-shaped valley surrounded by mountains, opening towards Hanoi. The *Weasels* were about 6000 ft ahead of us.

'I had a great view of the "Thuds", our flight and the area to the right and behind – my area of responsibility to clear visually. That is exactly where the SAM came from. I didn't see it launched but picked it up visually coming our way. It appeared to be tracking on Jack on the right of the formation, not on the *Weasels*. I did not feel threatened, but did not know how many more had been fired.

'When the *Weasels* had said "take it down" we pushed the nose down in a negative G "bunt" for the fastest change in altitude to take us nearer the ground and make it more difficult for the enemy radar to track us. Jack rolled inverted in a "split-S", keeping positive G. This alarmed me immediately because, with the turn rate of the F-104, I thought we were too close to the ground for this to be a safe manoeuvre. I could see that my "push" had taken me down to 300-400 ft, while Jack had only changed his aircraft's attitude but had not lost altitude. As a result the SA-2 scored a direct hit on his aircraft's fuselage behind the canopy and electronics bay, about where the fuel tank was located. A fireball began to expand outwards and Jack ejected through it. My impression was that he didn't have time to eject himself, but was blown out of the aircraft. However, he separated from the seat and the parachute opened automatically.

'Charlie and "Howie" headed north "in the weeds", but I wanted to check on Jack, so I made a tight circle around him at 5g or more in afterburner and at near-supersonic speed. I saw that two or three sections of his 'chute

were scorched black. In my turn I saw one more SAM pass pretty close to my right, but it did not detonate and was past me in a flash. Jack was getting close to the ground, but he never waved or deployed his survival kit. When he landed I could not see him in the tall grass or bamboo, so I memorised the spot and “bugged out” as low as I dared to rejoin with “Howie” and Charlie near the Red River.

‘After we got back to Udorn and we were walking into debriefing, we were informed that Lt Col “Al” Finney, our operations officer, had also been shot down by a SAM [in 57-0925 *Smoke II* from “Dagger-0” flight ten miles from the target at 4000 ft] in the same area where we had been. He was leading a two-ship *Iron Hand* escort with Capt Sam Cottrell, who said the missile went right up “Al’s” tailpipe. It probably came from the same site that got Jack.’

‘Al’ Finney was killed and Jack Kwortnik’s remains were returned in 1985. Both pilots were comparatively new to the F-104, joining the squadron at a time when many experienced TAC pilots had completed their tours and were being replaced by aircrew from various backgrounds, including bomber and transport units. On one occasion two pilots fresh from conversion courses after C-130 Hercules tours were sent on F-104C missions over North Vietnam the day after they had arrived in Southeast Asia! These losses to the small F-104C detachment, and a reduced estimation of the MiG threat to *Wild Weasels*, prompted the withdrawal of the 435th TFS from most *Iron Hand* escort sorties. Subsequently, the F-105D/F crews sometimes managed to MiG CAP their own *Weasel* missions, shooting down several MiGs in the process.

MiG sightings were actually very rare. A MiG-21 was seen during a 7 August *Iron Hand* mission, flying at 11,000 ft and 550 knots. The F-104C escort, at 1500 ft and 425 knots, was heading in the opposite direction and stayed with its F-105F charges.

The 435th TFS’s battle with SAMs was not over as, on at least one occasion, they were called in to bomb a SAM site. This was a challenging task for an aircraft that Maj Gen Stan Hood conceded ‘was not an air-to-ground aeroplane – it was a tremendous high-altitude interceptor’. Former F-104 instructor pilot Lt Gen Tom Baker added that it was ‘low on combat loads and on-station time’.

On one attack in October 1966 Maj Rock saw two sets of M117s ‘hit at least a mile from the target’. Without precision bombing gear in such a high-threat environment the F-104Cs were clearly at a disadvantage. Mainly, it was obvious that they required ECM protection if they were to continue operations over North Vietnam, or areas where radar-directed AAA might be encountered. Project *Pronto* equipped F-104Cs with the Applied Technologies AN/APR-25 radar warning receiver and AN/APR-26 launch warning receiver, similar to the ECM devices in F-105s, F-4s and some F-100s.

The *Pronto* equipment was tested at Edwards AFB in F-104C 56-0912 later in 1966. Installation required a blade fairing beneath the radome and two fairings just ahead of the engine tailpipe, housing four spiral antennae in total. Information on the (*text continues on page 67*)



Jim Trice of the 435th TFS with his F-104C-5-LO 56-0910 *PUSSYCAT*. Starfighter pilots often had the opportunity to discredit popular myths about their aircraft’s lack of range compared with the F-4. For *Wild Weasel* escort the F-104C demonstrated a range 1.15 times greater than the F-4C, and it flew 1.05 times further in the medium altitude strike escort role (*Jim Trice*)

COLOUR PLATES

1

F-104A-25-LO 56-0842 of the 83rd FIS/13th Air Task Force, Tao Yuan AB, Taiwan, September 1958



2

F-104C-5-LO 56-0892 of the 476th TFS/479th TFW, George AFB, California, 1958



3

F-104A-15-LO 56-0769 of the 83rd FIS/78th FG, Hamilton AFB, California, 1959



- 4 F-104A-15-LO 56-0782 of the 157th FIS/169th FIG, South Carolina ANG, Congaree AB, South Carolina, 1960



- 5 F-104A-20-LO 56-0795 of the 197th FIS/161st FIG, Arizona ANG, Skyharbour Airport, Arizona, 1961



- 6 F-104A-25-LO 4201 (56-0839) of the 8th TFS/3rd TFW, Ching Chuan Kang AB, Taiwan, 1961



7

F-104C-5-LO 56-0929 of the 434th TFS/479th TFW, George AFB, California, 1961



8

F-104A-25-LO 56-0834 of the 151st FIS/134th FIG, Tennessee ANG, Ramstein AB, West Germany, May 1962



9

F-104A-20-LO 56-0851 of the 319th FIS/32nd AD, Homestead AFB, Florida, 1963



10

F-104C-10-LO 57-0915 of the 434th TFS/479th TFW, George AFB, California, July 1963



11

F-104C-5-LO 56-0886 of the 479th TFW, Da Nang AB, South Vietnam, April 1965



12

F-104C-5-LO 56-0883 of the 476th TFS/479th TFW, Da Nang AB, July 1965



13
F-104A-25-LO 56-0874 of No 9 Sqn 'Griffins', Sargodha AB, Pakistan,
September 1965



14
F-104A-25-LO 56-0877 of No 9 Sqn 'Griffins', Sargodha AB, Pakistan,
September 1965



15
F-104B-5-LO 57-1312 of No 9 Sqn 'Griffins', Sargodha AB, Pakistan,
September 1965



16

F-104C-5-LO 56-0919 of the 435th TFS/479th TFW, Da Nang AB, December 1965



17

F-104C-10-LO 57-0925 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, July 1966



18

F-104C-10-LO 57-0928 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967



19

F-104C-10-LO 57-0923 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, 1967



20

F-104C-5-LO 56-0910 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, 1967



21

F-104C-5-LO 56-0891 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, 1967



22

F-104C-5-LO 56-0938 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, 1967



23

F-104C-5-LO 57-0927 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, 1967



24

F-104C-5-LO 56-0892 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, 1967



25

F-104C-5-LO 56-0902 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, spring 1967



26

F-104C-5-LO 56-0936 of the 435th TFS/479th and 8th TFWs,
Udom RTAFB, Thailand, spring 1967



27

F-104G-LO 4347 (64-17773) of the 427th TFS/3rd TFW, Ching Chuan
Kang AB, Taiwan, 1967



- 28 F-104A-15-LO 910/H (56-0788) of No 9 Sqn Prince Hassan (Safawi) AB, Jordan, 1969



- 29 F-104G-LO 62-12344 of 141 Flot, Murted AB, Turkey, July 1974



- 30 TF-104G-LO 63-8455 of 2. Deutsche Luftwaffen-Ausbildungsstaffel F-104 USA/58th TTW, Luke AFB, Arizona, 1981





435th TFS F-104Cs lined up at Udorn RTAFB in an area which they shared with the RF-101C Voodoos of the 20th TRS/432nd TRW in the summer of 1966. Mobile shelters helped to keep out the sun and rain while aircraft were prepared for flight (432nd TRW/USAF)

direction and range of radar threats was displayed on a small screen mounted above the instrument panel. Warnings were shown by a strobe whose length indicated the strength of the hostile signal and a series of rings measuring the distance from the threat. The latter was very approximate, but pilots quickly learned that a 'three ringer' was a real problem, showing that a missile was heading their way.

The AN/APR-25 could identify signals from C-, X- and S-band radars, and a small panel near the strobe cathode ray tube displayed coloured lights to show whether the threat was

from an SA-2's 'Fan Song' radar control unit, a 'Firecan' radar for guided AAA or the fire control radars in MiG-17PF and MiG-21 fighters. These lights were accompanied by various aural tones in the pilot's headset corresponding to the threat type.

This ECM fit, with associated electronics added ahead of the cockpit instrument panel, was installed in all Udorn-based F-104Cs towards the end of 1966, but it was only a partial solution. In combat, the addition of aural warnings and lights to the already-crowded background of radio traffic, warnings from airborne and naval radar patrol sources and emergency calls on the radio Guard channel merely overloaded the pilots. They tended to ignore or switch off the ECM and RHAW (radar-homing and warning) signals and focus on more relevant radio warnings from other pilots who could see the imminent threats from missiles, MiGs and guns.

The only real chance the pilot had of avoiding being hit by an SA-2 was to see it in time to perform the hazardous last-second dive and turn needed to out-maneuvre the missile. In any case, the RHAW was itself susceptible to many irrelevant transmissions that merely created distracting false alarms. ECM jamming pods like the QRC-160-1 carried by F-105s and F-4s were a much more effective method of defeating SAMs electronically, and one F-104C was test-flown at Korat RTAFB with a QRC-160-1 on the centreline pylon. The pod was designed for use on all four aircraft in a strike flight, where the jamming power could be multiplied sufficiently in a carefully measured 'pod formation' to provide real protection. It was far less effective on a single aircraft and, in any case, the pods were scarce, with F-105 units taking precedence in the supply chain. No extra pods were funded to protect F-104Cs, and the AN/APR-25/26 units were removed when the aircraft returned to the USA.

Escort missions over the North continued into the summer of 1966. On 22 August a 388th TFW F-105 flight approached its target near Hanoi along 'Thud Ridge', dropped its M117 bombs and was met by two MiG-17s as it egressed. Two F-105Ds were damaged by gunfire, and they called in the F-104C MiGCAP flight. Apparently, the Starfighters did not receive the call and their CAP orbit placed them too far north of the target area to see the MiGs, whose controller had clearly vectored his interceptors past the MiGCAP for a clear shot at the F-105s.

From late August 1966 F-104C pilots were given direct air support, CAS and armed reconnaissance tasks for many of their missions. Some of the other duties were temporarily allocated to an F-4 wing instead, as Col George Devorshak noted;



'In the autumn of 1966 the 12th TFW [equipped with F-4Cs at Cam Ranh Bay, in South Vietnam] was tasked with performing the MiGCAP missions over the Gulf of Tonkin that were being flown by the 479th TFW at Udorn. A representative from the 12th went to Udorn to get briefings on the mission, whose primary aim was to counter potential MiG attacks on support aircraft codenamed *Silver Dawn* and *Big Eye* flying off the North Vietnamese coast east of Haiphong Harbour. The CAP track's northeast point was no closer than 15 nautical miles from Hainan Island, with the closest leg of the orbit to North Vietnam being just outside of SA-2 range and abeam Haiphong Harbour. These were Route Pack VI missions.'

F-104C-10-LO 56-0914 was one of the first Starfighters to be camouflaged, the jet subsequently deploying to Udorn with the 435th TFS. Capt Mike Korte, who flew the squadron's 1000th combat mission on 30 September 1966, poses in the cockpit of the aircraft at George AFB, with an SUU-21 practice bomb dispenser on the centreline pylon (via T Panopolis)

GROUND ATTACK MISSIONS

The F-104C's ground attack targets were in South Vietnam or on the Laotian supply trails network, pilots seeking out and bombing truck parks or suspected supply dumps and troop concentrations. Korean War veteran Maj Norman Schmidt, flying F-104C 57-0913 ('Fresno 1') on 1 September 1966, was flying an armed reconnaissance sortie and bombing a truck park near Troc when AAA struck his aircraft as he climbed away. The F-104C survived for only ten miles and Schmidt had to eject over coastal flatlands south of Dong Hoi. He was captured and sent to the notorious 'Desert Inn' section of Hoa Lo prison, where he was apparently beaten to death by a brutal guard in August 1967.

On a more positive note, on 30 September 1966 Capt Harold Alston of the 435th TFS became the first Starfighter pilot to fly 100 missions over North Vietnam. That same day squadronmate Capt Mike Korte (nicknamed 'The Mellow Indian') flew the 1000th F-104 sortie over the North. Capt Alston was presented with a commemorative plaque by Lockheed test pilot Tony Le Vier, the first person to fly the F-104. A State Department recommendation that the unit should receive an award for this achievement was not acted upon by Seventh Air Force commander Gen William Momyer, however, and the statistics were then logged incorrectly in official records as figures for the entire tour.

Two more losses in October underlined the hazardous nature of missions in the *Barrel Roll* campaign over Laos, and also signalled the end of the F-104Cs' participation in them as 1966 drew to a close. Pilots had gained a reputation for considerable accuracy with their limited ordnance loads at a time when the USAF tended to assess fighters by their ordnance-carrying capacity. Although less concentrated than in the Hanoi-Haiphong complex, AAA and mobile SAM launchers were still a serious threat in Laos.

Capt Norman Lockard, on his 83rd combat mission, flying F-104C 56-0904 in 'Fosdick 0' flight with Capt Chuck Tofferi on an armed reconnaissance, dropped BLU-1 napalm canisters on a truck park near

The financial advantages of using a single type (the F-4 Phantom II) for fighter, bomber, reconnaissance and escort duties were increasingly persuasive by late 1966. However, TAC was concerned about the modest success of F-4Cs during their early MiG encounters and alarmed at the low reliability of their missiles, particularly the AIM-7 Sparrow, in combat. Up to September 1966 only one of the F-4Cs' 12 MiG kills had been scored with a Sparrow. Delivery of improved AIM-7Es in November 1966 increased the jet's lethality, and by the time the F-104Cs were withdrawn from Udorn in July 1967 13 more Sparrow kills had been achieved. It was decided to preserve the small number of F-104Cs with the proven accuracy of their guns as a reserve asset in case the MiG threat increased again, pending delivery of the first gun-armed F-4E Phantom IIs to Korat RTAFB in November 1967. The second aircraft (57-0928, with a 'zap' from F-4B squadron VMFA-115 just forward of its engine intake) in this 1967 photo was used by Capt Harold Alston for his 100th F-104 combat sortie. He was the first pilot to achieve this milestone (MSgt Manuel A Collaso/USAF)



Sam Neua on 2 October. In his dive from 10,000 ft both 37 mm and 57 mm AAA rounds hit his F-104C. Tofferi broke left and Lockard swung away to the right as his engine power dropped to 80 per cent – insufficient for continued flight. He tried a quick engine stall clearance procedure but the Starfighter continued to slow, and he was forced to bail out at 4500 ft as it neared stall speed. Lockard landed intact in tall bamboo and he was soon recovered by a HH-3 Jolly Green Giant rescue helicopter.

Capt Tofferi, winner of the September 1962 William Tell Gunnery Meet at Nellis AFB, was also shot down in similar circumstances on 20 October, but it seems that he was wounded as no ejection was seen. A burst of AAA during an armed reconnaissance over the Plain of Jars may have hit his 'Bango Bravo 0' flight F-104C (56-0918 *Hog Wild*) in the rear fuselage as Tofferi pulled up from an attack run on a suspected storage area. Once again, he was flying in the same element as Norman Lockard, with M117 bombs on Tofferi's jet and napalm on Lockard's No 3 aircraft. However, Lockard judged that Tofferi's throttle had stuck, for as he pulled out very low and possibly hit trees.

By the time that Capt Gene West arrived at Udorn in December 1966 *Big Eye* missions were once again the 435th TFS's main task. Although the RHAW installation programme was underway at that time ('rather quickly', as he recalled) *Iron Hand* escorts were reduced;

'During the six months I was at Udorn the MiGs never came out to play. None of our missions included any type of escort [for F-105Fs, RF-4Cs, EB-66s or other tactical types]. *Silver Dawn/Big Eye/College Eye* protection was our assigned mission during my tour, and we continued to fly these sorties until we departed the area in July 1967.'

However, semi-official attacks on targets of opportunity continued, as Gene explained;

'Road and river armed reconnaissance activity was accomplished after the boring *Big Eye* "fragged" [assigned] missions. Generally, we never reported to Brigham control [management centre for air strikes] that we had performed this mission on our own as they would most likely have forwarded this information to the 8th TFW or Seventh Air Force HQ.

'On one occasion, two of our pilots joined up after a test flight for their F-104Cs and they did some reconnaissance over an area known for the movement of vehicles and equipment through the jungle. They found what they were looking for and made a low pass to check it out. This brought

ground fire from the vehicles. The procedure we used to deal with such a situation called for one aircraft to attack the lead vehicle and the other to attack the rear truck, hopefully blocking in the remaining vehicles. Both F-104s expended all their 20 mm ammunition, and smoke and flames could be seen for miles. When they returned to Udorn and reported the incident, word reached HQ in Saigon. Because this activity was not officially 'fragged', both pilots were sent back to the USA and discharged

from active duty! After that, all ammunition had to be removed from the aircraft for test flights.'

In December 1966 the 435th TFS's 18 aircraft achieved extraordinarily high mission rates, flying 506 combat sorties. One aircraft completed 156.4 hours and nine exceeded 100 hours in that month. Availability was

also excellent. Gene West observed that 'the F-104C was put through many situations where it exceeded its g-limits. When you are being shot at you can yank and bank the aircraft more than it's ever been through'.

BOLO AND AFTER

The 8th TFW's 'MiG trap' Operation *Bolo* was successful in destroying almost half of the VPAF's active MiG-21 force when seven were shot down by F-4C Phantom IIs on 2 January 1967. Plans for the operation, in which the Phantom IIs posed as F-105 Thunderchiefs to lure MiG-21s into battle, also included three flights of F-104Cs from the wing's 435th TFS – the 'Wolf Cubs' in Col Robin Olds' 'Wolf Pack'. Some Starfighter pilots looked forward to a belated chance of showing their prowess against MiG-21s in the main MiG interception wave, but the overall fighter force was also required to cover all MiG airfields and possible escape routes to China. The F-104s were therefore given their usual CAP role, although their radar cross section was actually more like that of a Thunderchief, and could have helped with the deception if they had flown in the F-105-type pod formations used by the F-4Cs carrying QRC-160-1 ECM stores to simulate F-105s. The shortage of ECM pods ruled this out, however.

Fighters from the 355th, 388th and 366th TFWs joined 8th TFW units to provide a full strike package, complete with *Iron Hand* and support aircraft. The F-104C component was led by the unit's final commander at Udorn, Col Robert Preciado, whose jet later bore the nickname *Helloo Dolly*. The aircraft were configured with underwing drop tanks and wingtip Sidewinders.

Heavy cloud cover rising to 7000 ft and a slow response by the VPAF reduced the possibilities for air-to-air successes. Four of the MiG-21s that attempted to intercept the force were able to escape into the undercast. Some timing problems, combined with the effects of the weather, reduced the size of the strike force and kept the 366th TFW's F-4Cs from their designated patrol areas over Kep and Cat Bi airfields. The ECM-equipped F-104Cs were



F-104C-10-LO 57-0928 leads a second Starfighter in formation off the right wingtip of a KC-135A over Laos in October 1966. After its transfer to the Puerto Rico ANG in August 1967 and service with that unit until 1975, this Starfighter was destroyed in a terrorist attack at Muñiz ANGB, Puerto Rico, in January 1981 (via *T Panopolis*)

F-104C-5-LO 56-0891 over Thailand, the jet adorned with the nickname *NANCY J* in honour of pilot Herb Drisko's wife. "*SNOOPY SNIPER*" artwork appeared on its right side. Decorating aircraft with personal names did not appeal to all pilots, with Gene West claiming that it was 'a dumb thing to do'. 56-0891 later became the personal aircraft of the 198th TFS deputy commander at Muñiz AFB (via *T Panopolis*)



given a CAP area over the Black River, southwest of Hanoi, where they were on call to support the F-4 flights if needed. Thick cloud covered this area, which was known to be populated with SAM sites and frequented by MiGs.

Arriving on station, the aircraft divided into pairs, orbiting at 8000 ft for around 45 minutes until they were released for the homeward journey after a disappointing wait for some action. Only one MiG had approached them, turning away at four miles range when the CAP aircraft turned to intercept it. The F-104s' secondary role, protecting the 'strike' force's egress route, was also unnecessary as the VPAF did not chase the out-bound Phantom IIs.

Capt Gene West participated in *Bolo*;

'*Bolo* wasn't a big surprise. I often wondered why it took so long! It was scheduled for 1 January but the weather was really bad, so it was re-scheduled for the next day. We were first advised of the mission, which was to provide top cover for the F-4Cs, that morning. Only high-time F-104 pilots were assigned to this operation, and I was one of the 12 to go. Bob Preciado led and I flew as No 3 in Col Ed Gaines' flight. He was at Seventh Air Force HQ and often came down to fly missions with us – a real fighter pilot's pilot!

'We got airborne and headed north. When we crossed into Laos, Preciado had a navigation problem so Gaines took the lead. The weather was not the greatest, with visibility of one to two miles and cloud up to 8000 ft. The red SAM-warning "launch" light on my RHAW equipment stayed on during the entire mission. A number of MiGs "bought the farm" and I would say we had a very good day up North.'

Capt Gene West's 28 January 1967 *Silver Dawn/Big Eye* flight with Lt Travis Harrell proved to be less successful, however. Indeed, West endured a sequence of mishaps that he was lucky to survive. While refuelling he saw a Master Caution 'low oil' indication and the oil pressure rapidly dropped to zero. West's unsuccessful attempts to close the jet nozzle with the emergency 'T' handle in order to gain some power were followed by a brief afterburner re-light and a complete failure of the engine oil system, causing it to rumble, grind and soon seize up. As Gene ruefully commented, 'when the engine quits, there is silence'. He was obliged to eject as the F-104 descended to 8000 ft around 25 miles offshore.

The survival kit attached to West's seat was damaged so that his liferaft was not inflated when he hit the water. Disentangling himself from his parachute meant swallowing a lot of Tonkin water. A ship passed quite close to him, but he could not contact it on his RT-10 radio, and then the radio was lost in a heavy wave while he tried to get hold of a flare – which did not ignite. After 20 minutes' effort West managed to retrieve his second radio, deploy his seat survival kit and inflate the raft by placing both feet on the kit and pulling mightily on the release handle. His second radio malfunctioned, but he was able to indicate his position to an A-1E RESCAP aircraft by keying the microphone. After several close passes a rescue helicopter located him visually, although the cloud ceiling was at 50 ft and visibility was restricted to 200 ft. He returned to Udorn uninjured.

The 435th TFS deployment continued until 19 July 1967 and the majority of the missions flown saw the unit providing *Silver Dawn/Big Eye* cover. To relieve the tedium pilots might indulge in a little air combat manoeuvring practice on the return journey, or look for road or river

targets to strafe. The alternative, as Gene West commented, was just to 'hurry home. Missions of three hours plus were tough on the rump!' However, some sorties over the North still took place in the spring of 1967, with Gene logging a night mission near Haiphong. On 10 March – the day that the Thai Nguyen steel mill was cleared for attack, and also the day of 'Pardo's Push', in which one F-4 literally pushed another powerless F-4 to a safe bail-out area – an F-104C came close to having an aerial collision with 469th TFS strike flight F-105D 60-0423. The pilot of the latter machine, Capt Clint Murphy, recalled;

'I came as close today as I ever will to having a mid-air. We were manoeuvring around a tanker. As I looked back, in front of me an F-104 passed head-on not more than 20 ft below me. With about 1200 mph closure rate I didn't even have time to react.'

Withdrawal of the F-104Cs after 5306 combat sorties, and their replacement by the F-4D Phantom IIs of the 4th TFS 'Fightin' Fuuajins' from Eglin AFB's 33rd TFW, was justified by the general phasing out of USAF Starfighters (and a consequent shortage of parts, particularly M61 spares, and support) and the need to preserve the remaining F-104Cs in case they were required to meet a MiG threat in future. The aircraft's small ordnance load compared with the F-4 was also cited.

Kelly Johnson's verdict on its wartime service was that 'the F-104 went to Vietnam but did not get into [air-to-air] combat. It was not directed to get into combat. It was directed to stay high and provide protection. The people who flew it over there thought very highly of it, but there were many pressures to get a new aeroplane going in the US – the F-15 in particular.'

With the ending of F-104 operations at George AFB in July 1967, the returning 435th TFS aircraft needed a new home. They flew to San Juan, in Puerto Rico, via the Philippines, Guam and Hawaii for service with the 198th TFS 'Flying Buccaneers' – the only ANG unit to fly the F-104C.

For the remainder of the Vietnam War, particularly the *Linebacker* phase in 1971-72, it was planned that many of the 70 surviving F-104C/Ds could be quickly recalled to combat if required, as enough would have been kept at operational readiness by Puerto Rico's 'Bucaneros'. Four 'PRANG' pilots were trained by the 436th TFS at George AFB in April 1967 and F-104C/D Starfighters were sent to Muñoz, in Puerto Rico, in August for continuation training of 23 pilots. Refuelling support for the unit came from the KC-97Ls of the 181st Air Refueling Squadron at NAS Dallas, Texas, and KC-135As from McCoy AFB, Florida. They were often loaded with an extra pair of AIM-9Bs on the centreline fuselage rack, as well as underwing fuel tanks and wingtip missiles. The 198th finally replaced its F-104Cs with A-7D Corsair IIs in 1975.

After its Udon tour F-104C-5-LO 56-0892 *My Darlin' Dorothy* joined the 198th TFS of the Puerto Rico ANG in August 1967 (via *T Panopolis*)



ASIA GOES DOUBLE-SONIC

The Pakistan Air Force (PAF) began F-104 operations on receipt of the first of 12 F-104As in August 1961 for No 9 Air Superiority Squadron at Sargodha, Pakistan's main fighter base. These jets had been refurbished prior to delivery under US Military Defence Mutual Assistance Programme (MAP) funding, having been fitted with C-2 upward-firing ejection seats, arresting hooks and more fuel-efficient J79-GE-11A engines. This uprated powerplant gave these F-104As the best thrust-to-weight ratio of any Starfighter variant (until the USAF's 318th FIS's F-104As subsequently received the even more powerful J79-GE-19), the jets having dazzling speed and climb performance. M61 guns, removed during the aircrafts' USAF usage, were also re-installed. Removing the gun had enabled an optional 100-gallon fuel tank to be installed. AIM-9B missiles could be carried for the air superiority role on 'Red Dog' wingtip launchers. Two F-104Bs (57-1309 and 57-1312) were also supplied.

In 1948, after the partition of India and Pakistan and the initial clash over Kashmir between the two countries, the Indian Air Force (IAF) became the first Asian air arm with jet fighters (de Havilland Vampires), but the PAF's acquisition of F-104As made it the first with Mach 2 aircraft. Pilots for the elite No 9 Sqn 'Griffins' came with considerable flying time on the F-86F Sabre. Groundcrews, with long experience on Sabres, achieved 80 per cent in-commission rates during the F-104's first five years of service with US assistance.

The first three PAF pilots chosen to fly the F-104 received conversion training in the USA, and one, Sqn Ldr M Sadruddin (the designated No 9 Sqn commander) was attached to the 434th TFS at George AFB and became the first Pakistani to fly at Mach 2. Flt Lts Alauddin 'Butch' Ahmed and Mervyn Middlecoat were trained with the 157th FIS, South Carolina ANG, at McEntire ANG base. This unit was also tasked with training F-104 pilots from the RoCAF and Japan. Stan Hood had the son of the famous Adm Isoroku Yamamoto among his students, and rated him an excellent pilot, although he could never beat Capt Hood at ping-pong, Japan's national game.

USAF test pilot Maj Swart Nelson checked out the three US-trained PAF officers before they flew their F-104As home to Sargodha air station, where a team of Lockheed 'tech reps' provided maintenance advice while two ANG pilots oversaw conversion training.

The 'Griffins' safety record was also excellent compared with some other Starfighter users. One F-104A (56-0802) was lost on 13 November 1963 when Flg Off Asghar Shah entered a spin during air-to-air training and had to eject at high speed. A second F-104A (56-0803) crashed ten

months later when Flt Lt Tariq Masood misjudged his pull-up from a practice ground-strafting run and the tail of his aircraft impacted the ground with fatal consequences. Both jets were replaced under MAP, by 56-0773 and 56-0798.



Gunnery comprised a major part

of the training, including air-to-air practice against dart and banner targets. Dissimilar air combat training (DACT) against F-86Fs was also included in a programme, which emphasised low and high altitude interception.

When hostilities with India began in August 1965 the PAF had 150 aircraft. Of these, 102 were well-tries F-86Fs, 25 of which were AIM-9B capable. The IAF could field about 460 frontline aircraft including ten MiG-21F-13s, 118 Hawker Hunter Mk 56s (from the 160 ordered in 1957) and 80 Folland Gnat 'midget' fighters, mostly licence-built by Hindustan Aircraft Ltd and extremely manoeuvrable, as well as hard to see due to their mere 22 ft length and 28.5 ft wingspan. Also, 80 Dassault Mystère IVAs, most effective as ground-attackers, were available.

While the Gnat had the best power-to-weight ratio of any IAF aircraft, and could take on most of the other fighters, the MiG-21F-13s were the only direct supersonic opposition to the PAF Starfighters. However, the Hunters and Mystères would prove to be relatively dangerous adversaries at lower speeds in manoeuvring combat situations unless a Starfighter pilot was able to make a high-speed surprise attack, taking advantage of his fighter's low head-on visibility and the lack of a smoke trail when in afterburner.

India's MiG-21s had been ordered in 1962 specifically to counter the PAF's F-104As, and they became the only option after France opposed Indian licence-production of Mirage IIIs and the USA refused to supply Starfighters to the IAF without political constraints on their use. The USSR agreed to deliver ten MiG-21F-13s in 1963-64 (and trained a core of seven pilots), followed by two MiG-21PFs in March 1965, all for No 28 Sqn. It gave the IAF a fighter with similar performance and ground-attack capability to the F-104A, although the squadron was still completing training and had just eight serviceable aircraft at the outbreak of war, so its role in that conflict was only small. In training, MiG-21 pilots of No 28 Sqn concentrated on the high-altitude interception for which the aircraft was conceived, rather than the low and medium altitude dogfighting that they would later encounter.

PAF Starfighters initially saw action during the fighting over the Rann of Kutch, a mineral-rich area of India bordering Pakistan which Pakistani forces entered in March 1965. Hostilities began in April, and two F-104As were detached to Mauripur air base, from where they flew CAPs near the Indian border. An IAF Canberra PR 57 on a reconnaissance mission was intercepted and followed for ten minutes, but the Starfighter pilot was told not to shoot at it.

By September the focus had shifted to the disputed Kashmir area, and the 'Griffins' began to fly CAPs at around 30,000 ft, many of them at night against Canberra interdiction. The F-104's ASG-14 short range,

F-104A 56-0803, delivered to No 9 Sqn on June 1961, was lost on 3 September 1964 when it struck the ground while pulling out of a low-altitude strafing pass. The pilot, Flt Lt Tariq Masood, was killed. Part of the impetus for supplying F-104s was the USA's interest in establishing the South East Asian Treaty Organization and Central Treaty Organization as strong NATO-type defence alliances in the area. This political initiative helped Pakistan in negotiating for supersonic F-104s rather than the essentially subsonic F-100 Super Sabres that the USA had preferred to supply. India had also requested F-104As in 1961 to counter the Chinese threat, but the US government withheld approval (PAF)



F-104A 56-0798 was delivered to the PAF in 1964 after use by the RoCAF, and it was later preserved at the PAF Museum at Faisal AB, in Karachi (John Fricker via Simon Watson)

spin-scan fire-control radar was optimised for high-altitude daytime use, and pilots were heavily reliant on ground-control information to make interceptions. Pakistan had incomplete early warning radar coverage, as there were only two main radar installations, one at Badin in the southeast of western Pakistan and a second atop Sakesar Peak – from here, a General Electric FPS-6 gave operators a good view across many IAF airfields. There was also the approach radar at Karachi airport and an AR-1 radar at Pir Patho. As Air Vice Marshal Farooq Umar recalled;

‘In 1965 there were gaps within the radar systems, and we had to rely on our Mobile Observer Units deployed all along the border, five miles apart, as the outer belt. They would report somebody breaching the border. Then the inner belt [observers], closer to the bases, would confirm speed and direction of the intruder. Next, the terminal low-level radars would pick up intruders at a range of 25 miles to vector the fighters to a successful interception.’

This complex system caused inevitable delays in setting up interceptions. Canberra pilots could evade the CAPs by flying below 1000 ft, where they were lost in ground clutter to the F-104As’ radars. PAF pilots had to fly even lower, hoping to acquire the Canberras visually against a sky background, but knowing that their radars could set off the tail warning radar, code-named *Orange Putter*, fitted to Canberras to alert their crews to imminent interceptions.

The F-104 was greatly respected by IAF crews, who had nicknamed it ‘badmash’ (‘hooligan’) before the war. On 3 September 1965 eight No 23 Sqn Gnats were detached to Pathankot to escort four Mystères that were supposed to be medium altitude ‘bait’ for PAF Sabre pilots. The Gnats, led by RAF-trained Sqn Ldrs William ‘Johnny’ Greene and Trevor Keelor, trailed the Mystères at an altitude of 100 ft, ready to spring an ambush. A CAP of two F-86Fs was sent to intercept them. Keelor damaged Sqn Ldr Yousuf Khan’s F-86F (claimed as the first IAF aerial victory of the war, but denied in PAF records), and Flt Lt V S Pathania was about to fire on a second when two Starfighters, scrambled to rescue the F-86Fs, dived into the Gnat formation at 500 knots and had to overshoot, scattering the IAF fighters but presenting a fleeting target for Flt Lt Krishnaswamy’s 30 mm guns.

The Gnat pilot hesitated, admitting he was overawed by the sight of the Starfighter, and the IAF pilots then headed for base as their fuel was low. One Gnat (IE 1083), flown by a Sikh senior flight commander, Sqn Ldr Brij Pal Sikand, had become separated over unfamiliar territory and

headed for an airfield that came into view as both Starfighters entered the area. Flg Off Abbas Mirza, piloting F-104A 56-0877, reflected that 'had he known that neither Flt Lt Hakimulah [in the second F-104A] nor I had visual contact with him until he was about to land he may have got away safely, but I guess personal safety got the better of him. The Gnat is an extremely small aircraft and difficult to spot from a distance even in good visibility'.

Visibility that day was poor. According to Sikand the Gnat's radio, compass and hydraulics had apparently failed (requiring manual

undercarriage extension) and it entered the barrier at the end of the disused Pasrur airfield, where Sikand was captured, apparently unaware that he had landed in Pakistan, but very conscious of the presence of F-104As near him at the time. PAF inspectors found no faults in the Gnat.

Starfighters made several high-speed, visual reconnaissance runs over IAF bases in the following 48 hours to establish the disposition of Indian aircraft ahead of planned air base attacks by Sabres. Several IAF Hunter and Mystère pilots had their first glimpse of the Starfighter, but they also observed that pursuit of the aircraft with their subsonic jets was futile. Later, F-104As provided top-cover for the PAF's few No 20 Sqn RT-33A photo-reconnaissance flights, safeguarding them from interception at their typical speeds of less than 400 knots, but having to cope with the considerable speed differential by constant manoeuvring.

The two PAF F-104Bs also flew reconnaissance missions, roaring along at very low altitude close to 600 knots, often in advance of bombing missions by B-57Bs. Photographs were taken with a hand-held camera by the back-seat crew member, sometimes with the aircraft inverted. Although the results were not equal to those obtained with the type of reconnaissance pod used with the F-104G, the Starfighter's inherent stability at low altitude produced some useful images nevertheless. Farooq Umar was one of the pilots who flew these hazardous missions;

'Hand-held cameras were resorted to, and Flt Lts Middlecoat, Aftab Alam and I did the photography in the two-seat jet. A camera with image motion compensation was also tried out, and I was one of the pilots who tested it. It worked very well, with clear resolution at high speed and at altitudes as low as 150 ft above the ground.'

One of the F-104Bs was later modified with three TA7M cameras (one vertical and two oblique) in the rear cockpit, although this installation was not used operationally as three Mirage IIIRPs had been acquired by then. These missions led to proposals for low-altitude strafing attacks on IAF airfields up to 350 miles from Sargodha by F-104As equipped with four external fuel tanks to provide adequate range. However, most of the IAF aircraft destroyed by the PAF were bombed during night attacks on their bases by B-57B Canberras of No 31 Wing.



Air Marshal Nur Khan, Commander-in-Chief of the PAF, prepares for his first flight in an F-104A in 1966. On the crew entry ladder to assist him is his instructor, Flt Lt Farooq Umar (via AVM Farooq Umar)

Generally, 'Griffin' pilots noticed that the IAF would allow Starfighters to overfly its bases without attempting interception. There were many reports circulating at Sargodha of Hunter and Mystère pilots avoiding engagement with the No 9 Sqn fighters and denying them the chance of aerial combat. F-104As had their best fighting opportunities at lower altitudes. Higher up, the larger wing area of IAF Hunters and Gnats gave them superior manoeuvrability.

When the conflict spread to a full-scale war on 6 September the PAF was accused of bombing the Sikh spiritual centre at Amritsar, but the damage was later attributed to a reconnaissance F-104's sonic boom. The 'Griffins' reverted to air defence and a dawn CAP that day, and Flt Lts Aftab Alam and Amjad Hussain, patrolling at 15,000 ft, detected four Mystère IVAs from No 1 'Tiger' Sqn, led by Wg Cdr 'Omi' Taneja, attacking railway fuel tankers near Ghakkhar. Mystère pilot Sqn Ldr 'Paddy' Earle saw the F-104As diving on them and his formation headed for the border on the deck at 450 knots, with Aftab Alam (who had jettisoned his wing tanks) closing rapidly behind them, guided by the Sakesar radar air defence controller. His wingman remained at altitude as radio relay. Aftab Alam later reported;

'Manoeuvring started at treetop level. I kept my eyes glued on the target. I could feel the strain under high-gs, looking over the tail of the aircraft as I skimmed the trees at high speed. One mistake and I would have hit the ground. The F-104A, with afterburner blazing, was responding very well. I used the flaps to improve the turning capability as required. The "stick shaker" was a big help in flying the aircraft to its limits. I armed my weapons and decided to shoot the first two [Mystères] with missiles and the next two with guns. I fully realised that a confidential order prohibited me from using the missiles below 10,000 ft. However, I was confident that they could be used at any height provided the target could be discriminated from background heat sources. A distinct increase in missile tone ensured this.'

Flt Lt Aftab Alam set up a Sidewinder, firing at the nearest Mystère and then zoom-climbing, dazzled by the flash of the missile launch so that he did not see an impact. Although Pakistani forces detected a fire and explosion on the ground near Rahwali airfield, and radio transmissions from the Mystères reported a shoot-down, the IAF stated that the fire resulted from burning fuel in drop tanks jettisoned from Earle's Mystère. However, the PAF claimed this as its first kill by an F-104A, as well as Pakistan's first missile kill.

Attacks on PAF air bases began on 7 September when Flt Lt Amjad Hussain's F-104A was in a CAP over Sargodha, which turned to meet six 'Tigers' Mystères that were about to attack the 'Griffins' home base. He pursued the formation, caught them about eight miles from Sargodha and fired an AIM-9B at an altitude of about 100 ft. The missile dived into the ground and he closed to within gun range, seeing 20 mm hits on a Mystère. A second Mystère turned towards Amjad Hussain, forcing him to climb to around 15,000 ft in a high-speed yo-yo manoeuvre so that he could re-attack. Closing once again to gun range (3000 ft), he fired his Vulcan cannon at the IAF fighter, which exploded about 1000 ft ahead of him.

The IAF acknowledged the loss of its pilot, Sqn Ldr Ajamada B Deveyya, but also claimed that he had forced Amjad Hussain into a low-altitude turning fight in which the French fighter's better turn radius

allowed Deveyya to fire his DEFA cannon, fatally damaging the F-104A (56-0877). From Hussain's viewpoint, and that of other witnesses, the explosion of the Mystère after his gun attack left debris immediately ahead of him, which hit the F-104A, forcing him to eject at an altitude of 100 ft as the fighter began to roll to the left. His parachute opened just above the ground and he returned to base by horse, jeep and finally a helicopter.

It is thought that Amjad Hussain was making a high-speed yo-yo manoeuvre at the time in order to turn with Deveyya, and the closure rate at the bottom of the yo-yo would have made it impossible for him to avoid fragments of disintegrating Mystère. In the opinion of Air Vice Marshal Farooq Umar, a member of the same squadron at the time, it is quite possible that Amjad's controls were damaged;

'Even if the Mystère was flying at nearly 500 knots Flt Lt Amjad might have closed in with a very high rate of overtake of 150 to 200 knots. Firing his burst of bullets might have ended when he was too close to the Mystère, at say 1000 ft, and thus he may well have picked up debris in his engine. Also, if an F-104A overshot a Mystère at such a high overtake speed, even if the IAF pilot had opened up to full power he could not have caught up with the F-104A for a guns kill.'

The 'Griffins' contact with IAF aircraft was sporadic during the following week. Flt Lt Hakimullah, patrolling over Indian terrain on 11 September but low on fuel, was warned that two sections of IAF fighters, including two Gnats, had sighted him. He dived towards them and lined up for a Sidewinder shot, but he was outside AIM-9B launch parameters. Re-positioning his aircraft behind one of the Gnats, he received a positive launch tone for the missile, but he also heard a further warning of two more IAF interceptors.

Hakimullah identified the new threats as MiG-21s and turned towards them, but he noticed that his fuel was worryingly low and so briefly engaged afterburner to evade them. Another pair of MiG-21F-13s was then seen approaching him head-on, and he used the F-104A's formidable acceleration to dive beneath them at supersonic speed, out of Indian airspace. Indian records suggest that he may have mistaken Hunters for MiG-21s, and it would certainly have been unusual for four of the modest IAF MiG-21 force to have been patrolling one small area simultaneously. Hunters were very active that day and two other F-104 pilots passed close to No 7 Sqn jets attacking the Lahore-Kasur road bridge in their CAP area. Hakimullah landed at a forward airfield at Risalawala and his fuel-starved engine flamed out whilst the jet was rolling down the runway.

PAF F-104As were the only aircraft capable of intercepting IAF Canberras at night. Although the twin-engined bombers were large and limited to speeds of around 380 knots, they could operate below Pakistan's radar coverage and enter its airspace along the 1000-mile border separating West Pakistan and India. Intruders could only be met by standing two-aircraft CAPs from Sargodha or Peshawar, and engagements were often



A four-ship formation of No 9 Sqn F-104As over Lahore in 1965
(via Usman Shabbir)



Flt Lt Amjad Hussein Khan shot down an IAF Mystère IVA on 7 September 1965 and then had to eject after his F-104A was hit by debris from the Indian fighter. This was the only Starfighter loss to enemy action conceded by the PAF in 1965, and the IAF pilot, Sqn Ldr Deveyya, was subsequently awarded the IAF's highest honour, the Mahar Vir Chakra, posthumously in 1988. If Deveyya's gunfire was in fact the true cause of the loss, it was also the only aerial victory for the Mystère IVA. Starfighter pilot Flt Lt Arif Iqbal also had the chance to claim a Mystère IVA kill in that battle, but an F-86 pilot jumped in behind the enemy jet and shot it down before Iqbal could pull his trigger (PAF)

fortuitous. At the same time the PAF was using its two squadrons of similar B-57B Canberra light bombers for night attacks on Indian bases.

The F-104A was the only PAF type with a radar/fire-control system capable of assisting with night interceptions, while the IAF would have had radar interception capability with its two MiG-21PFs using their RP-21 'Spin Scan' radar had they not both been destroyed by strafing F-86 Sabres at Pathankot on 6 September 1965. The MiG-21F-13 pilots had no experience of night interceptions, and they were unable to stop any of the B-57Bs or pursue them to their distant base at Peshawar, 600 miles from the Indian border. The IAF Canberra units did have the required range, however, and on 13 September six B(I) 58s from No 5 Sqn attacked the Peshawar base, flying to the limits of their range with light bomb-loads and braving the F-104As that they knew were likely to engage them.

Their standard tactic was to cruise at around 25,000 ft, descending to 1000 ft to cross the Pakistani border and climbing to bomb from around 8000 ft, before escaping at low altitude. Pairs of F-104As were detached to other bases to provide night coverage for northern Pakistan, while F-86F pilots joined the night CAPs hoping to detect Canberras visually or even by using the seeker heads of their AIM-9B missiles – a last-ditch option offered in the pilot's 'Dash-1' manual. They would then have attempted interception from below the Canberras.

During the 13 September raid the No 5 Sqn force deposited its ordnance on Peshawar, hitting the runway and fuel storage area after first illuminating the target with flares. Bombing from as low as 200 ft, the Canberra crews came close to destroying an entire flightline of B-57Bs. As Sqn Ldr J C Verma led the force off target, a No 9 Sqn F-104A flown by Sqn Ldr Mervyn L Middlecoat closed in behind Sqn Ldr Vivian C Goodwin's Canberra and fired an AIM-9B. The darkness was split by a brilliant explosion, which Middlecoat and several other Canberra crews assumed marked the demise of Goodwin's aircraft. Although PAF authorities awarded Middlecoat a kill, all the Canberra crews reportedly returned intact to their base. The Sidewinder had apparently exploded prematurely, and confirmation of the shoot-down was impossible because the incident occurred over Indian terrain.

This attack convinced the PAF that none of its bases were safe, and it also motivated the USAF to evacuate personnel from an electronic surveillance 'spy' base near Peshawar.

Another Canberra interception was achieved by Flt Lt Amjad Hussain near Lahore that same night, but the armament switches in his F-104A short-circuited as he prepared to fire, leaving him with gun-camera film as the only evidence of the encounter.

An undisputed Canberra kill occurred on the night of 20/21 September, however, which proved to be the only loss of this type conceded by the IAF. Four aircraft from No 5 Sqn, led by Sqn Ldr Vivian Goodwin, had been sent to bomb Sargodha airfield. The No 2 aircraft was crewed by Flt Lt Manmohan M Lowe (pilot) and Flg Off Kewal Krishnan Kapur (navigator/bombardier). As they turned away from a successful bomb-run the Canberra was hit by AAA and Lowe descended to 1000 ft, noticing heavy fuel loss from his main tank. As he started to climb back up to 13,000 ft to prolong his fuel consumption, he did not realise that his *Orange Putter* tail-warning radar, which was relatively ineffective at low



altitude and had been switched off, had not re-engaged. Canberra pilots thought that the *Orange Putter* emissions could be detected by an F-104A's AN/ASG-14, but these fears were unfounded. Lowe was therefore unaware of Wg Cdr Jamal Ahmed Khan's F-104A (56-0874) closing behind him.

As Lowe continued to climb and Kapur in the Canberra's glazed nose area sought visual confirmation of their location, Khan fired an AIM-9B. The missile exploded in the bomber's starboard Avon engine, Khan's previously unseen target rolling to the right with flames streaming from its wing. As the bomber began a long death-dive, Lowe, who first reassured himself that 'Raj' Kapur had already bailed out through the crew hatch, ejected and was taken prisoner. Sadly, Kapur perished.

The *Orange Putter* could be a lifesaver, as Wg Cdr Peter Wilson discovered during a 16 September night attack by five Canberras on Sargodha. As the IAF formation departed for home Wilson's aircraft was intercepted by a Starfighter, but his *Orange Putter* gave him adequate warning and he was able to lose the F-104A in a spiralling dive in which the fighter pilot was unable to keep the target within his AN/ASG-14 TI radar's 'vision'.

During the third and final week of the war the Sargodha Starfighters maintained their very effective defensive posture. Rather like the USAF's F-104Cs in Vietnam, their presence as a deterrent to enemy interdiction or interference with air strikes was more important than the actual combat engagements that they participated in. A typical example occurred on 16 September when four Mystères attacking Pakistani gun emplacements near Pasrur with T10 rockets curtailed their strike when warned of an approaching F-104A.

The 'Griffins' flew 254 sorties in all, of which 246 saw them flying day and night air defence CAPs and the rest were escort missions of various types. Most engagements were fought between the dominant F-86F force and Indian Hunters, Gnats and Mystères. The fledgling IAF MiG-21 squadron undertook only 80 sorties, its pilots being limited by their lack of training and experience to flying morale-boosting 'fighter screen' sorties that were meant to deter the Starfighters, which were seen as the greatest threat to IAF interceptors. No direct contact between MiG-21s and F-104As took place, although several mutual sightings were reported. Three of the IAF's ten MiG-21s were lost to other causes during the 1965 conflict.

Pakistan's two wartime F-104A losses were not replaced by the USA due to an arms embargo that was applied to both sides once hostilities began, although Indian forces were relatively unaffected as they used little US equipment compared with Pakistan. The combat loss of 56-0877 was followed on 17 September by a landing accident to F-104A 56-0868, Flt Lt Ghulam Omar Abbasi being caught in a sandstorm as

F-104A 56-0875 (right) with a PAF Shenyang J-6 (MiG-19) and pilots – flying both fighter types – of No 9 Composite Squadron in 1966. In the back row, from left to right, are, Flt Lt Amanullah, Flg Offs Salim Sandal and Arif Manzoor, Flt Lt Rashid Bhatti, Flg Off Mushtaq and Flt Lt Abbas Mirza. Seated, from left to right, are Flt Lts Ghulam Omar Abbasi and Farooq Umar, Wg Cdr M L Middlecoat, (squadron commander), Sqn Ldr Hakimullah Hussein and Flt Lts Ishaq and Aftab Alam. Pilots much preferred the F-104A's logical cockpit layout to the rather confusing flying controls and instrument arrangements in the MiG-19. This F-104A was later preserved at the PAF HQ in Chaklala (via AVM Farooq Umar)



Sargodha Starfighters are parked alongside B-57Bs of No 31 Wing. Both the PAF and IAF operated versions of the Canberra bomber during the conflicts between the two nations, complicating the process of identification and interception for F-104 pilots (PAF)

he attempted to land at Peshawar in near-darkness. Sargodha's inadequate runway lights and the lack of an instrument landing system compounded the problems and he undershot the airstrip. The Starfighter disintegrated and Abbasi was fortunate to be thrown clear, still in his seat and suffering only minor injuries. He was not so lucky on 15 April 1968 when he crashed 56-0807 during low-level aerobatics and was killed. Three months later a ground fire in the starter unit of 56-0805 became uncontrollable and burned out the aircraft, although the pilot, Sqn Ldr Arif Iqbal, escaped unscathed.

With only half of the original force of 14 F-104s airworthy by 1970, No 9 Sqn pilots had to supplement their flying time with sorties in F-86Fs and Shenyang J-6s as spares levels for the remaining Starfighters ran down and they were progressively placed in storage. However, assistance eventually reached the PAF via another F-104A/B user, the Royal Jordanian Air Force (RJAF).

Jordan requested 32 ex-RoCAF F-104As and four F-104Bs in 1966. The two-seat jets and a pair of F-104As, along with a USAF training detachment, reached Prince Hassan Air Base (H-5) in June 1967 just as the Arab-Israeli Six-Day War broke out. America then interrupted deliveries and withdrew the H-5 aircraft to Turkey, while 18 RJAF pilots who were training on the type in the USA with the USAF's 4760th Combat Crew Training Wing (which had formed with the personnel and aircraft of the 331st FIS and led by Howard Kidwell) returned to Jordan in October to fly Hunters instead.

By then the majority of Jordan's other combat aircraft had been destroyed at their bases by Israeli air strikes. F-104 deliveries resumed in 1969, and conversion training recommenced in the USA. Only 22 F-104As were eventually supplied to No 9 Sqn at H-5, including five that had been returned to the USA following service with the RoCAF.

In 1968, in anticipation of their arrival, the RJAF asked Pakistan for assistance with interim training, and No 9 Sqn was operational by September 1969. This relationship proved mutually beneficial when Jordan loaned some of its Starfighters to the PAF during the second Indo-Pakistani conflict, which began with the civil war between West and East Pakistan following the latter's declaration of independence as Bangladesh in March 1971.

Following the 1965 conflict with India over Kashmir, the F-104As' underwing fuel tank pylons were modified to carry LAU-3B AIM-9B launchers so that the wingtip stations could be used for fuel tanks, which caused less drag than the underwing type. Air Vice Marshal Farooq Umar recalled;

'I joined the F-104 squadron in 1962 as part of the second batch of PAF pilots. Following the 1965 war the underwing pylons were adapted to carry Sidewinders for operational benefits. This was done primarily by Sqn Ldr Hakim Ullah, a combat pilot who later led the F-104 unit. Although not an engineer, Ullah, who was a remarkable innovator, did all the planning and drawings for this modification, and it performed well.'

Umar was also impressed by the F-104A's radar, which in his opinion was perfectly adequate for the jet's air superiority role;

'It had a range of 25 miles [the same as the terminal radars protecting PAF bases], but being a helical scan type it would pick up a lot of clutter at lower altitudes. In spite of the best tuning it was difficult to identify targets from the ground clutter, however.'

Attempts were also made by both local technicians and a PAF F-104 pilot to improve the half-mile operational range of the Starfighter's inbuilt infrared detection set. Using liquid nitrogen to cool its indium antimonide infrared cell, the set's detection and lock-on range was apparently increased to ten kilometres, but icing problems forced the abandonment of this project. Another initiative involved the testing of a short-range, low-altitude radar detection (SLARD) set to improve target acquisition nearer the ground. It used two nose-mounted sensors to give detection capability at a range of up to ten miles and 30 degrees on either side of the jet. The equipment occupied too much cockpit space, however, and further development was curtailed.

THE THIRTEEN-DAY WAR

Lacking intervention from the international community, India (which had signed a defence pact with the Soviet Union in August 1971) decided in December 1971 to intervene in the conflict over Bangladesh and began a massive build-up of forces along the East Pakistan border. Air action had actually begun on 16 October when IAF Canberras attacked targets in West Pakistan. Around 40 Canberras remained in Indian service, and they had been joined by around 175 MiG-21FL 'Trishuls' (Tridents) equipping nine squadrons. There were also 150 Sukhoi Su-7BMs, 160 Gnats, 100 Hunters, domestically produced HF-24 Maruts and 40 Mystères, allowing the IAF to field around 730 combat aircraft. Of these, the Hunters and Su-7BMs would sustain the majority of the casualties during the renewed conflict.

Pakistan, deprived of Western re-armament by embargoes, had acquired 70 Sidewinder-armed Chinese Shenyang F-6s (MiG-19s) and 90 ex-Luftwaffe Canadair Sabre 6s via Iran. The Chinese deal was allegedly connected to the loan (and eventual return) of a disassembled PAF F-104A to China for technical examination. France also supplied 18 Mirage IIIEPs to replace F-104As, and in September 1970 the USA offered a 'one-time exception' arms deal that included seven more F-104As, but at that stage the PAF was already looking to phase out its Starfighters. Indeed, the pilots of No 9 Sqn had been preparing for new aircraft, as Farooq Umar explained;

‘The first step was to make two flights out of No 9 Sqn, one flying the Chinese F-6s and the other flying F-104As, and in some cases alternating so that each pilot could fly both types of aircraft. Several years earlier, in 1967, six pilots out of the entire air force had been selected to become instructors in the first Mirage squadron, which was to be commanded by M M Alam, the ace and legend of the PAF. I was one of those hand-picked pilots.’

However, the partial US relaxation of spares supplies in 1971 enabled No 9 Sqn to resurrect its surviving F-104As, and their contribution during the fighting in 1971 temporarily suspended their retirement. Pilots who had been involved in training Jordanian F-104 crews returned to Sargodha, having regained currency on the Starfighter. They brought with them some useful experience of DACT with RJAF Hunters. They also trained hard in low-altitude nighttime interceptions, using an improved ground radar network. Eleven pilots were serving with No 9 Sqn by 3 December, and three days later the unit's eight F-104A/Bs were moved to a new base at Masroor, with veteran pilot Wg Cdr Arif Iqbal in command.

Despite possessing only 210 combat aircraft, the PAF attempted a massive pre-emptive strike known as Operation *Chengiz Khan* in the hope of achieving results identical to those of the Israeli Defence Force Air Force during Operation *Focus* at the start of the Six-Day War in June 1967. On 3 December 1971 Sabres and B-57s struck eight IAF airfields, but the operation was spread too thinly with too few aircraft to inflict much damage. Four PAF jets were lost, adding to the two Sabres that had been shot down by Gnats the previous month.

The ‘Griffins’ role was to attack radar installations at Amritsar and Faridkot, shooting out their antennae with 20 mm gunfire. Wg Cdr Arif Iqbal and Sqn Ldr Amanullah hit the P-35 ‘Bar Lock’ site at Faridkot at 1735 hrs local time on 3 December in a low altitude strafing attack in which Amanullah fired 400 rounds in two passes, destroying the radar vehicles and a HAL HAOP-27 Krishak light observation aircraft parked in a shelter. Arif kept watch at higher altitude. On his return to Sargodha, Amanullah was fired on by the base's AAA batteries, forcing him to repeat his approach.

The Amritsar radar had been attacked at the same time as the site at Faridkot by Sqn Ldr Amjad Khan (testing the SLARD in F-104A 56-0804) and Sqn Ldr Rashid Bhatti. The Amritsar antenna was damaged but quickly repaired, and with the site active again by the next day, a second attack was undertaken by Sqn Ldrs Bhatti (in 56-0804) and Amanullah. The Indian defenders clearly expected a re-attack as the two Starfighters were met by heavy AAA and a Gnat flown by Wg Cdr Johnny Green, who closed in behind 56-0804 and opened fire. Warned by Amanullah, Bhatti accelerated away and his wingman closed in behind the Gnat and fired an AIM-9B that exploded close to the Gnat's underside. Although the IAF fighter was damaged no crash

F-104A pilots from No 9 Sqn pose in front of F-104B 57-1312 during the autumn of 1967. In the back row, from left to right, are Sqn Ldrs Hatif and Farooq Feroze Khan, Wing Cdr M M Alam, Sqn Ldr Arif Iqbal and Flt Lts Abassi and Arif Manzoor. Kneeling, again from left to right, are Sqn Ldrs Farooq Umar and Rao Akhtar and Flt Lt R A Bhatti. They were the first group of F-104 pilots to be sent to France for conversion to the Mirage III (PAF via AVM Farooq Umar)



was seen, and IAF records did not include a Gnat among the 20 aircraft lost that day. However, an Indian source stated that Green had indeed been forced to eject from the aircraft.

Bhatti, with Sqn Ldr Amjad Hussein in 56-0804, was obliged to return to Amritsar later on 4 December after this aborted attack, where they were once again met by IAF aircraft. Two Su-7 'Fitters' appeared ahead of them, and as they turned to engage the Starfighters Amjad managed to get onto the tail of one of them, while the second Su-7 positioned itself behind him. To complete the chase, Bhatti closed up behind the second Sukhoi and called for Amjad to 'break right'. This enabled Bhatti to set up and fire an AIM-9B from 56-0879, and he claimed that the IAF fighter was hit and crashed – one of seven Su-7s destroyed in the day's fighting. Bhatti then accelerated ahead for a shot at the lead 'Fitter' and fired his second AIM-9B, but he did not see any result as he was simultaneously attempting to jettison his underwing fuel tanks. Only one tank released, throwing the Starfighter into a dangerous asymmetric roll at around 600 knots. Bhatti was able to climb, shed some speed and restore control for a safe recovery at Sargodha.

The Amritsar radar was still operational, so a fourth attack was organised. Sqn Ldr Amjad Hussain led the strike once again, with Flt Lt Samad Changezi as his wingman. On his first strafing pass Amjad was distracted by a decoy radar van, which he destroyed with 20 mm fire before seeing the real antenna operating nearby. Both pilots reefed their jets around for a second pass and took out the radar with gunfire, although Hussain's 56-0804 was hit by a burst of AAA from the 27th Air Defence Regiment. The long-suffering Starfighter began to burn and Amjad Hussain was forced to eject. He duly became a prisoner-of-war, and had the dubious distinction of being the only F-104A pilot to be shot down in both the 1965 and 1971 conflicts.

F-104As continued to fly ground attack missions alongside Sabres, including 30 sorties on 10 December against the strategically important Indian railway line that ran through the Sind desert.

Two more aerial victories for the dwindling No 9 Sqn Starfighters followed as the war intensified. The first victim was an Indian Navy Breguet Br 1050 Alizé anti-submarine aircraft of No 310 Sqn, which had waged a successful campaign against gunboats of the Pakistan Navy. Wg Cdr Arif Iqbal and Sqn Ldr Manzoor Bokhari were searching for Indian Navy vessels near Okha Port on 10 December when they encountered the Alizé and downed it with gunfire, killing the three-man crew. The following day Iqbal and Sqn Ldr Amanullah, who was flying the same F-104A (56-0874) that Jamal Khan had used to shoot down a Canberra in September 1965, destroyed an IAF HF-24 Marut fighter that was about to take off from Utterlai air base. Its pilot, Sqn Ldr Mickey Jatar, was lucky to escape with his life when 20 mm cannon shells raked his fighter. Wg Cdr Arif Iqbal claimed a second Marut and Amanullah strafed other aircraft on the flightline. Starfighter 56-0874 was subsequently preserved at Peshawar.

Air Marshal Nur Khan, commander of the PAF, just before his first F-104A solo. Flt Lts Middlecoat and Umar stand near the cockpit. Ironically, F-104A 56-0773 was the aircraft in which Middlecoat was shot down on 13 December 1971 (PAF via AVM Farooq Umar)



HASHEMITE STARS

The 'Griffins' move to Masroor air base near Karachi was followed by the loan of Jordanian aircraft. King Hussein had apparently received urgent Pakistani requests for this aid, and he sought American approval, which could not be given legally. On 12 December 1971 Gen Alexander M Haig, Deputy Assistant to the President for National Security Affairs, told Ambassador Huang Hua, the Permanent Representative to the UN of the Peoples' Republic of China, that Jordan 'has sent six fighter aircraft to Pakistan, and intends to send others up to a total of 14'. The latter would be replaced within the RJAF by fighters sent from Iran. On 19 January 1972 US National Security Advisor Henry Kissinger told President Richard Nixon that Jordan had sent 17 aircraft in all.

Kissinger regarded India as a 'Russian satellite' at the time, and he therefore sought covert ways of assisting Pakistan. The US Ambassador in Pakistan had also warned that the 'trickle' of Chinese MiG-19s and Jordanian F-104s arriving in Pakistan would be inadequate if India made an all-out attack on its neighbour.

Five RJAF F-104As were flown in by Jordanian and PAF pilots on 13 December and another five followed a short while later. Maj Ihsan Shurdom (later the commander-in-chief of the RJAF) led the first RJAF Starfighters into Masroor on 13 December, and the aircraft were to remain in Pakistan with PAF markings until three weeks after the war. Political constraints officially prevented them from crossing the Indian border on combat missions, however.

On 26 December three camouflaged Northrop F-5As in PAF markings passed through Tehran airport en route to Pakistan. They were part of a batch of eight originally destined for Libya but diverted from storage in California to Pakistan, via Iran and Turkey. It is conceivable that they were used in combat since Indian troops claimed to have discovered camouflaged wreckage that they stated was from an F-104. PAF and RJAF Starfighters were not camouflaged, unlike F-5As.

Pakistan's 1971 air war was essentially defensive, as the PAF's inability to secure air superiority over India due to a lack of fighters severely limited the number of air attacks it could mount on enemy forces. Airfields were hit whenever possible to try and reduce the IAF's considerable numerical advantage, and F-104As were used for their speed advantage, although their ordnance limitations confined them to strafing attacks, with AIM-9Bs for self-defence.

One such mission was against the heavily defended Jamnagar air base on 13 December. Thirty-one-year-old Wg Cdr Mervyn L Middlecoat (in F-104A 56-0773) led Flt Lt Tariq Habib at low level towards the target before climbing to 3000 ft to begin a strafing dive. Middlecoat, who had already made three reconnaissance flights over Indian airfields in recent days and completed 17 air defence sorties in 1965, had become the CO of No 9 Sqn during 1971. Positioning for the strafing run took several minutes, giving the defences additional preparation time. A missile was fired at Habib's aircraft as the pilot lined up for his strafing run, but a more sinister threat lurked above in the form of two MiG-21FLs. They had been alerted by coastal observation posts to the Starfighters' approach, giving the IAF pilots time to scramble from Jamnagar's Operational Readiness Platform and commence a CAP at 10,000 ft immediately overhead the air base.

Middlecoat, with two underwing Sidewinders and two wingtip tanks, thundered along the course of the main runway through 40 mm AAA, firing at parked aircraft that the IAF later claimed were decoys and egressing over the sea. A No 47 Sqn 'Black Archers' MiG-21FL (Type 77), flown by Flt Lt Bharat Bushan Soni, dived after the Starfighter in afterburner and fired a K-13 'Atoll' missile from an altitude of 100 ft, which missed its target.

Unfortunately for Middlecoat the MiG (coded C750) was one of a handful that had been fitted with the Soviet twin GSh-23 gun installation in a GP-9 gondola beneath the aircraft. Soni, some 900 ft behind the F-104, fired three bursts from the short-ranged gun pack and the Starfighter began to burn. Middlecoat told Habib that he could not reach land and ejected. His parachute just had time to deploy before his aircraft crashed inverted into the sea, witnessed by Habib. A rescue boat was sent to search for the pilot but no trace of Middlecoat was ever found in the shark-infested Gulf of Kutch. Moments later Habib saw a MiG-21 off his right wing, and he tried to move in behind it and engage the IAF jet, but a fault developed in his auto-stabilisation system and he had to break off and escape.

Middlecoat had earlier flown for King Hussein of Jordan during the Six Day War, and the King asked for the Jordanian flag to be buried with the pilot if he was ever found, together with the Pakistani flag.

As in 1965, India's Canberra fleet still posed a formidable threat to Pakistan's air bases at night in 1971. Being home to the F-104s, Masroor was particularly at risk of attack, so a CAP flight of Starfighters was detached to the nearby Faisal air base in an effort to catch incoming Canberras. It was during the course of one such CAP flight on 17 December that No 9 Sqn suffered its last combat loss. Sqn Ldr Bhatti and Flt Lt Samad Changezi, flying borrowed Jordanian F-104As 56-0839 and 56-0767, respectively, had been patrolling near the Indian border when they were ordered to orbit around Mirpur Khas and Naya Chor during a return flight from Faisal to Masroor. Two Utterlai-based MiG-21FLs had been detected escorting a flight of four HF-24 Maruts on an army support mission, the six IAF jets heading straight for the F-104s at an altitude of 10,000 ft.

MiG-21 pilots often flew seek-and-destroy missions at high altitude, with two Hunters or Maruts as bait to lure PAF interceptors into action. However, they usually reported little reaction from the Pakistani bases. This time, both pairs met in a head-on 'blow through' and then turned in behind each other. Changezi, who saw the MiGs first, broke away from Bhatti and turned in behind the lead MiG. He then closed to within gun range.

The need to use wingtip tanks for increased endurance had ruled out Sidewinder carriage on the RJAF F-104As, which had not been modified to carry missiles on their underwing pylons. External fuel tanks were not usually carried in combat due to the extra drag they created.

The second MiG-21FL (flown by Flt Lt Aruna Datta) moved in behind Changezi and fired an 'Atoll', but it was launched outside correct parameters. Moments after Changezi had ignored Bhatti's order to jettison his tanks and disengage, a second 'Atoll' smashed into 56-0767 and the jet exploded, killing the young pilot. The MiGs headed back to the border and Bhatti, low on fuel, had to return to Masroor. Two more F-104As were damaged by MiG-21FL gunfire and missiles during the closing hours of the war but both jets survived. For the remainder of



At the end of their service lives several RJAF F-104As were used at Prince Hassan AB as airfield decoys, painted in Mirage F1 camouflage colours, although some retained their original silver-grey finish. The M61 gun and other parts have been removed from this example, 910/M. PAF M61s were set to fire at 66 rounds per second (via T Panopalis)

the war the Jordanian aircraft were used for night air defence, carrying wingtip tanks for increased endurance and relying on their gun armament only.

The small PAF Starfighter force had been worked hard in both wars, flying 104 sorties in December 1971 alone. No 9 Sqn's limited opportunities for aerial combat showed that the unit was both a formidable opponent and deterrent. In its long-anticipated fights with the MiG-21 circumstances militated against it every time. A longer conflict might well have given the 'Griffins' a chance to face the MiGs on more equal terms, although one of the encounters showed the Soviet fighter to be a little more manoeuvrable than the F-104A at lower speeds.

Continued US restrictions on providing support for the PAF's small F-104 force eventually drove availability down to unfeasible levels, and the type was withdrawn a year later after 12 years of service and 11,690 flying hours. A total of six original PAF F-104As had been lost in accidents during this time. Six surviving airframes were preserved for display.

The return of Jordan's F-104 contingent (minus 56-0767, which Pakistan offered to replace) enabled the RJAF to resume operations with No 9 Sqn until July 1977, when the Starfighters were replaced by Northrop F-5E/F fighters. On 18 November 1972 the world's media were told that King Hussein, a keen pilot himself, came close to falling victim to his own air force. A rebel officer, allegedly Maj Talal al-Khatib, was involved in a coup attempt inspired by Palestine or Libya and he used an F-104A to strafe the King's Puma helicopter as he prepared to take off. The pilot's aim was poor and his Starfighter was apparently shot down by other F-104As, killing Khatib, who was one of the RJAF pilots trained by the USAF at Homestead AFB in 1968. The identity of his aircraft was said to be F-104A 56-0767, but it is likely that this serial was used to cover the loss of the jet in such controversial circumstances. Of course the original 56-0767 had been shot down during its loan spell with the PAF in December 1971.

Although Pakistan's F-104As were initially supplied to provide border defence against Soviet bombers, they were involved in a very different type of action on two occasions. During the conflicts of both 1965 and 1971, PAF Starfighter pilots experienced the most intense air-to-air combat situations faced by the F-104 anywhere. In these engagements they had achieved the kind of air superiority over much larger aerial forces that the Lockheed designers had originally envisaged for their Century Series fighter.

TAIWAN TENSIONS

The October 1958 ceasefire did not mark the end of F-104 operations in Taiwan, for the aircraft eventually became the RoCAF's longest-serving fighter. The release of Starfighters from USAF ADC duties allowed the supply of 22 F-104As, eight F-104Bs and six F-104Ds to the RoCAF's 8th TFS in 1960 under Project *Alishan 1* – an eleven-stage programme that imported F-104s from various sources. The 3rd Air Group operated these from Ching Chuan Kang AB, near Taipei.

Many F-104A pilots were trained by the 157th FIS at Congaree AFB, where there were occasional language-based problems, as Maj Gen Stan Hood recalled;

'One student commenced his take off without the use of afterburner as he had not activated the correct switch. His instructor told him to "Stop! Abort!" from the moment his brakes released, but he didn't know the Chinese word for "abort". The pilot went off the end of the runway and crashed in a field, where he was pulled from the burning aeroplane by Congaree's chief of maintenance. He was taken to Shaw AFB hospital and we bought him a radio, but the senior RoCAF officer told us not to give the pilot this present as he had "lost face". He returned to Taiwan, and seven years later when I asked about him, I was told he would have been made a social outcast and might have even committed suicide.'

When later Starfighter models became available some F-104As were returned to the USA, many for re-delivery to Pakistan and Jordan. Their replacements were 69 F-104Gs and nine TF-104Gs supplied directly from Lockheed and Canadair production, delivered between 1963-65.

Renewed hostility in 1967 brought these aircraft into combat with communist Chinese fighters, and on 13 January four F-104Gs from a batch of 27 Canadair-built aircraft tangled with eight J-6/MiG-19s. In the ensuing melee air-to-air victories were claimed by Maj Shih-lin Hu and Capt Bei-puo Shih, flying F-104Gs 64-17770 (coded 4344) and 64-17773 (coded 4347), respectively. Each claimed a communist aircraft destroyed, but one F-104G and its pilot were lost in return.

For Kelly Johnson this clash revealed the strengths and weaknesses of the F-104 in aerial combat. 'We had them on acceleration and we had them on steady-state altitude, but we could not turn with them', he noted. Johnson continued working on a big-wing, more manoeuvrable version of the F-104 known as the Lancer.

By 1970 Taiwan's need for improved interception capability prompted an order for another 22

RoCAF pilot Chien-shuo Chang in a semi-pressure suit poses with F-104G 62-12222 (coded 4322) at Ching Chuan Kang AB in the 1960s (*Chien-shuo Chang via Jing-ping Fu*)



refurbished F-104As. These ex-ADC aircraft, with uprated J79-GE-19 engines, were initially flown by the 41st TFS at Hsinchu AB prior to being transferred to the 8th TFS in 1974, where they served alongside the surviving F-104B/D two-seaters until 1988.

In due course the RoCAF obtained a variety of surplus Starfighters including F-104Ds, RF-104Gs and, in 1987, 31 ex-Japan Air Self-Defense Force F-104Js and five F-104DJs (with another ten time-expired airframes for spare parts), bringing the total used to at least 217 aircraft. Their arrival allowed the RoCAF to withdraw earlier versions from service by 1983. AIM-9J Sidewinders and Chung Shan Institute Sky Sword 1 all-aspect infrared missiles (similar to the AIM-9G Sidewinder) were available as armament upgrades. The largest batch of 'pre-owned' aircraft came in the form of 39 ex-Luftwaffe F-104Gs (including 11 modified as RF-104Gs and 27 TF-104Gs), delivered from 1984 onwards. Another 18 F-104Gs and TF-104Gs were added from Royal Danish Air Force surplus stock from 1988.

Attrition of the earlier Lockheed/Canadair-supplied aircraft had been high, with 58 aircraft allegedly lost in accidents. Six squadrons plus a reconnaissance unit remained on strength in 1990, with the intention of operating them until 1996. In 1993 the 427th TFW began to retire its aircraft, followed by the 499th TFW in 1997. On 22 May 1998, 38 years of service (totalling 380,000 flying hours) by RoCAF F-104s ended at Ching Chuan Kang AB with the disbandment of the 12th TRS. Several aircraft received special tail markings at this time to commemorate the eight units that had flown the aircraft and maintained RoCAF air superiority against a far larger communist threat.

CYPRUS

The Turkish Air Force (*Türk Hava Kuvvetleri*, or THK) took delivery of a batch of 46 brand new Lockheed and Canadair F-104Gs and TF-104Gs under MAP in 1963-66, followed by 350 others passed on by NATO partners Germany, Belgium, Spain, Norway, Canada and Holland. They were supplemented by a final batch of 40 F-104Ss from Aeritalia/Alenia production in 1974-76. Two squadrons (*Filos* 141 and 142) and an

Capt Shih-lin Hu points to the MiG kill marking on his F-104G 64-17773/4347. He scored one of the RoCAF Starfighters' two confirmed MiG victories on 13 January 1967 (*Family of Shih lin Hu via Jing-ping Fu*)



operational conversion unit flew the F-104Gs from Murted AB, while the AIM-7 Sparrow missile-firing F-104Ss were issued to *Filo* 142 and the squadrons of 9 *Ana Jet Us* at Balikesir.

When hostilities between Greece and Turkey flared up in 1963 over their respective areas of occupation of the newly-independent island of Cyprus the F-104G units were still training. Fighter-bomber duties were therefore undertaken by Turkey's F-84G Thunderjet and F-100C/D Super Sabre squadrons. An uneasy truce was then maintained for ten years, but the appointment of the Greek Nicos Sampson as president of the country in July 1974 re-ignited the fighting. Turkey dropped paratroops around the capital, Nicosia, and mounted a seaborne invasion of the island. Air strikes were flown by three squadrons of F-100s and the F-104Gs of *Filo* 141, hitting targets around Nicosia including the National Guard barracks. Balikesir's *Filo* 191 F-104Gs were moved to Murted to supplement *Filo* 141 and a second wave of amphibious forces was launched on 21 July, ostensibly to protect Turkish Cypriot citizens.

As the conflict burgeoned and peace talks stalled, many anticipated clashes between THK Starfighters and the 170+ examples of the Lockheed fighter flown by the Greek Air Force – it too had also received batches of newly-built MAP F-104Gs from 1961 and others passed on by fellow NATO air forces. However, events for the THK took a different turn. The Greek Navy sent the landing craft HS *Lesvos* (L-176) to Paphos, giving rise to reports that a Greek convoy was being assembled to counter the Turkish amphibious landing. Grumman S-2E Tracker aircraft were sent aloft to investigate, and their radar reports seemed to indicate a substantial enemy naval force, although a daytime reconnaissance flight by a THK RF-84F Thunderflash pilot found no supporting evidence. Radar sweeps by a coastal station and further S-2E flights still showed a potential threat, however, with intercepted Greek radio traffic indicating that three Hellenic navy destroyers were attempting to join up with HS *Lesvos*.

A strike force of fighters, including the F-100Ds of *Filo* 111 and the F-104Gs of *Filo* 141, was armed with M117 750-lb bombs and cleared to attack at 1300 hrs on 21 July after the Turkish Navy confirmed that it had no vessels in the target area. Sixteen F-104Gs took off from Murted at 1443 hrs and 28 F-100Ds departed Eskisehir and Antalya, with F-104Gs from *Filo* 191 and F-100Cs from *Filo* 112 on alert for follow-up

TF-104G 5911, one of 36 ex-Luftwaffe two-seaters supplied to the Turkish Air Force (*Türk Hava Kuvvetleri*) and operated here by 9 *Ana Jet Us* at Balikesir AB. F-104 units were deployed from Balikesir AB to Murted AB during the 1974 conflict with Greece over Cyprus (*Author's Collection*)



attacks. At 1530 hrs the tragic news was received that the fighters had actually attacked the Turkish destroyers *Kotacepe* (D-354), *Adatepe* (D-353) and *Maresal Fevzi Cazmak* (D-349). The intercepted Greek radio traffic (originating from the Greek Cypriot Naval Command) had been a ruse, which described the Turkish vessels as Greek, relying on the fact that both navies operated similar-looking destroyers.

All three ships were hit by F-104Gs and F-100s in no fewer than 203 attack sorties, and *Kotacepe* sank with the loss of 54 of its crew. Reconnaissance flights by *Filo* 191 F-104Gs were undertaken to check for other Greek intruders, but they located only four small, unthreatening boats. One F-104G (64-17783) from *Filo* 191 was lost, and confirmation of the sinking of *Kotacepe* was received the following morning. Later that same day 17 F-104Gs from *Filo* 141 hit Nicosia airport (a particular hot-spot in the conflict) with M117s, shortly after which an armistice was declared. American intervention forestalled attacks on Turkish forces by Greek F-4E Phantom IIs of 117 Wing, but sporadic air attacks by THK fighters began again on 14 August, leading to a ceasefire the following day and the imposition of the 'Attila' line, partitioning Cyprus into two regions.

LIBYA, IRAQ AND KOSOVO

The Starfighter's final wartime scenarios both involved F-104s of the Italian Air Force, which operated the aircraft longer than any other NATO air arm, and considerably longer than the jet's 15 years of USAF service. Its first F-104Gs, selected in preference to the F-4 Phantom II and Dassault Mirage III, were received in March 1963, and 11 *Aeronautica Militaire Italiana* (AMI) *Stormos* eventually flew 136 Fiat-built F/R/T-F-104Gs (plus 12 Lockheed TF-104Gs) and 206 of the Aeritalia-produced F-104S 'Spillone' version. With a versatile air-to-air upload of AIM-7 or Aspide radar-guided missiles and AIM-9L Sidewinders (but no gun), and a performance that was still as spectacular as it had been in 1954, the F-104S was limited only by the relatively short range of its look-down/shoot-down radar and the lack of in-flight refuelling equipment.

These shortcomings were overlooked when Col Muammar al Gaddafi's forces fired several 'Scud' missiles towards the Italian island of Lampedusa following USAF and US Navy air strikes on Libya on 14 April 1986. AMI Starfighters were immediately moved to bases in southern Italy, their mission being to protect the US Navy's LORAN (long range navigation) installation on Lampedusa and deter any further hostile action by Gaddafi's air force or missile batteries. Armed with live weapons, the F-104s flew CAPs and escorted civilian airliners in areas that were vulnerable to Libyan intervention.

From 5 January until early March 1991, six Villafranca-based 3rd Stormo F-104Gs with Orpheus reconnaissance pods were detached to Erhac air base, in Turkey, for Operation *ACE Guard*, which was part of Operation *Provide Comfort* – the Coalition's campaign to protect the Kurdish population of northern Iraq. Their daily missions with the ACE Mobile Force took the Starfighters over some hostile terrain in challenging winter weather, but the imagery taken during these flights was very useful when it came to monitoring the threat posed to the local population by President Saddam Hussein's forces. The F-104s' primary task, however, was the air



defence of Turkish territory, rather than engaging with Iraqi aircraft. The Mobile Force also included Hawk ground-to-air missile batteries.

F-104G/S units from 9°, 10°, 18°, 20° and 23° *Gruppi* deployed to Amendola and Gioia del Colle, in Italy, for quick-reaction alert duties during Operation *Allied Force* (the Kosovo campaign) in 1999. They also flew CAP missions to replace other NATO fighters that were involved in intercepting Serbian Air Force aircraft. In all, the F-104s flew more than 300 hours in *Allied Force*.

The Starfighter's Cold War service in other air forces, while beyond the remit of this book, drew much media attention when loss rates in some air arms reached unacceptable levels. The USAF's role in training many NATO pilots was inevitably examined as part of the several subsequent enquiries that were convened, particularly in West Germany.

Lt Gen Tom Baker, a former instructor pilot at Luke AFB, emphasised that the F-104 was essentially 'easy to fly. It had some "squirrely" characteristics, but it was okay if you were aware of them and stayed out of those situations. There is no doubt that the inexperience of some senior Luftwaffe officers in the early 1960s caused them to make decisions regarding the operation of this very high-performance aeroplane in Europe, with its associated poor weather conditions, that pushed a lot of guys into situations that were high risk. A high accident rate was the end product of these decisions.

'Gen Johannes Steinhoff [Luftwaffe Chief of Staff] came to Luke AFB searching for an answer. The accident rate here was not high, and it was obvious that conditions in Western Europe were very different which meant that pilots were not being prepared for frontline flying. One of their "fixes" was to make pilots [in Europe] fly the maximum number of hours per sortie, but numbers of hours was not the real answer. It was a safe aeroplane if flown in the correct environment by experienced people. If you degraded one of those factors then you were probably set up for a high accident rate.'

Apart from sporadic combat encounters within NATO, the F-104's role was principally as a deterrent, as it had been in Vietnam, Taiwan and Pakistan. In Air Vice Marshal Farooq Umar's view, 'my 1000 hours on the Starfighter proved to my colleagues and I that the jet had an awesome effect on its opponents. The F-104 could climb like a rocket and accelerate like a bullet. Anytime the opponents saw or heard of a Starfighter in the vicinity they hastily retreated, and didn't engage the F-104.'

F-104S 5-42 in 1985 with the distinctive 5° *Stormo* 'Diana Cacciatrice' fin badge and 23° *Gruppo* insignia on its flank. The first F-104Ss were Lockheed conversions of F-104Gs, and they entered service as interceptors in 1969. In 1999 this unit flew CAP and quick-reaction alert missions during Operation *Allied Force*, although its markings were more subdued by that time (Author's Collection)

APPENDICES

COLOUR PLATES

1

F-104A-25-LO 56-0842 of the 83rd FIS/13th Air Task Force, Tao Yuan AB, Taiwan, September 1958

This aircraft was assigned to Capt M C Crompton, and TSgt D M Mayfield was its crew chief, during Operation *Jonah Able*. Note that the aircraft's 20 mm gun was faired over for this deployment and a tailhook had yet to be installed. 56-0842 was subsequently transferred to the RoCAF as 4205, and it was lost on 10 October 1964 in a mid-air collision.

2

F-104C-5-LO 56-0892 of the 476th TFS/479th TFW, George AFB, California, 1958

The first markings displayed on the Starfighters of TAC's only F-104C wing reflected the exuberant personality of its commander, Col George Laven. His own aircraft, 56-0891, was famously trimmed with numerous gaudy decorations that raised some questions about the cost of their application, and maintenance. The remaining aircraft in the wing used slightly more restrained schemes based on their squadron colours. 56-0892 completed Vietnam service with the 435th and 476th TFSs at Da Nang AB and Udorn RTAFB. Fitted with the tail unit from an F-104G, the jet has been on display at Luke AFB, Arizona, for many years. It wears the serial 61-3243, which was the identity of an F-104G based at Luke and used by the 58th Tactical Training Wing (TTW) to train German pilots for service with the Luftwaffe from the early 1960s through to the 1980s. Amongst the last German F-104s at Luke, the real 61-3243 was supplied to the RoCAF upon retirement by the 58th TTW and written off in a crash in 1990.

3

F-104A-15-LO 56-0769 of the 83rd FIS/78th FG, Hamilton AFB, California, 1959

The 83rd FIS was the first frontline unit to operate the Starfighter, receiving jets from February 1958. Part of the 78th FG, its primary mission was high-altitude interception. 56-0769 (as 4208) was one of the first Starfighters to be transferred to the RoCAF in 1960, and it crashed during an emergency landing at Hsinchu AB in December 1963.

4

F-104A-15-LO 56-0782 of the 157th FIS/169th FIG, South Carolina ANG, Congaree AB, South Carolina, 1960

'SCANG' was the first ANG unit to receive the F-104A, and this particular jet has wingtip tanks and a faired-over gun port 56-0782, which also deployed to Ramstein AB in with the 197th FIS, Arizona ANG, in 1961 for the Berlin Wall crisis ended its service life with the RJAF's No 9 Sqn as '922'.

5

F-104A-20-LO 56-0795 of the 197th FIS/161st FIG, Arizona ANG, Skyharbour Airport, Arizona, 1961

Adorned with *BAD DOG 1* nose art, this jet was among the unit's 20 aircraft that deployed to Ramstein AB, during Operation *Brass Ring*. Whilst in Europe the unit came under the control of the Seventeenth Air Force's 86th Air (Defense) Division during the Berlin Wall Crisis in November 1961. 56-0795 was withdrawn from service in March 1968 and eventually scrapped at Mojave in 1982.

6

F-104A-25-LO 4201 (56-0839) of the 8th TFS/3rd TFW, Ching Chuan Kang AB, Taiwan, 1961

This fighter was included in the first batch of F-104As to be delivered to the RoCAF's 3rd Air Group in 1960. It was returned to the USA in 1966 and re-allocated to the RJAF in October of that same year for service with its No 9 Sqn as 914/L. Finally, the veteran fighter was used as a decoy at Al Shaheed Muwaffaq Salti AB.

7

F-104C-5-LO 56-0929 of the 434th TFS/479th TFW, George AFB, California, 1961

This aircraft deployed to Hahn AB and Morón AB during the Berlin Wall Crisis in August 1961. Following combat in Vietnam, the jet finished its operational career with the Puerto Rico ANG. It has been on display in the Southern Museum of Flight in Birmingham, Alabama, for many years.

8

F-104A-25-LO 56-0834 of the 151st FIS/134th FIG, Tennessee ANG, Ramstein AB, West Germany, May 1962

This aircraft was among the unit's 18 F-104As and two F-104Bs that were deployed to Ramstein AB in May 1962 during the Berlin Wall crisis. Three jets were lost in accidents over Germany during the deployment. 56-0834 was transferred to the RJAF in 1973 and concluded its career as a gate guard at Sawafi AB.

9

F-104A-20-LO 56-0851 of the 319th FIS/32nd AD, Homestead AFB, Florida, 1963

Following the Cuban Missile Crisis the 319th FS, formerly equipped with F-102A Delta Daggers, transitioned to F-104As withdrawn from the 159th FIS, SCANG. It resumed alert duties at Homestead AFB in April 1963, thus returning the F-104A to ADC control once again. 56-0851 was destroyed by fire during a maintenance mishap in January 1965.

10

F-104C-10-LO 57-0915 of the 434th TFS/479th TFW, George AFB, California, July 1963

Equipped with a bolt-on flight-refuelling probe, this aircraft flew across CONUS from California to Myrtle Beach, South Carolina, in July 1963 to participate in Operation *Swift Strike III*. The latter was an annual joint-services operation in South

Carolina that provided training in the kind of counter-insurgency warfare that would be experienced in Southeast Asia. Following its retirement 57-0915 was preserved at the Joe Davies Heritage Airpark at Palmdale, California.

11

F-104C-5-LO 56-0886 of the 479th TFW, Da Nang AB, South Vietnam, April 1965

Christened *fannie*, this aircraft was one of the few F-104Cs to bear a nickname during the 479th TFW's time at Da Nang. It was flown by both the 435th and 436th TFSs whilst in-theatre, spending seven months in South Vietnam. 56-0886 returned to Southeast Asia in June 1966 to operate from Udorn RTAFB, before eventually being passed on to the Puerto Rican ANG's 198th TFS. Preserved at Holloman AFB Museum since the late 1970s, it has worn the serial 56-0764 for many years. The latter machine was an F-104A that crashed near Edwards AFB on 15 June 1964 after entering a flat spin.

12

F-104C-5-LO 56-0883 of the 476th TFS/479th TFW, Da Nang AB, July 1965

Capt Phillip E Smith was shot down in this, the first production F-104C, near Hainan Island on 20 September 1965. Two communist Chinese Shenyang J-6s (MiG-19s) dived on his aircraft while he was attempting to establish his location after the loss of his navigation indicators. Smith subsequently spent more than seven years in solitary confinement in a Chinese prison.

13

F-104A-25-LO 56-0874 of No 9 Sqn 'Griffins', Sargodha AB, Pakistan, September 1965

This aircraft was flown by Wg Cdr Jamal A Khan on the night of 20/21 September 1965 when he shot down an IAF Canberra B(I) 58. On 11 December 1971 Sqn Ldr Amanullah Khan was at its controls during an attack of Utterlai AB that saw him destroy an IAF HF-24 Marut light strike aircraft with 20 mm gunfire. The Starfighter, which first served with the USAF's 56th FIS at Wright-Patterson AFB, was subsequently preserved at Sargodha AB.

14

F-104A-25-LO 56-0877 of No 9 Sqn 'Griffins', Sargodha AB, Pakistan, September 1965

This aircraft was flown by Flt Lt Amjad Hussain when he shot down Sqn Ldr A B Devayya's IAF Mystère IVA on 7 September 1965, but the aircraft was lost in the same engagement, probably due to damage by debris from the French jet. Previously, it was involved in the forced landing of an IAF Gnat fighter on 3 September 1965, the F-104 being flown by Flg Off Abbas Mirza on that occasion.

15

F-104B-5-LO 57-1312 of No 9 Sqn 'Griffins', Sargodha AB, Pakistan, September 1965

One of two F-104Bs delivered to the PAF in June 1961, this aircraft was used not only for pilot training but also for reconnaissance flights over Indian bases in September 1965. Both F-104Bs survived the Indo-Pakistan wars and were put on permanent display at PAF installations following their retirement. 57-1312 is presently displayed wearing the serial 57-1309 at the PAF Academy at Risalpur.

16

F-104C-5-LO 56-0919 of the 435th TFS/479th TFW, Da Nang AB, December 1965

This aircraft is illustrated in the four drop-tank external fuel configuration used for many of the *Big Eye/Silver Dawn* escort missions flown by the wing in 1965-66. 56-0919 also flew with the Air Force Flight Test Center, and following its retirement the jet has been on display atop a pole at NAS Glynnco Jetport in Brunswick, Georgia, wearing the markings of the Georgia ANG. Although the latter never operated the Starfighter, F-104s from the 'PRANG' visited Georgia annually for the 198th TFS's summer training camp during the aircraft's eight years with the unit.

17

F-104C-10-LO 57-0925 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, July 1966

This Starfighter arrived at Udorn on 22 July 1966 and was soon nicknamed *Smoke II*. It lasted little more than a week in-theatre, however, as on 1 August it was hit by an SA-2 missile while escorting an F-105F *Wild Weasel* flight during an attack near Thai Nguyen. Lt Col Arthur 'Al' Finney was killed in the explosion.

18

F-104C-10-LO 57-0928 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

This aircraft was amongst the first 12 F-104Cs to arrive at Udorn RTAFB on 22 July 1966 for the Starfighter's second wartime deployment – it had been in-theatre with the 436th TFS from September to December of the previous year. Note the AN/APR-25/26 RHAW equipment installed in the nose and rear fuselage of the jet. The unit's management by the 8th TFW is indicated by the blue and yellow 'Wolfpack' flash on the yaw damper at the base of the rudder. Capt Harold Alston flew this F-104C for his 100th combat mission, thus becoming the first Starfighter pilot to pass the century mark. A veteran of the 436th TFS's Berlin Wall Crisis deployment to Hahn AB in January 1962, 57-0928 was one of the first F-104Cs to be re-assigned to the Puerto Rico ANG in June 1967. Transferred to museum status and stored at Muñiz airport from June 1975, the aircraft was destroyed on 12 January 1981 during the *Ejército Popular Boricua* attack on the Muñiz ANG Base that also saw nine A-7D Corsair IIs blown up. The remains of the Starfighter were used for practice by firefighting crews on base until eventually scrapped.

19

F-104C-10-LO 57-0923 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

This aircraft was assigned to the squadron commander, Lt Col Robert Preciado (who led the F-104C contribution to Operation *Bolo*), and named *HELLOOO DOLLY* after his wife, with a nod to the Barbra Streisand musical also. Its crew chief for the tour was SSgt Bennett. In July 1967 57-0923 was re-assigned to the 198th TFS, Puerto Rico ANG, and it crashed on takeoff at Savannah, Georgia, during a training camp in August 1968.

20

F-104C-5-LO 56-0910 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

Lt Jim Trice organised the eye-catching *PUSSYCAT* artwork that adorned the nose of his assigned F-104C, which was in the

first batch of Starfighters to arrive at Udorn on 6 June 1966. Passed on to the 'PRANG' following its service in Southeast Asia, the jet was eventually retired in 1975 and subsequently became a display item in the Wings over the Rockies Museum in Denver, Colorado.

21

F-104C-5-LO 56-0891 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

Piloted by Maj Herb Drisko this aircraft was named *NANCY J* (after his wife) on its left side and sported some of the squadron's most sophisticated nose-art, along with the name "*SNOOPY SNIPER*", on the right. The jet's crew chief was Sgt Joel Swanson. Following its withdrawal from service with the 'PRANG' in August 1975, 56-0891 became another museum artefact – albeit in the markings of an Arizona ANG F-104A – at Sky Harbor Airport in Phoenix, Arizona.

22

F-104C-5-LO 56-0938 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

The nickname of this jet was derived from the home state of its pilot, Maj Robert Bedsworth, who hailed from Missouri – the 'show me' state. Reinforcing his connections with home, Bedsworth also adorned the jet with a Missouri mule, the state animal. 56-0938 carries a typical ground attack load of two M117 750-lb General Purpose bombs. Yet another museum exhibit, this F-104C has been on display atop a small plinth within the Keesler AFB Air Park, in Mississippi, since the late 1970s, still in its 'PRANG' markings.

23

F-104C-5-LO 57-0927 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

Named *Debbie Sue* after the daughter of its pilot, Capt Rodney Trimble, this aircraft features the Texas state flag on its yaw damper. The Starfighter survived the war and joined the 198th TFS at Muñiz airport in June 1967. It was destroyed on 19 June 1969 when the jet suffered an engine flame-out over the Poinsett bombing and target range and crashed during the 'PRANG's' annual summer training camp in Savannah, Georgia.

24

F-104C-5-LO 56-0892 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, 1967

Maj Floyd Totten flew this F-104C, nicknamed *MISS BEVIE J. VI* after his first girlfriend – his five previous aircraft had also been *MISS BEVIE J.s* as well. Paint erosion reduced the 'B' in her name to a 'P', and the jet was then re-named *My Darlin' Dorothy* after Totten's new bride, with crew names displayed behind the canopy area. This aircraft is also the subject of Profile 2.

25

F-104C-5-LO 56-0902 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, spring 1967

Assigned to Lt Addison 'Ace' Rawlins and named (on both sides of its nose) after his wife Judy, this aircraft had SSgt Patton as its crew chief. In later service with the 198th TFS it was destroyed during a high speed, high-altitude flight control test on 23 April 1968 when it became unstable.

26

F-104C-5-LO 56-0936 of the 435th TFS/479th and 8th TFWs, Udorn RTAFB, Thailand, spring 1967

The aircraft was assigned to Tom Mahan and christened *Lil' Poo II*, which was the pilot's nickname for his wife, Shirley. It had been delivered to the 479th TFW on 4 October 1959, deployed to Ramstein AB, West Germany, with the wing from 19 September 1961 to 22 January 1962 and then seen combat from Da Nang AB in 1965. The jet returned to Southeast Asia in late 1966, operating from Udorn from 21 November. In August 1967 56-0936 was reassigned to the 'PRANG', with whom it remained until retired in August 1975 to the Edward J Peterson Air and Space Museum within Peterson AFB, Colorado. Placed on static display shortly thereafter, the aircraft was repainted as F-104A 56-0808 (written off in RoCAF service on 16 April 1984) in honour of the Starfighter's service with ADC.

27

F-104G-LO 4347 (64-17773) of the 427th TFS/3rd TFW, Ching Chuan Kang AB, Taiwan, 1967

This Canadair-built aircraft was being flown by Capt Shih-Lin Hu when he shot down a communist Chinese Shenyang J-6 (MiG-19) with an AIM-9B Sidewinder on 13 January 1967. The fighter was subsequently written off after an aborted takeoff on 29 April 1971 saw it come to rest on a golf course.

28

F-104A-15-LO 910/H (56-0788) of No 9 Sqn Prince Hassan (Safawi) AB, Jordan, 1969

The RJAF sent nine F-104As to reinforce PAF numbers in December 1971. Previously, this aircraft was among the first batch of F-104As to be delivered to the RoCAF, as 4217, in the 1960 Project *Alishan 1* following its ADC service. Upon the aircraft's retirement, 56-0788 was stored for many years at Safawi AB.

29

F-104G-LO 62-12344 of 141 *Filo*, Murted AB, Turkey, July 1974

Part of the original batch of 46 Canadair F/TF-104Gs delivered to Turkey under MAP in 1963-66, this aircraft took part in the tragic THK attack on Turkish naval vessels on 21 July 1974 during the Greek-Turkish conflict over Cyprus. Since its retirement, the jet has been displayed at the Yesilkoy aviation museum in Istanbul.

30

TF-104G-LO 63-8455 of 2. *Deutsche Luftwaffen-Ausbildungsstaffel* F-104 USA/58th TTW, Luke AFB, Arizona, 1981

USAF instructors prepared German pilots for their Cold War duties using Lockheed-built TF-104Gs and Luftwaffe F-104Gs in USAF markings. Built in 1963, this fighter served with the 4510th CCTW/58th TTW until it was sold to the USAF in July 1983 after completing 3082 flying hours. Supplied to the RoCAF (as 4183) in Project *Alishan 8* during 1984, this F-104G was lost in a fatal takeoff accident at Ping Tung AB in August 1986.

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Front Cover

Escort duties for *Big Eye* and *Silver Dawn* electronic reconnaissance aircraft occupied much of the mission time for 479th Tactical Fighter Wing F-104C pilots based at Da Nang and Udorn during the Vietnam War. Their presence was extremely effective in deterring North Vietnamese MiGs from attacking these invaluable but vulnerable aerial vantage points. Their lengthy sorties were occasionally enlivened by threatening approaches by MiGs, but the Starfighter pilots knew that their missions had been a success when they and their piston-engined charges returned safely to base.

If the escort fighters were unavailable for any reason both the *Big Eye* flights and the day's air strikes on North Vietnamese targets were usually cancelled.

These 476th Tactical Fighter Squadron F-104C-5-LOs are loaded with underwing fuel tanks for the long patrol and a pair of AIM-9B Sidewinder missiles in case the MiGs get too nosey. The 476th was the first of four Starfighter units to deploy to Southeast Asia, flying from Da Nang between 19 April and 11 July 1965 (Cover artwork by Gareth Hector)

Previous Spread

F-104C 56-0910 (foreground), loaded with M117 bombs for a November 1966 ground attack mission, was later nicknamed *PUSSYCAT*. It is seen here before the installation of ECM antennae (1352nd Photo Group/USAF)

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